ScienceDirect

Journal of Manufacturing Processes

Date: 17 January 2025

Show more

Review article Full text access Get rights and content

Review article

Review of in-situ process monitoring for ultra-short pulse laser micromanufacturing

Kerim Yildirim [a], Balasubramanian Nagarajan [a], Tegoeh Tjahjowidodo [b], Sylvie Castagne [a] Show more

View PDF Download full issue Cite Add to Mendeley Share 10.1016/j.jmapro.2024.12.011

Article Recommended articles Cited by (21) Metrics

Outline

Abstract

Ultrashort pulse (USP) lasers enable machining of a wide range of materials, including hard and brittle materials such as steel, glass, silicon, and ceramics. Despite ongoing technological advances in USP laser micromachining systems, the lack of process repeatability and stability continues to hinder industrial adoption. This necessitates the development of innovative in-situ monitoring techniques capable of maintaining process stability and detecting early onset of anomalies. Besides, tight tolerances in micromanufacturing strongly require an automated, controlled, and regulated process. On the one hand, USP laser micromachining systems must be outfitted with in-situ sensing devices capable of measuring essential process signatures. On the other hand, in-process data analytics and statistical techniques are required to facilitate closed-loop process control. This paper provided a detailed literature review of in-situ monitoring of USP laser micromachining and other laser-based manufacturing processes. The mechanisms, advantages, limitations, signal processing methodologies and application in laser-based manufacturing techniques of different sensing techniques including acoustic, optical and image-based techniques are discussed. The techniques are reviewed with special focus on adaptability for USP laser micromachining towards automated process control, understanding process mechanisms and quality control.

Keywords

Ultra-short pulsed laser ablation; Femtosecond laser micromachining; In-situ process monitoring; In-line measurement; Multi-sensor monitoring; Real-time process monitoring

Previous article in this issue

Next article in this issue

Nomenclature

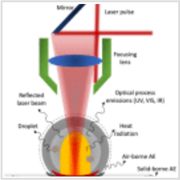
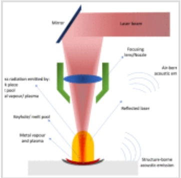
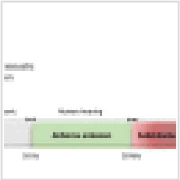
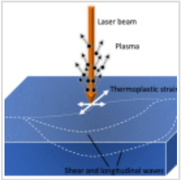
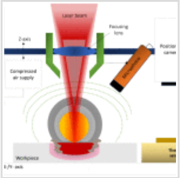
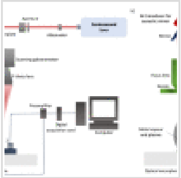
ABAE
: Air-Borne Acoustic Emission

AE
: Acoustic Emission

AFM
: Atomic Force Microscopy

AM
: Additive Manufacturing

Figures (28)

Show 22 more figures

Tables (12)

ANN
: Artificial Neural Network

ASW
: Acoustic Shock Waves

CCD
: Charge-Coupled-Device

CMOS
: Complementary Metal-Oxide Semiconductor

CNN
: Convolutional Neural Networks

CUP
: Compressed Ultrafast Photography

DBN
: Deep Belief Network

DED
: Direct Energy Deposition

DLMS
: Direct Laser Material Sintering

DOF
: Depth of Field

DP
: Double Pulse

DRC

Depth Response Curve

FBG
: Fibre Bragg Grating

FD-OCT
: Frequency Domain Optical Coherence Tomography

FFT
: Fast Fourier Transform

FINCOPA
: Framing Imaging based on Noncollinear Optical Parametric Amplification

FIRE
: Fluorescence by Radiofrequency-Tagged Emission

fs
: femtosecond

FTOP
: fs Time-Resolved Optical Polarography

Ge
: Germanium

HDR
: High Dynamic Range

ICI
: Inline Coherent Imaging

InGaAs
: Indium-Gallium-Arsenide

IR
: Infrared

LBR
: Laser Back Reflection

LCI
: Low Coherence Interferometry

LIBS
: Laser Induced Breakdown Spectroscopy

LMD
: Laser Material Deposition

LPBF
: Laser Powder Bed Fusion

LPP
: Laser Produced Plasma

LSP
: Laser Shock Peening

LSW
: Laser Shock Waves

MARS
: Multivariate Adaptive Regression Spline

MEMS
: Microelectromechanical Systems

Mfps

Million frames per second

NCOPA
: Noncollinear Optical Parametric Amplification

ND
: Neural Density

NDT
: Non-Destructive Testing

NIR
: Near Infrared

ns
: nanosecond

OCT
: Optical Coherence Tomography

OES
: Optical Emission Spectroscopy

PC
: Polarization Controller

PES
: Plasma Emission Spectroscopy

ps
: picosecond

PSD
: Power Spectrum Density

PZT
: Piezoelectric Transducer

RF
: Random Forest

RMS
: Root Mean Square

SAW
: Surface Acoustic Waves

SBAE
: Structure-Borne Acoustic Emission

SD-OCT
: Spectral Domain Optical Coherence Tomography

SEM
: Scanning Electron Microscopy

SLD
: Super Luminescent Diode

SMT
: Single-Shot Multispectral Tomography

SNR
: Signal-to-Noise Ratio

SP
: Single Pulse

SS-OCT

Swept Source Optical Coherence Tomography

STAM
: Sequentially Timed All-Optical Mapping Photography

STD
: Standard Deviation

STEAM
: Serial Time-Encoded Amplified Microscopy

STFT
: Short Fourier Transform

STMV
: Short Time Mean Value

SVD
: Singular Value Decomposition

SVM
: Support Vector Machine

TD-OCT
: Time Domain Optical Coherence Tomography

Tfps
: Trillion frames per second

UHR-OCT
: Ultra-High Resolution Optical Coherence Tomography

USP
: Ultra-short pulse

UV
: Ultraviolet

VIS
: Visible

WAAM
: Wire Arc Additive Manufacturing

XRF
: X-ray Fluorescence

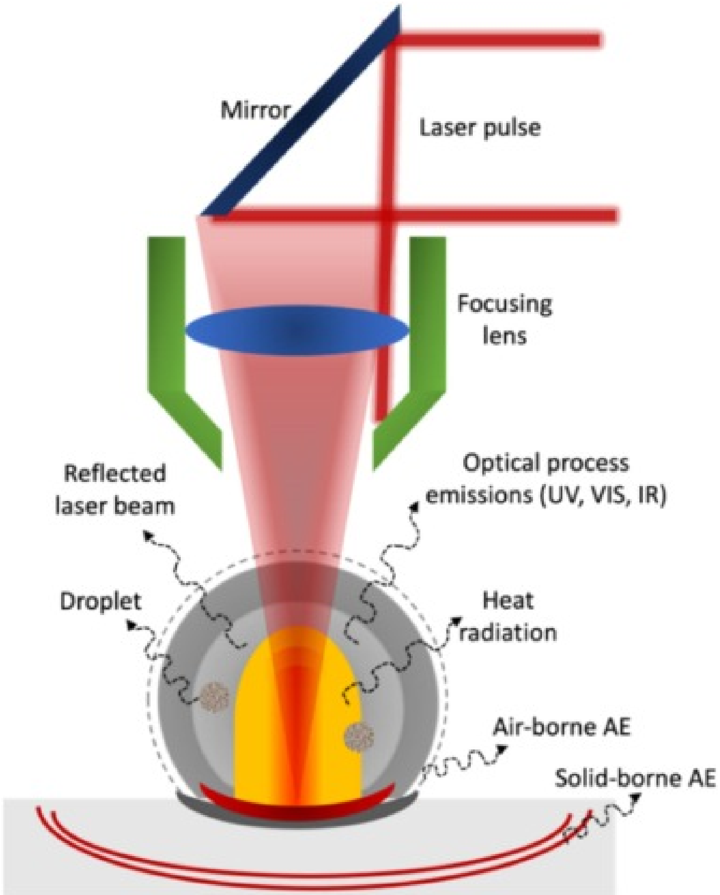
Introduction

Laser micromanufacturing has gained significant attention due to its potential applications in various fields such as space technology, medical science, nanotechnology, and electronics which involves the fabrication of small-scale components and devices [1,2]. Current needs in high-technology industries, such as semiconductor, optics, MEMS, etc., have necessitated the development of manufacturing techniques that can reliably produce ever-smaller features with extremely tight tolerances [3]. The development of ultrashort pulse (USP) lasers, which has pulse duration in the range of femtoseconds (10^{-15} s) to picoseconds (10^{-12} s), offers high precision and quality necessary in micromachining applications [4,5]. The primary benefits of material processing with USP lasers are: i) efficient, rapid, and localized energy deposition; ii) well-defined ablation thresholds [6]; and iii) negligible or minimal thermal damage to the substrate [7], and iv) ability to process almost any material (e.g., metals, ceramics, glass, polymers, organic tissue, etc.) [8]. The machined features are extremely smooth and free of burrs, and the surface is clear of redeposited particles [9]. The industrial use of these laser systems for micromachining continues to grow in relevance across a variety of sectors and applications and has become one of the most essential microfabrication tools [10].

However, USP laser micromachining has certain limitations that prevent its complete implementation for processing complex geometry at high throughput. Firstly, USP laser processing is influenced by a number of process parameters related to laser beam (e.g. wavelength, pulse duration, laser fluence), scanning (e.g. speed, hatching strategy), workpiece (e.g. absorption coefficient, roughness), and process environment [11]. Typically, trial-and-error experimentation is conducted to optimize the process variables to achieve the required feature dimensions and quality, which is laborious and time-consuming [12]. Secondly, prediction of material ablation behaviour is typically achieved by physics-based, numerical and empirical modelling of the USP laser process. USP laser micromachining lacks a reliable physics-based ablation model due to the complex process dynamics and the large number of process variables [13,14]. However, there are several numerical modelling approaches for femtosecond (fs) laser ablation, but they are limited by the complexity of the ablation mechanisms involving material removal at different spatial (atomic level to microscales) and temporal (fs to ns) scales [15]. As different theoretical models have their own temporal and spatial scale constraints, it is challenging to describe the complete process using a single theoretical model. Due to the limitations of the physics-based models, a multiscale theoretical model is proposed to describe and predict the interaction between a fs laser and a material [16]. The simulation of these physical processes has been comprehensive, but numerical methods may have reached their practical limits [17]. Thirdly, USP laser micromachining requires iterative offline characterization techniques such as surface profilometry, atomic force microscopy (AFM) and scanning electron microscopy (SEM) to achieve the desired surface topography and morphology, which significantly increases the production time and cost [18,19].

To overcome these challenges, in-process monitoring of USP laser micromachining is a promising approach to understand the process mechanics and to achieve real-time process control. By continuously monitoring key parameters, manufacturers can ensure that the laser-based micromanufacturing process is operating within the desired range and producing consistent, high-quality products [20]. Sensor-based monitoring provides useful information about the manufacturing process that can serve the dual function of process control and quality monitoring and will eventually be included into every fully automated production environment. For any sensor to be deployed as a monitoring device, however, a high level of confidence and accuracy in characterizing the manufacturing process is necessary [3]. Furthermore, the collected data from process monitoring can be analysed to identify trends, optimize process parameters, and develop improved process control strategies. Monitoring the advancement of the laser ablation allows real-time feedback adjustment of the laser micromachining processing parameters. Therefore, it is possible to process unknown materials without prior knowledge regarding appropriate processing settings [12,18,21].

Laser material interaction during USP laser ablation involves different stages including ablation, vaporization, plasma formation and redeposition based on the applied energy density. For detailed understanding of the laser-material interaction during USP timescales, the readers are referred to [[22], [23], [24], [25]]. Fig. 1 shows an example of the process signals emitted from the micromachining zone. The detection of acoustic shock waves, plasma plume, and laser reflection during the processing requires the integration of sensors such as acoustic emission sensor, photodiode, spectrometer, pyrometer and thermal cameras, depending on the specific dynamics present during the micromachining process [[26], [27], [28]]. Additionally, the time scale of the process, characterized by ultra-short pulses, necessitates the use of sensors capable of capturing and processing data with sufficient speed [29].



Download: Download high-res image (157KB)

Download: Download full-size image

Fig. 1. Schematic representation of the process signals emitted during laser micromanufacturing. Due to their origin directly from the interaction zone they form the starting point for process and quality monitoring - adapted from [36,37].

Laser-matter interaction is a complex phenomenon with various process dynamics. When a laser beam interacts with a material, it typically leads to absorption, melting, vaporization, plasma formation, and shock wave propagation. Depending on the material, energy is transferred from electrons to the lattice via carrier-phonon scattering on a period of several hundred fs to a few ps. Initially, the laser energy is absorbed by the material through inverse Bremsstrahlung (IB) absorption within the pulse duration [22]. Absorption of laser energy by a material occurs through the interaction between laser photons and electrons in the material. As the absorbed energy accumulates, it leads to the melting of the material, typically within a few ns [24]. At higher laser intensities, vaporization happens, leading to recoil and melt expulsion phenomena, which occur on a similar timescale as the melting process [27,28]. Subsequently, the material undergoes plasma formation and expansion, as the laser-induced energy further ionizes the material. This plasma generation takes place within tens to hundreds of ns [29]. Finally, shock wave propagation emerges as a result of the rapid material expansion and can be observed within ns to a few μs [31]. During the laser-matter interaction at long and short pulse laser processing (>10 ps), most of the above-mentioned process dynamics occur during the pulse duration itself, making it challenging to evaluate the relative importance of each. For USP laser processing, the material removal typically occurs by non-thermal ablation modes such as spallation, phase explosion, fragmentation and coulomb explosion, occurs within the pulse duration whereas the subsequent plasma formation and shockwave propagation occurs after the end of the laser pulse [14]. By understanding the process dynamics and the associated timeline, the appropriate sensors can be implemented to ensure accurate and real-time monitoring of the USP laser micromachining process [30,31].

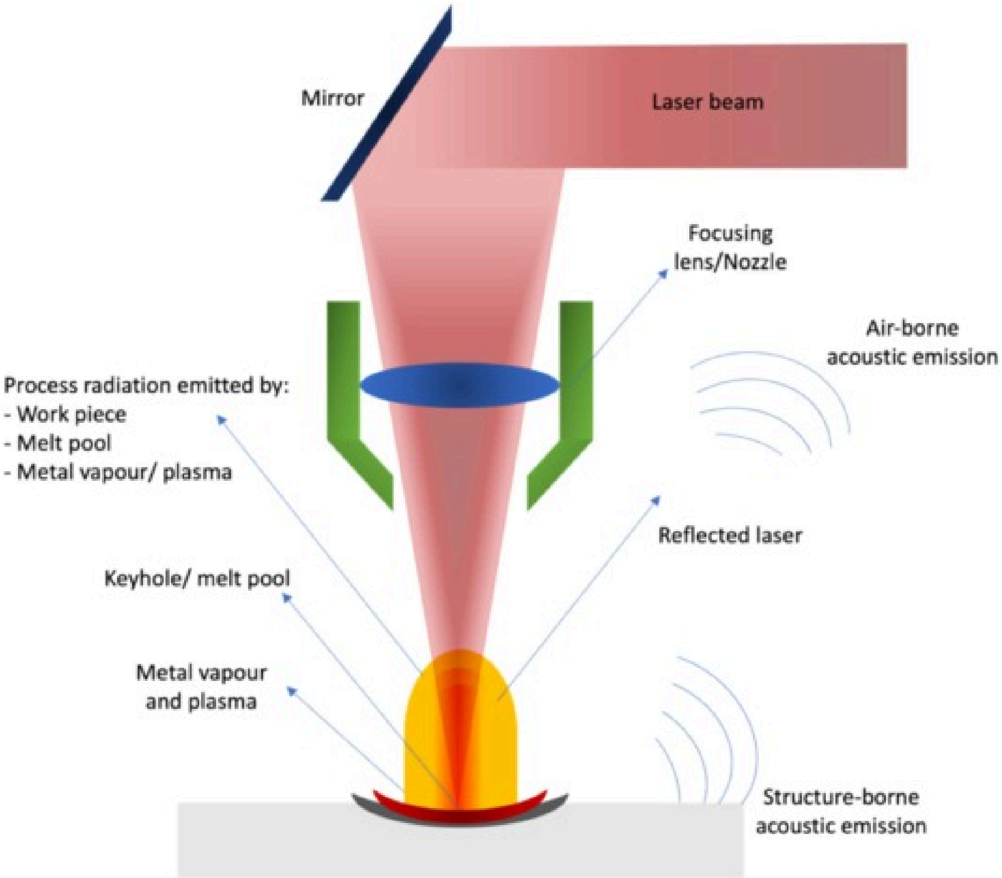
The research and development efforts in this field, driven by urgent industrial needs, are getting more intensive and widespread each year, but the outcomes are not yet at the same level of maturity as other manufacturing applications [32]. Few research works have developed automated procedures and analytic techniques for in-process defect detection. Moreover, a relatively few researchers proposed control systems to conduct corrective or reactive actions after detection of defects [32]. An ideal metrology for fs laser micromachining must be capable of overcoming numerous optical and acoustic challenges inherent to sensing intense light-matter interactions, such as large variations in backscatter due to rapidly changing geometry and physical state, simultaneous multiple reflections in the machined cavities, blinding scatter of the intense machining beam, and strong, broadband optical emissions from induced plasma [33]. In addition to having a resolution down to 10μm and a high acquisition rate, an effective system must also have a high dynamic range and sensitivity [34]. Due to the nature of the process, regardless of the type of sensor employed in USP laser, these sensors must have a quick response time and a high spatial resolution. Understanding the information included within sensor signals and resolving them will assist to identify the physics, hence facilitating the development of in-situ monitoring and virtual verification systems. In-situ sensing must be performed at sufficient rates to resolve the rapidly changing shape, with adequate sensitivity, dynamic range, and robustness to filter out intense process noise [35].

Different process monitoring approaches have been developed for traditional laser applications including laser welding and additive manufacturing (AM). In contrast to other laser-based techniques, monitoring for USP laser micromachining in the literature is limited. It could be attributed to the characteristics of the USP process which involves high repetition rates, rapid processing speeds and finer feature resolution [38]. In this paper, we have conducted a detailed review of the state of the art in in-situ process monitoring of firstly, pulsed laser micromanufacturing and then other laser-based manufacturing processes such as laser welding, laser cutting and AM in the following sections. Finally, other non-destructive testing (NDT) processes, that are yet to be implemented for in-situ monitoring, but that have been identified as being potentially suitable, are reviewed.

Acoustic emission

AE is an effective NDT technique capable of measuring transient elastic emissions generated by materials when they undergo stress, deformation, cracking or other structural instabilities. Unlike ultrasonic NDT, which is a form of active scanning (i.e. generation, transmission, and collecting of signal), AE is mostly a passive form of scanning, comparable to a microphone or other similar sensors ‘listening’ to various occurrences [3]. AE generally refers to the propagation of elastic waves in the ultrasonic frequency range (20–2000 kHz). Due to the nature of the high frequency range of the sensor (usually in the kHz/MHz range), which is much higher than the characteristic frequencies associated to the machining or natural structural modes in the machine tool structure, it insulates the process generated AE signal from the corresponding voices. At the frequencies most commonly used for AE testing, 100–300 kHz, the AE sensor can detect surface movements as small as 10^{-13} μm, which is a thousand times smaller than the size of an atom [31]. Due to its ability to detect microscale deformation in a relatively "noisy" machining environment, AE is hence ideally suited for process monitoring [3]. Air-borne AE (ABAE) sensors detect sound waves that travel through the air, capturing and analysing auditory signals such as those generated by a laser-induced plasma or laser beam. On the other hand, structure-borne AE (SBAE) sensors monitor vibrations propagating through solid materials or structures, enabling continuous assessment of precision, stability, and potential mechanical deviations.

Fig. 2 schematically illustrates the typical AE signal formation and its propagation during laser-material interactions. When a laser interacts with a medium, optical and acoustic signals are typically generated [39]. There are different ways through which AE signals are generated during laser-material interactions. As seen in Fig. 2, both ABAE and SBAE is generated due to the changes in the surface and bulk of the material, melt pool dynamics (formation and collapse), vapor and plasma dynamics (generation and propagation). In addition, the photons in the laser beam are not completely absorbed by the material during laser-material interactions depending to the laser wavelength, the surface characteristics and optical properties (refractive index and extinction coefficient) of the material. The reflected laser beam can interact with the laser beam guiding optics, such as mirrors, resulting in the generation of acoustic waves within these optics [40].

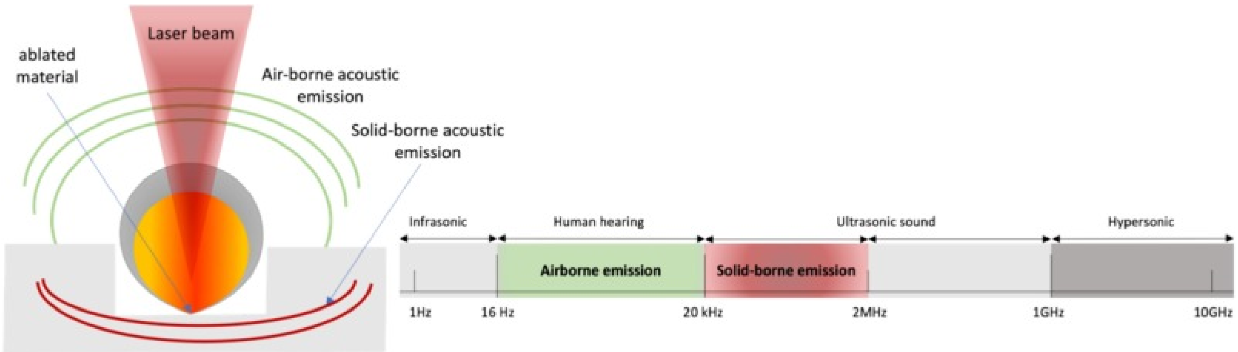


Download: Download high-res image (157KB)

Download: Download full-size image

Fig. 2. AE signal generation during laser processing of materials - reproduced from [39].

Pulsed laser-material interaction produces coherent electromagnetic radiation, which includes ultrasonic sounds with a wide frequency band in the material. Different power densities of incident laser beams generate ultrasonic waves via distinct mechanisms. At lower laser energy, the material experiences thermal-stress induced deformation without melting or ablation, which produces ultrasonic waves (2–12 MHz) [41]. When the material is irradiated with a high-power-density laser, the surface of the material may be melted and/or vaporized, generating sonic (20Hz to 20kHz) and/or ultrasonic emissions in the ablative regime (Fig. 3) [42,43]. During the laser micromanufacturing, the major mechanism for the formation of ultrasonic waves is ablation.

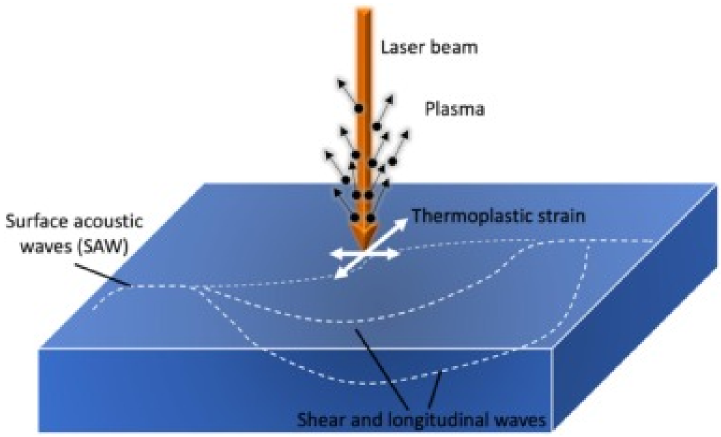


Download: Download high-res image (134KB)

Download: Download full-size image

Fig. 3. Sound ranges of AE emission – reproduced from [30].

As described in Fig. 4, laser-material interactions can generate all acoustic and elastic vibration modes, including longitudinal waves, shear waves, and surface acoustic waves. During the ablative regime, all waves are produced [44]. Acoustic modes refer to the various patterns of mechanical oscillations or vibrations within the material caused by the interaction with laser energy. These modes include longitudinal waves, which involve compression and expansion of the material along the direction of wave propagation, and shear waves that induce perpendicular displacement within the material. Elastic vibrations encompass the oscillations of the material structure in response to transient changes in temperature induced by the laser. These vibrations are evident as surface acoustic waves (SAWs) propagating along the material surface and bulk waves (such as shear and longitudinal waves) propagating within the material. All of these elastic-wave amplitudes expand linearly as the volume of the material increases, as results of the increasing laser pulse energy [45]. Currently, longitudinal acoustic waves are mostly employed in USP laser processing for process monitoring. Shear acoustic waves are used less frequently since there are no simple shear wave transducers on the market [46].

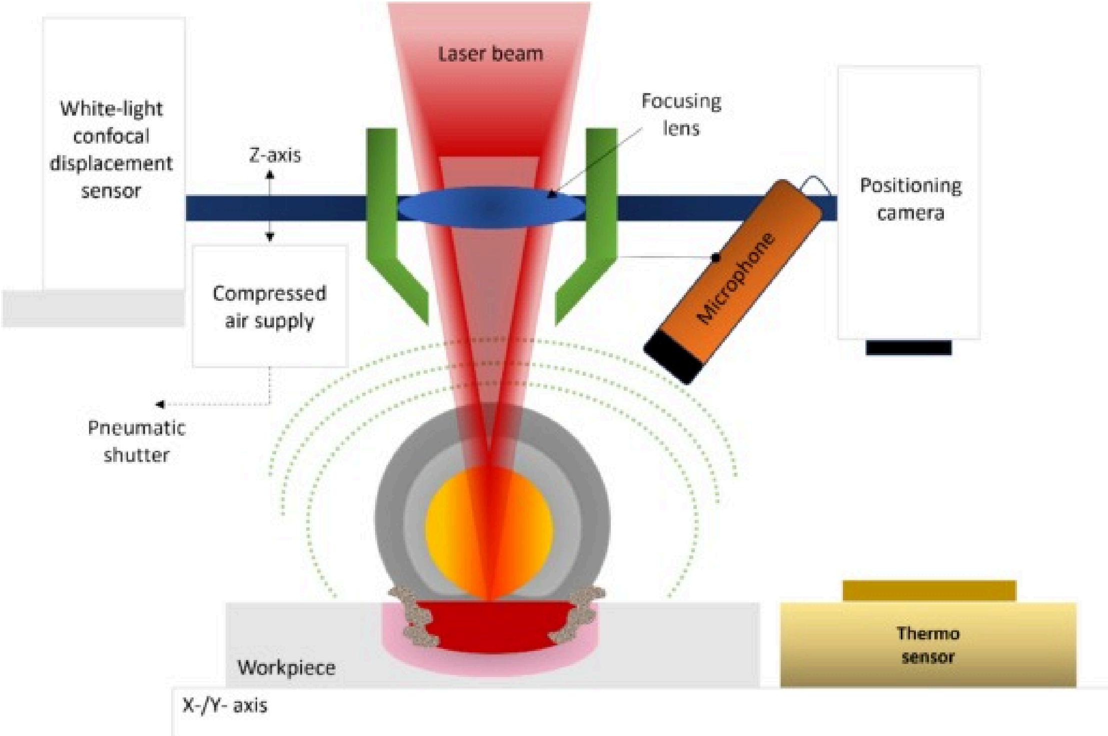


Download: Download high-res image (80KB)

Download: Download full-size image

Fig. 4. Laser-generated ultrasound waves – reproduced from [44].

AE signal monitoring systems transform an acoustic signal into an electrical signal. The resulting electrical signal is then amplified to better receive signals that can be used for processes involving melting, vaporization, plasma generation, and keyhole formation [47]. Vallejo et al. [48] employed microphones to monitor air-borne emissions, whereas piezoelectric detectors were used to detect AE in solid structures. Typically, the frequency of laser-induced audible AE sounds is between 20Hz and 20kHz. Fig. 5 illustrates a typical experimental setup [49]. Microphones consist of a capacitor where its capacitance changes in response to the variations in the observed sound field.

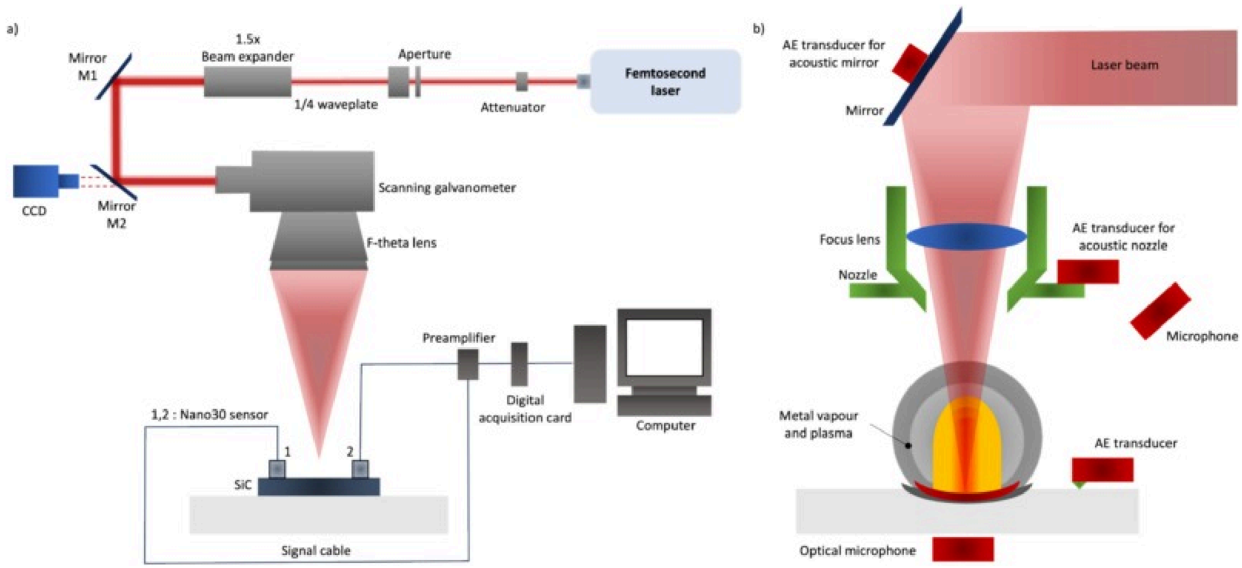


Download: Download high-res image (183KB)

Download: Download full-size image

Fig. 5. Schematic test setup and arrangement of the integrated ABAE sensor technology – reproduced from [50].

SBAE frequencies in the ultrasonic spectrum range from 20kHz to 2MHz can be detected with piezoelectric transducers [30]. In nature, certain non-conductive piezoelectric ceramics, and crystals (such as quartz) exhibit the piezoelectric effect. If a material is subjected to a mechanical stress, an electrical potential travel through the material. The polarization of the electric potential indicates whether the mechanical stress is compressive or tensile [39]. A piezoelectric AE transducer transforms acoustic energy (pressure waves traveling through a medium) rather than electrical energy. Piezoelectric transducers are commonly mounted on laser beam reflective mirrors, laser processing heads and gas nozzles, and workpieces. Fig. 6 illustrates several mounting sites for each type of sensor [51]. Wedges are often attached directly to the sample surface using a suitable medium such as silicone grease or magnetic holders [52]. Table 1 summarizes the comparison between SBAE and ABAE as well as optical microphones.



Download: Download high-res image (202KB)

Download: Download full-size image

Fig. 6. Schematic diagram of a) monitoring setup of SBAE system implemented in the fs pulsed laser micromanufacturing machinery, b) possible SBAE sensor implementation regions in the laser-based manufacturing chamber – reproduced from [51,52].

Table 1. Comparison of various AE measurement technologies.

AE sensor technology	AE bandwidth	Advantages	Disadvantages	Typical application areas
Microphone with amplifier for audible sound	20Hz to 20kHz	Low cost, Non-contact	The signal amplitude in air is greatly attenuated with increasing frequencies and highly affected by external noise	- Laser ablation rate monitoring [55] - Laser welding defects detecting in AM processes [56]
Structure-borne, Piezoelectric transducer with preamplifier and coupling	100kHz to 1MHz	Entails a more complex analysis, Cost-effective hardware	The necessity for mounting the sensor directly on the workpiece, making it unsuitable for production environment	- In-process defects/micro-features monitoring - Real-time laser focus position adjusting - Process status monitoring
Optical microphone	5Hz to 1MHz in air; and up to 25MHz in liquid	Very sensitive Non-contact Well protected against the external disturbance by optical fibre Operational in high electromagnetic	Expensive maintenance, Performance strongly depends on the optical properties (e.g., complex interferometer setups)	-In-process defects monitoring -Real-time laser focus position adjusting

AE sensor technology	AE bandwidth	Advantages	Disadvantages	Typical application areas
		fields		

The working principle of AE sensors is similar to that of accelerometers, which is operating at considerably lower frequencies, typically in the tens or hundreds of Hz rather than hundreds of kHz. The AE sensor inspects a localized part of the structure and is extremely sensitive to a point of disturbance. They can, in particular, be used to detect aberrant processes as they occur. A sound-based diagnostic system would have to be able to identify damage-related sounds and extract the information from the background noise. Its high-frequency characteristic provides the advantages of a high signal-to-noise ratio [53]. Any acoustic sensing technology is sensitive to adverse disturbances, but it has the potential to be a highly powerful tool if the operating conditions are acoustically repeatable and the diagnostic system can distinguish the acoustic patterns accurately [54]. Despite the fact that SBAE monitoring is gaining a lot of interests in academia and industry, noncontact ABAE sensors have also gained momentum attributed to the main drawback in SBAE monitoring related to the difficulty to implement this technique in harsh environments [38].

In the past, AE monitoring technology was often used in the traditional processing fields such as grinding and milling [3]. With the advancement of AE technology, the use of this method in the laser-based processing has gained popularity. Stafe et al. [57], for example, employed a piezoelectric transducer (PZT) coupled to a digital oscilloscope to measure shockwaves propagating through irradiated material during hole drilling with an ns pulsed laser. In-situ AE monitoring demonstrated that after a number of laser pulses hit a certain threshold, the depth of the crater ceases to increase detectably, even after a large number of extra laser pulses. Table 2 summarizes the findings of AE monitoring in laser-based manufacturing.

Table 2. Summary of literature review on in-situ sensing of laser-based manufacturing via AE.

Reference	Object of investigation	Type of AE	Sampling frequency	Data processing method
AM				
D. Ye et al. [58]	Detecting defect patterns	ABAE	0–100 kHz	Time-domain Frequency-domain
V. Pandiyan et al. [59]	Lack of fusion, balling, pores, and keyhole	SBAE	0–100 kHz	RMS and FFT
N. Eschner et al. [60]	LBPF part quality	SBAE	5 Hz–2 MHz	STFT-Spectrogram
Laser welding				
B. Kuo et al. [61]	Low-strength joints after micro welding	ABAE	0–10 kHz	RMS, STD, time-based evaluation
M. Bastuck et al. [62]	Laser welding process quality factors	ABAE - SBAE	<100 kHz for ABAE <500 kHz for SBAE	STFT-Spectrogram
A. Sun et al. [38]	Defects during keyhole-mode welding	SBAE	Up to 300 kHz	RMS
Laser shock peening (LSP)				

Reference	Object of investigation	Type of AE	Sampling frequency	Data processing method
T. Takata et al. [63]	Quantitative evaluation of LSP in water	SBAE	200–750 kHz	Time-based statistical evaluation
J. Wu et al. [64]	Online detection of acoustic waves during LSP	ABAE	0–150 kHz	Time-based evaluation
Pulsed laser micromanufacturing, laser ablation				
S. Conesa et al. [19]	Plasma expansion dynamics during ns-pulsed laser ablation	ABAE	30–15,000 Hz	FFT
X. Xie et al. [65]	Ps-pulsed laser derusting	SBAE	125–750 kHz	RMS and FFT
M. Zuric et al. [66]	Monitoring high-speed processes in real-time	ABAE	10–20 kHz	FFT
A. Stournaras et al. [67]	Monitoring the percussion laser drilling process	ABAE	15–20 kHz	Short time mean value (STMV)
X. Xie et al. [51]	Fs-laser modified SiC monitoring	SBAE	125 kHz–750 kHz	RMS, MARSE, Peak frequency
V. Schulze et al. [68]	Focal position monitoring	ABAE	6.3 Hz–20 kHz	FFT
A. Kacaras et al. [69]	Focal position monitoring	SBAE	5 Hz–2 MHz	STFT

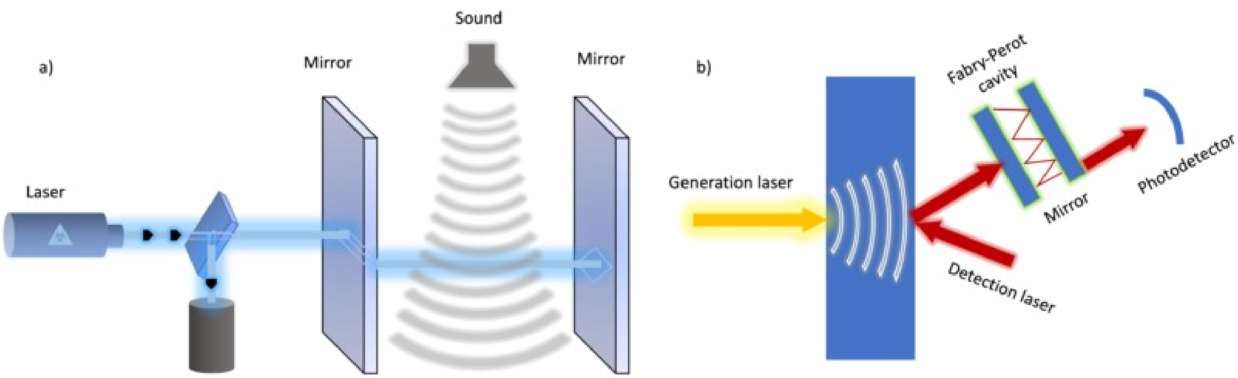
Several studies have recently been conducted on defect and process monitoring approaches using AE sensors in laser subtractive manufacturing. The ultrasonic signal produced by pulsed lasers can be several orders of magnitude more powerful than that produced by a contact piezoelectric transducer [40]. Palanco et al. [70] used a microphone to detect acoustic emissions from four targets (copper, aluminium, tin, and air) that were investigated in terms of energy transferred and spectral components. Acoustic wave spectral analysis proved to be a valuable, easy, and low-cost tool for diagnosing laser plasma dynamics. The spectral analysis results indicated the correlation with each material plasma formation threshold. However, due to the low sample rates provided by the analog-to-digital converter, most activities that occur in the plasma evolution are undetected by the monitoring system. Conesa et al. [71] studied the AE of laser-induced plasmas (ns pulsed laser) further by focusing on the spectrum content within the ABAE range from aluminium, copper, and silicon samples obtained at different irradiances. Acoustic spectra of laser-induced plasmas have demonstrated a remarkable sensitivity for detecting actual plasma expansion dynamics that occur after the laser pulse irradiates. Additionally, Stournaras et al. [67] researched the use of ABAE signals for monitoring the percussion laser drilling process with ns pulsed lasers. A quantitative relationship between acoustic sensor output and hole geometrical properties (depth and upper diameter) was investigated. The study demonstrates that AE signals recorded during the drilling procedure contain process information and can be used for process monitoring. There is a strong association between the amplitude of the signal and the depth of the hole.

Many process parameters influence the performance of the laser-material removal process, beginning with the laser beam characteristics, which include wavelength, pulse energy, repetition rate, beam intensity profile, workpiece material properties, and the overlap between two consecutive laser pulses. Manikanta et al. [72] presented AE measurements of a ns pulsed laser on the propagation properties of laser-generated acoustic shock waves (ASWs) and vibrations inside metal targets, with the goal of understanding the laser-solid interaction in real-time. As a result, the shock arrival time with respect to incident laser energy decayed exponentially, but acoustic shock velocities increased linearly with incident laser energy. The arrival time of the ASWs was observed to increase with the microphone distance, whereas the pressure of the AWS dropped.

In addition, the location of the focal plane in relation to the ablated surface, the diameter of the focal spot, and the beam waist determine the impinging laser beam energy into the laser-material interaction zone due to the changes in the beam diameter and spatial energy distribution (beam density) during propagation [73]. As described in [30,68], Schulze et al. established control units for monitoring and regulating significant process parameters such as focal positioning and laser power (ps pulsed laser) by recording the ABAE for the focal positioning with a precise adjustment of the focal depth to within 20 μm of the workpiece surface. Acoustical focused positioning for micromachining is accomplished via a rough positioning followed by a fine positioning. Kacaras et al. [69] proposed a SBAE sensor with a wider bandwidth and a lower sensitivity to environmental sounds for an in-situ identification of focal position during laser ablation (ps-pulsed laser) utilizing SBAE measurement with time-frequency analysis. A significant relationship was observed between the focal position and the sum of the estimated intensity values. They discovered that the intensity levels of processed AE were linearly proportional to the laser power. In addition, the strongest AE signal was found when the laser beam focused on focal point, where the deepest ablation depth was measured. Based on the AE signals, this study indicates that it is possible to establish the focal position during the laser micromachining. Bordatchev et al. [73] investigated the impact of the focal position on the laser-material removal process by analysing the AE produced by laser-material interactions. It was discovered that the amplitude-frequency distribution of power spectrum densities changed dramatically in combination with the focus position. A pattern recognition analysis of provided informational process signatures gives an accurate statistical classification of the actual focus location, which can be utilized in monitoring and control systems of laser processes.

Xie et al. [65] explored AE for real-time monitoring of laser derusting process as AE signals are generated simultaneously with the removal of the rust layer and its frequency far exceeds that of mechanical noise. They have discovered that by monitoring AE signals in the time and frequency domain, the changes in rust removal conditions and the effect of derusting parameters can be precisely mapped. The same group of researchers demonstrated the association between fs laser induced material removal, surface structure characteristics, surface defects, and AE signals of SiC using RMS, MARSE, and peak frequency. It has been discovered that when the laser fluence increases, the removal in the depth direction increases, and the acoustic emission signals in the time-domain demonstrate an increase in AE events and MARSE, while the shift in RMS becomes more apparent. They presented a time-frequency analysis method based on the short-time Fourier transform (STFT) of AE signals to quantify surface form defects in laser-modified SiC [51].

Conventional AE techniques face inherent limitations when it comes to detecting and capturing acoustic signals at exceptionally high repetition rates beyond the design specifications of sensors. This limitation becomes increasingly evident in scenarios involving rapid, high-frequency events, such as in laser ablation processes with elevated repetition rates. This is possible with an optical microphone, which works on the principle of interferometry and it consists of a control unit containing a laser which sends its light over an optical fibre to a sensor head. This sensor head, represented in Fig. 7, contains a pair of parallel, partially reflective, mirrors known as a Fabry-Pérot etalon. A passing sound wave between the two mirrors creates small alterations in the laser wavelength. This alters the amount of light reflected back from the interferometer, which is converted into an electrical signal by a photodiode [74]. The optical microphone has the following benefits over traditional acoustic microphones and piezoelectric transducers: i) As there are no deformable components in this sensor, the acoustic detection bandwidth of the optical microphone is wider (5 Hz to 1 MHz in air and up to 25 MHz in liquid) [74], ii) the optical microphone can tolerate high electromagnetic radiation and is therefore particularly useful for monitoring AE signals generated in laser processing. In-situ process monitoring using an optical microphone was demonstrated for non-contact detection of crack formation during laser cladding and AM [96]. Analysis of AE signals above 350 kHz has shown strong correlation in crack counts with offline NDT counting methods [75].



Download: Download high-res image (145KB)

Download: Download full-size image

Fig. 7. a) The working principle of the optical microphone - reproduced from [74], and b) laser-in/laser-out on the opposite surface (using a Fabry-Pérot interferometer).

Due to its high-frequency features and sensitivity, AE has been utilized in a variety of processes for precise productions. Acoustic monitoring has already been implemented in laser welding, where ABAE and SBAE are distinguished. In laser welding, the thermal expansion of solid material resulting from the application of laser energy to a workpiece will induce thermal strain in the laser-affected region [53]. From these investigations, it has been demonstrated that physical parameters can be inferred via AE signals during laser welding [76]. By studying AE signals, Lee et al. [76] explored the impact of welding settings on welds. The AE signals were sampled and analysed to identify AE-generating mechanisms, including phase changes, solidification, and potential crack initiation behaviour. Additionally, an in-situ monitoring and prediction strategy has been demonstrated for a laser spot welding using an artificial neural network (ANN) with specified features (LSW). From the spectral analysis of AE signals, it was found that high-frequency contents were predominantly influenced by welding properties which were determined from section-by-section frequency analysis of the time series of AE signals.

In a study on laser lap micro-welding, Kuo et al. [61] observed that the sound signal generated during the process has the potential to identify the low-strength joints. The signals being evaluated are audible sounds captured by a MEMS microphone with a frequency range <10 kHz. Sun et al. [53] investigated the relationship between sound-signal characteristics and defects produced during keyhole-mode welding. Based on the results, the RMS and gradient of signals produced during the initial millisecond of welding, as well as the 300 kHz frequency signal feature, are promising features for diagnosing low-strength connections. Chen et al. [77] employed a multi-spacing structured MEMS microphone array in combination with the established weld quality algorithm to detect weld joint defects caused by non-uniform contacts between two metal sheets in micro lap welding. The results reveal that noise will contaminate the signal collected from a single microphone, causing the classification rate to decrease. Bastuck et al. [62] explored the detection and evaluation of high-frequency SBAE and ABAE signals, as well as their relationships with laser welding process quality factors. The findings show that SBAE is only emitted during the phase of deep welding, before the sample is completely penetrated. Investigations indicate that ABAE are mainly released throughout all welding processes. SBAE can detect laser weld defects such as lack of penetration. ABAE has demonstrated its advantages for evaluating process conditions associated with plasma production, while SBAE has demonstrated its advantages for analysing internal defects such as lack of fusion or porosity.

In laser bed powder fusion (LPBF), laser interaction with powder particles results in complicated physical behaviour, including absorption, rapid melting, solidification, microstructure evolution, flow in a molten pool, and evaporation of materials [78,79]. More accurate monitoring and sensing oversight of the process utilizing sensors has been proposed as a strategy for understanding these mechanisms of the LBPF process. Pandiyan et al. [59] examined AE signal characteristics in three domains, including time, frequency, and time-frequency, for four processing regimes, including lack of fusion, balling, no pores, and keyhole, using an AE sensor with a frequency range of 0–100 kHz. It has been established that the defect mechanisms have a direct correlation with the energy density, highlighting the importance of selecting the optimal laser power and scanning speed for a quality build. Ye et al. [58] proposed using a deep belief network (DBN) to detect the impacts of LBPF. The DBN approach collected the training information from the raw AE signals without feature extractions and data processing and was used to detect five defect patterns. Wasmer et al. [80] and Shevchik et al. [62] coupled artificial intelligence (AI) and reinforcement learning (RL) for in-situ monitoring of LBPF. To detect the ABAE signals created during the AM process, a fibre Bragg Grating (FBG) optoacoustic sensor is mounted in the laser chamber. The approach appears to be a sufficient initial step towards in-situ quality monitoring in AM.

Ye et al. [81] captured AE signals to evaluate the process characteristics during direct laser material sintering (DLMS). The power spectrum density (PSD) of an AE signal that varies with different values of laser scanning speed was investigated, and the melt pool formation mechanism was predicted. The study demonstrates that there is a correlation between laser parameters, scanning condition, and AE signals which eventually can correlate with the quality of the subsequent laser sintering. By clustering AE signals, Gaja et al. [82] examined the defect types distinguishing the laser material deposition (LMD) and its associated key features. The AE technique is suitable for examining the causes of defects during LMD because it has plenty of defect-related data, including crack and pore formation, nucleation, and propagation. As a non-stationary process, it is difficult to get information on the development of defects using only the AE waveform in time-space; therefore, other parameters such as amplitude, energy, rising time, count, and frequency are extracted to qualitatively analyse defect processes. The objective is to first detect LMD defects in order to take the necessary corrective action, such as machining and remitting, and then to develop a reliable method for analysing AE data during LMD.

LSP is one of the surface treatments that induces compressive residual stress on the surface of the material to improve its fatigue resistance [83]. LSP is a laser ablation process involving an elastic wave. Consequently, the AE approach has been effectively applied to the LSP process in order to monitor the peening phenomenon and impact force [84]. Takata et al. [63] implanted a couple of AE sensors in water at fixed locations independent of the sample position to give flexibility in specimen placement and online control capability for process monitoring. The results indicated that this new arrangement is able to identify AE waves. Wu et al. [64] proposed utilizing acoustic wave online LSP detection to address the drawbacks of existing offline LSP detection methods. Thus, the empirical relationship between the energy of the acoustic wave and the surface residual stress is established, providing a theoretical framework for the in-situ online monitoring of LSP. Based on the measured AE signals and simulated waveforms, Enoki et al. [84] calculated the impact pressure during laser ablation. Based on their findings, it is obvious that the AE approach detected the phenomenon of laser peening impacts caused by laser ablation and cavitation bubble collapse. For each laser irradiation, individual impact pressure could be estimated using inverted analysis.

Calibration of AE sensors is crucial to ensure accurate and consistent measurements in monitoring of laser-based manufacturing processes. Common calibration methods include the pencil lead break (PLB) test, which involves breaking a pencil lead on the sensor surface to simulate a controlled, repeatable acoustic signal [70]. Another approach is the Hsu-Nielsen source often used in laboratory settings to validate sensor sensitivity and signal fidelity. In addition, hydrostatic pressure tests apply a uniform pressure to the sensor to evaluate its response under varying conditions [81]. Advanced methods use laser-based sources or ultrasonic transducers to produce precise, broadband signals for high-resolution calibration, especially when sensors need to capture detailed frequency information in applications like micromachining or material testing. Each calibration method is selected based on the specific application requirements and the frequency range the AE sensor is designed to detect.

Data filtering in AE monitoring involve isolating relevant signals from noise to ensure accurate process analysis. AE signal processing seeks to analyse the measured signal with the goal of obtaining useful information necessary for a better understanding of the underlying physical phenomena [81]. The AE signal is a composite signal composed of contributions from many AE sources, various reflections, and background noise. It typically consists of a sequence of oscillating transient bursts with non-uniform amplitudes that occur at irregular intervals with overlapping or independent occurrences. The application of any noise reduction method, usually a filter, has to be preceded by a clear knowledge of which aspects of the signals need to be retained. Frequency-based filtering, such as bandpass filters, focuses on specific signal ranges, eliminating irrelevant frequencies. This can be achieved through advanced signal processing techniques such as wavelet transforms, and machine learning algorithms. Each filtering method relies on a specific means of representing the differences between the signal and noise. For example: Linear filters assume that the signal and noise occupy disjoint regions of the spectral space. Statistical filters (e.g. Wiener filters) assume that the signal and noise have different second-order statistics. Time-scale filters assume that the signal and the noise occupy disjoint regions of the time-scale domain (e.g. different wavelet coefficients) [80].By categorizing AE signals based on their source characteristics (e.g., laser-induced plasma, melt pool dynamics), the data can be more accurately interpreted, leading to improved process control and optimization.

In conclusion, contact-based AE sensors are used to evaluate structure-borne process emissions. They perform well for numerous laser-based processing applications, but always require uniform physical contact with the workpiece for all processes. This is typically impossible in a production context. In contrast, non-contacting AE systems utilize air-borne sound and ultrasound emission techniques. Due to the short measurement range of traditional capacitive microphones, which is typically restricted to 100 kHz bandwidth, these systems struggle with interference from ambient industrial noise. The current ABAE and SBAE sensors in the market are listed in Table 3. The limitation of monitoring pulsed laser-based processes is primarily due to the fact that the specifications of SBAE sensors are insufficient to monitor USP micromanufacturing, but ABAE sensors are sensitive to the interface of work noise from the environment. AE sensors detect shock wave propagation generated by rapid material expansion and ejection due to laser ablation which happens at the timescale of a few μs. Additionally, AE sensors capture material removal events, such as recoil pressure, which produce acoustic vibrations. Particularly relevant to USP laser processes, AE sensors identify the onset of ablation and quantify shock wave intensity, though high pulse repetition rates (in the range of MHz) challenge their ability to differentiate overlapping shock waves, necessitating high temporal resolution of the AE sensors [73]. Consequently, the majority of research works focus on enhancing the identification, precision and resilience of non-contacting sensors via intelligent identification algorithms.

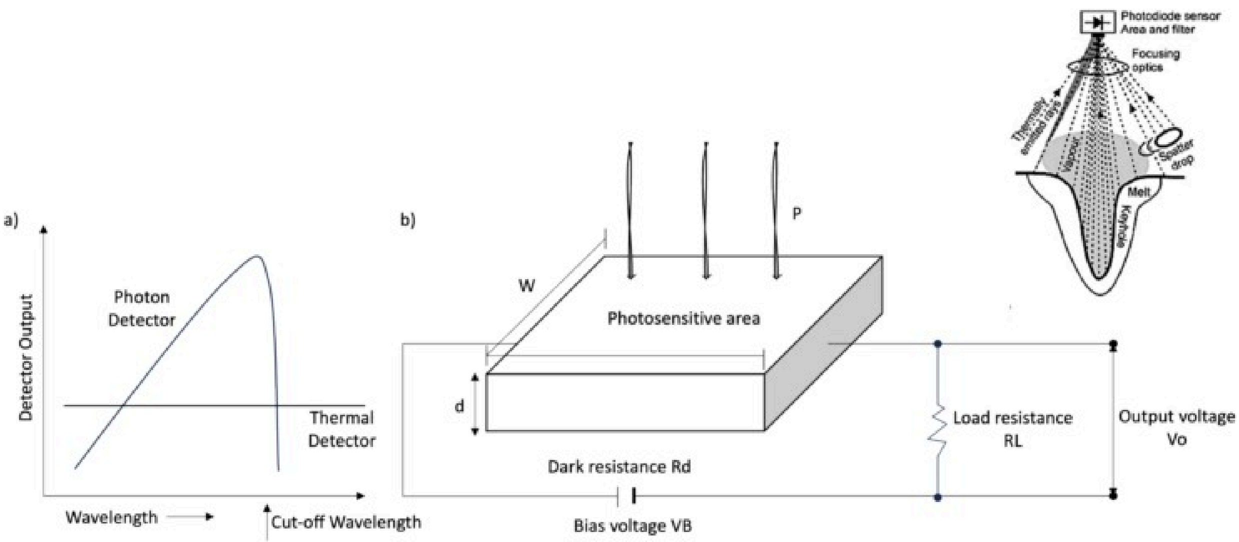
Table 3. Specifications of the existing AE sensors in the market.

Sensor type	Supplier	Frequency range	Sensitivity
SBAE	Vallen Systeme	30 kHz to 1500 kHz	–
SBAE	Kistler	100 kHz to 900 kHz	57 dBref 1 V/(m/s)
SBAE	Physical Acoustics	1 kHz to 1000 kHz	–
SBAE	Quass	20 kHz to 1500 kHz	–
ABAE	Sonelastic	70 Hz to 96 kHz	40 dB ± 8 dB
ABAE	Physical Acoustics	20 kHz to 45 kHz	–
ABAE	GRAS	3 Hz to 20 kHz	50 mV/Pa @ 1 kHz
Optical microphone	Xarion	10 kHz to 1000 kHz	10 mV/Pa @ 1 kHz

The physical phenomena of laser ablation and the mechanisms that generate ultrasonic waves are discussed. For monitoring AM, laser welding, LSP, and laser micromachining using a variety of AE sensors, a comprehensive literature review is conducted. The working principles, experimental setup, and benefits of AE detection systems, including microphones, piezoelectric AE transducers, and optical microphones are introduced. The AE monitoring techniques are compared to each other, and promising applications are highlighted. As current manufacturing trends strive for smaller and finer form and features, a dependable method of process monitoring with the right sensor technology is required to accurately characterize and monitor the manufacturing process. Using AE as an in-situ process monitoring and characterization tool can serve to closely link the manufacturing and quality control stages, and with further development, in a fully automated manufacturing environment, the quality control stage can be eliminated entirely with optimal in-situ monitoring and process control.

Photodiode

Photodiodes, which are spatially integrated single-channel detectors, are semiconductor devices that transform light (radiation) into electric current. They are frequently used in laser material processing since they are simple to integrate into various configurations [85]. They can be used to measure light intensities and often have a fast response time, which is essential for monitoring laser material processes [86]. The emitted radiation mostly depends on the process itself (the radiation that is used for monitoring), the processed material (spectral emissivity), the used laser (spectral sensitivity of sensor), and the optics (filters, lenses, fibres, having each a characteristic spectral transmission) [87]. The wavelength range that a photodiode can detect is frequently reduced on purpose via optical filtering. The light that passes through the filter is first detected by the photodiode sensor, then processed by a signal amplifier, and finally visualized on an oscilloscope. There are also commercially available laser-based manufacturing monitoring systems that employ photodiodes [88]. In the case of conventional laser applications such as laser welding and laser cutting, photodiodes have been well demonstrated for process monitoring. Nevertheless, optical emission monitoring of short pulsed and USP lasers is still a relatively unexplored field. By measuring the signal intensity of various spectral bands, it is possible, for instance, to identify certain processing regimes. When photons of sufficient energy (radiation) reach the active area of the photodiode or photodetector (Fig. 8a), it creates an electron-hole pair, thereby altering the effective resistance or conductance of the detector. As depicted in Fig. 8b, a bias voltage is applied to collect the charge carriers, and the signal (photocurrent) is monitored across a load resistor RL [89]. The photocurrent is directly proportional to the photon intensity reaching the active area. The reduction of the signal to a single point is related to data compression, which simplifies post-processing. They are useful for in-situ sensing of laser-based production processes due to their low cost, high sensitivity, and high sampling rates [90]. Table 4 summarizes photodiode-based monitoring in laser-based manufacturing.



Download: Download high-res image (160KB)

Download: Download full-size image

Fig. 8. a) Wavelength response of detectors, b) working principle of PIN photodiode – reproduced from [89,91].

Table 4. Summary of literature review on in-situ sensing of laser-based manufacturing via photodiodes.

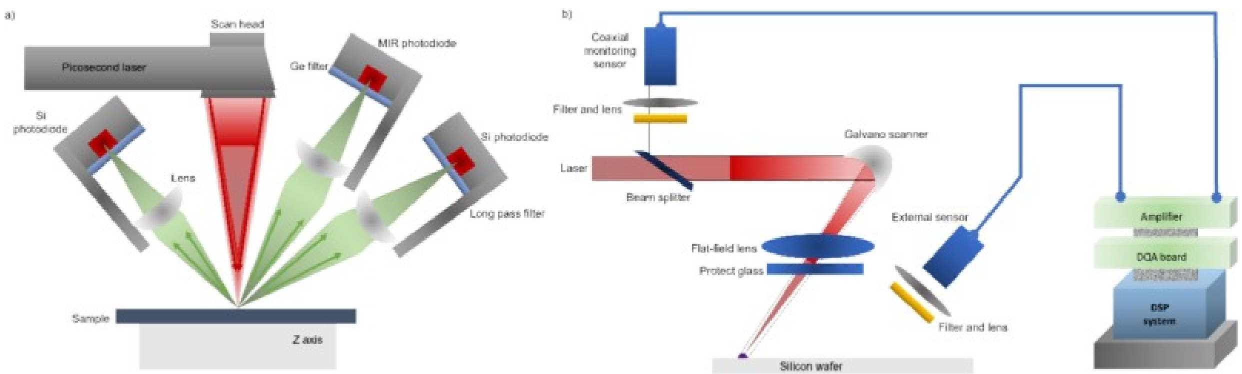
Literature	Object of investigation	Type of photodiode	Sampling frequency	Data processing method
AM				
S. Jayasinghe et al. [92]	LPBF build density	NIR, laser back reflection and IR	100kHz	K-mean clustering, Gaussian process

Literature	Object of investigation	Type of photodiode	Sampling frequency	Data processing method
Z. Mao et al. [93]	Melt pool temperature, plume characteristics	IR	4 kHz	Various neural network structures
Laser welding				
J. Yu et al. [94]	Plasma behaviour	VIS	10 kHz	Fuzzy multi-feature patter recognition
P. Norman et al. [91]	Melt pool and keyhole radiations	UV, back reflection and IR	8 kHz	Time-based analysis
Laser cutting				
De Keuster et al. [95]	Piercing and cut quality	VIS and NIR	50 kHz	Time-based analysis
Adelmann et al. [96]	Cutting defects	IR	20 kHz	Time-based analysis
Garcia et al. [97]	Contour cutting	VIS and IR	–	Quantitative time-based analysis
Pulsed laser micromanufacturing, laser ablation				
Gehrke et al. [37]	Focus position, surface roughness	VIS	50 kHz	Time-based analysis
Park et al. [98]	Plasma monitoring, width estimation	VIS	200 kHz	Regression analysis

Literature	Object of investigation	Type of photodiode	Sampling frequency	Data processing method
Martan et al. [99]	Heat accumulation	NIR, IR	–	Statistical time-based analysis
Ho et al. [100]	Drilling depth quality	VIS	1 MHz	Time-based analysis

There are three types of optical radiation signals that typically occur in laser-based manufacturing. The first type is that with ultraviolet (UV) and visible (VIS) light wavebands (200–750 nm) for which silicon-based photodiodes are used. The second sort of waveband corresponds to laser reflection. The third type is infrared (IR) radiation waveband (1100–1700 nm) which is acquired by photodiodes based on germanium (Ge) and indium-gallium-arsenide (InGaAs) [101]. In each sensor category, there are single sensor devices and array sensor devices. Often, single-sensor devices are referred to as "integrative" sensors which can only focus on a limited processing region. Nevertheless, in monitoring applications, the optics of the sensor front enable for it to see only a limited portion of the emitting object or surface. This 'integrative' single-sensor system simply monitors the overall radiation emitted by the surface with a defined region of interest (dependent of the area of view). It provides the total amount of radiation captured (i.e., intensity of the recorded radiation and its size), but does not provide spatial information about the radiation as provided by cameras.

Based on the configuration of the photodiodes in the laser processing setup, there exists two categories of photodiode-based process monitoring, namely off-axis and coaxial. Off-axis sensors can be placed relative to the principal laser beam (Fig. 9a). In this scenario, the sensor observes the process from a different direction and with a different set of optics than those of the primary laser beam. That is to say that the laser beam and the sensor both have their own optical paths [102]. Alternatively, the laser beam and the radiation sensor could share a portion of their optical path near the processing area. This configuration is known as coaxial as the laser beam and emitted radiation act along the same optical axis or "look" at the process in the same direction or along the same axis (Fig. 9b). The sensor always looks at the processing spot from the same (axis-symmetric) point and follows the spot while the laser spot scans across the object being processed [87]. Off-axis configurations may also be put on the laser head to track the movement of the laser spot regarding the processed object; however, this results in an asymmetric sight that generally varies as the scan direction of the laser beam moves.

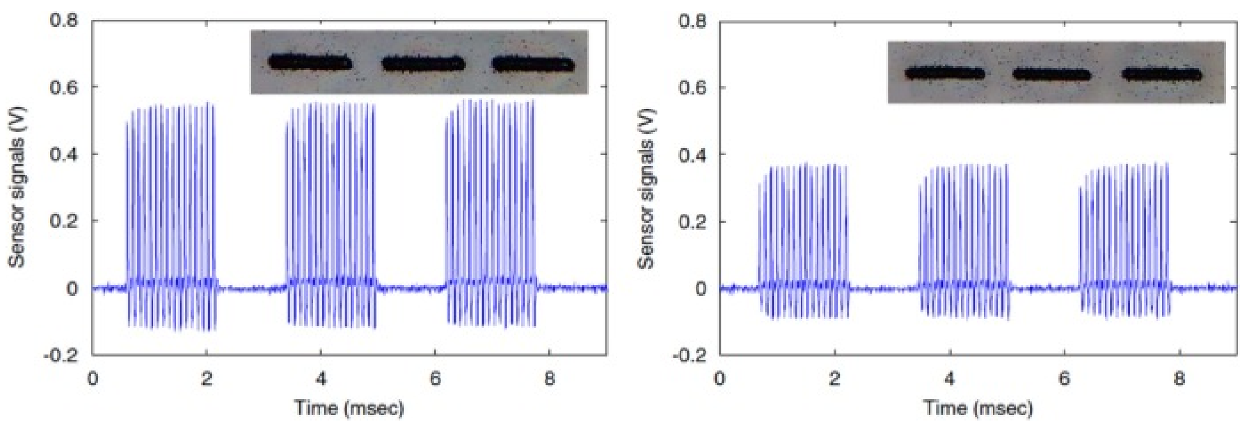


Download: Download high-res image (174KB)

Download: Download full-size image

Fig. 9. Schematic representation of a) off-axial and b) co-axial photodiode implementation – reproduced from [98,99,104].

The concept of monitoring VIS process emissions is in an incipient stage for laser micromachining with short pulses and USP [105]. By monitoring USP laser machining with photodiodes, Gehrke et al. [37] demonstrated a correlation between VIS process emission and focus position, laser power, ablated volume and the surface roughness. Park et al. [98] constructed a photodiode monitoring system that detects the plasma created during laser marking which can be used to develop marking-width estimation models. During the high-speed marking operation, the photodiode on the coaxial line was able to precisely record the light emission. Based on the relationship between the laser power, the line width, and the recorded sensor data, statistical regression and artificial neural network models were used to predict the width of laser marking (Fig. 10). It was demonstrated that the neural network model outperformed all other models in terms of both the coefficient of determination and the average estimation error rate. Zuric et al. [66] demonstrated a system for laser micromachining process and machine condition monitoring in real-time. By monitoring multiple process emissions (UV, laser back reflection and IR) concurrently, a relationship between the signals and the surface roughness was identified. Different signal pre-processing techniques were applied based on the required physical signal interfaces, sampling rates, and data types, where subsequently timestamped data was integrated for process analysis. Martan et al. [99] developed a high-speed IR photodiode-based monitoring system to investigate the heat accumulation during laser micromachining of lines. During laser micromachining, the IR measurement system was able to capture the thermal radiation changes at nanosecond and microsecond time scales from locally heated microscale locations. The proposed measurement method is ideal for evaluating various scanning strategies and material removal techniques towards selecting the optimal process settings.



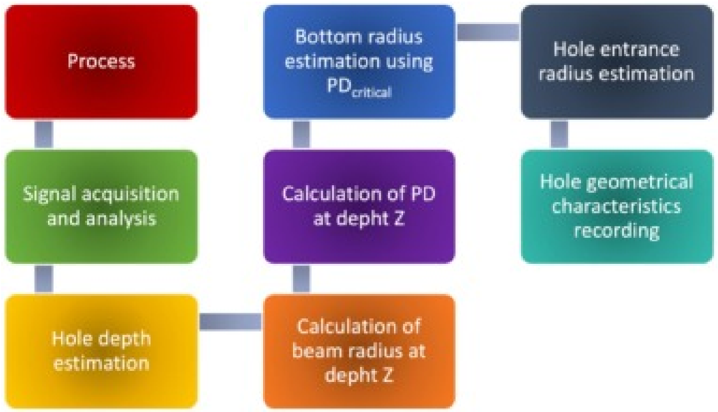
Download: Download high-res image (251KB)

Download: Download full-size image

Fig. 10. Laser marking shapes and signals from photodiode for different laser powers: a) 3W, b) 2W [98].

Optical process emission can also be utilized as a significant signal source for online process monitoring of laser drilling. It was demonstrated that the hole formation during ns laser drilling could be predicted from the coaxial recording of the process lighting. At the instant of breakthrough, the amount of laser-irradiated material at the bottom of the drill hole quickly decreases. Additionally, the vaporized material can leave in the direction of the beam outlet. Due to these factors, the breakthrough is indicated by a significant decrease in the coaxial optical emission intensity. This not only improves the quality of the hole, but also reduces the process cycle time while protecting any material behind the hole [106].

During laser percussion drilling, Ho et al. [100] used a photodiode to monitor the emitted light and alter the focus location accordingly to improve its quality while decreasing the process time. It was demonstrated that a coaxial photodiode is capable of sensing deviation of plasma emission in real time, which provides feedback for improving drilling depth via focus position control. The increase in plasma-induced brightness variation is associated with a reduction in drilling depth. In a different study, Ho et al. [104] demonstrated that the penetration depth is inversely proportional to the measured plasma light emission for a given laser pulse frequency. This approach of monitoring and controlling the drilling depth can therefore be successfully applied to a variety of materials and radiation intensities. Stournaras et al. [107] suggested a suitable system for monitoring the geometric properties of a pulsed laser drilling process in terms of the depth and upper diameter of the hole (Fig. 11). Photodiode-acquired optical signals were explored, and a link between sensor output and hole geometry was provided. A similar observation of reduced emission intensity drop was observed for drilling with both longer pulses and USP. However, the decline becomes weaker as the sample thickness increases. As the process continues, the borehole widens, causing the photodiode signal curve to climb at a steadily diminishing rate. For both helical and percussion drilling, the conclusion of the process is denoted by a constant signal value [108].



Download: Download high-res image (111KB)

Download: Download full-size image

Fig. 11. Concept of integrated optical-acoustic sensing for blind laser drilling with short-pulsed laser – reproduced from [107].

IR photodiode signals typically exhibit excellent signal to noise ratio, which helps to find the correlation between the signal characteristics and the cut quality variables in laser cutting processes. De Keuster et al. [95] explored photodiodes for a laser-assisted cutting of thick plates with high power lasers. For the most important cut quality features, correlational signal parameters were determined. This data was then fed into the designed control and optimization system, which specifies the steps that must be followed to always ensure optimal process conditions and cut quality. By ensuring a well-optimized weld quality and increasing the cutting velocity, it was possible to achieve satisfying results in terms of both durability and productivity. In another study, De Keuster et al. [109] analysed multi-sensor signals and found that the photodiode signal demonstrates superior adaptability for industrial applications. Photodiodes have shown to be a superior solution as even the raw photodiode signal provides an indicator of the process reliability. The average value of the photodiode signal demonstrated a strong correlation with the drag of the ridges and the likelihood of the dross development. The standard deviation of the signals could be associated with the surface roughness and the occurrence of burning defects.

Adelmann et al. [96] introduced a cutting interruption detection sensor based on an infrared photodiode for fibre laser cutting applications. The sensor is tested relative to typical application settings, and the signature of the sensor (light intensity) for cutting mild steel, stainless steel, aluminium, and laser flame cutting were identified. The piercing process, the laser on/off point, and the waiting interval were all distinguished in the measured signal. The high detection rate for a range of materials and thicknesses demonstrates the applicability of the sensor for cutting interruption detection in industrial applications. Levichev et al. [110] evaluated a coaxial photodiode-based monitoring system for fibre laser cutting of mild steel plates. The proposed monitoring framework enabled the determination of the maximum cutting speed at which cut quality does not decline as well as the detection of cutting in preheated areas. As the inclination angle of the cutting front changes, the InGaAs photodiode signal increases with the increasing speed. Garcia et al. [97] explored quantitatively the relationship between the cutting process parameters and monitoring signals, while analysing the results in four different cutting scenarios involving variations in the assist gas, material, and thickness, as well as each experiment in the VIS and IR spectra. The results contributed to the creation of monitoring systems that would enable accurate identification of cutting losses in a variety of operating settings, even when monitoring signals are either weak or too strong due to the differences in material thickness and emissivity. The measured intensity of the VIS signal from the cutting process was correlated to the creation of a vapor plume, while the IR signal was correlated to the attained temperatures.

Jayasinghe et al. [92] examined the possibility of estimating LPBF build density using a combination of multiple coaxial photodiode sensor information collected during the build process. The work extracts characteristics from databases of photodiode measurement data using Singular Value Decomposition (SVD) prior to feeding the recovered features into machine learning models. Low density specimens are detected using unsupervised clustering techniques (K-means and Gaussian mixture model) prior to using a supervised regression approach (Gaussian Process) to directly estimate the construct density from the same extracted attributes. It was demonstrated that, depending on the density, LPBF builds can be clustered with an accuracy of up to 93.54% using an unsupervised clustering approach. Mao et al. [93] proposed utilizing the benefits of continuous photodiode signals and extensive information on melt pool temperature fields, as well as a continuous online defect detection approach based on deep learning algorithms. After coupling the time corresponding to photodiode signal and melt pool temperature using a neural network, the developed correlation model and threshold setting enabled continuous online defect detection in LPBF. The neural network can build a strong link between the photodiode signal and the average melt pool temperature, with a maximum correlation error of 2.2%. Using plasma light signals, Yu et al. [94] created a technique to monitor the laser welding process. The relationship between processing parameters and monitoring signals were established, and an algorithm for predicting weld quality using monitoring signals was developed. Using the relationship between the weld strength and the monitored signals based on the process variables, an algorithm was developed to determine the quality of laser welds.

Calibration of photodiodes is crucial for ensuring accurate and reliable measurements in various applications, including real-time monitoring of ultra-short laser micromachining. There are several methods to calibrate photodiodes: Absolute calibration involves using a reference light source with a known intensity to measure the photodiode response, comparing it to the reference to determine its absolute responsivity. Relative calibration uses another already-calibrated photodiode as a reference when an absolute reference is not available [101]. Spectral calibration scans the photodiode with a monochromatic light source across different wavelengths to determine its spectral responsivity, essential for precise wavelength-dependent measurements. Each of these methods addresses different aspects of photodiode performance, ensuring that the sensor provides accurate and consistent data for the system requirements [102].

In optical emission monitoring for laser-based manufacturing, hardware filters are crucial in refining photodiode measurements by selectively controlling the light reaching the sensor. Spectral filters, for example, isolate specific wavelengths of interest, allowing the photodiode to detect emissions relevant to the laser process while blocking other wavelengths that may interfere with accuracy. This selective wavelength transmission is essential in applications where multi-wavelength emissions occur, helping to focus on specific laser characteristics or material reactions [106]. Neutral Density (ND) filters are also widely used to control the intensity of light entering the photodiode. By attenuating light uniformly across wavelengths, ND filters prevent saturation of the photodiode and enable measurements within a linear range, which is especially valuable in high-intensity laser environments. These hardware filters enhance the reliability of in-situ monitoring, ensuring that photodiode readings remain consistent and precise under varying laser conditions [108].

Overall, it can be perceived that optical emission using photodiodes is a promising process monitoring approach for pulsed laser micromanufacturing and other laser-based processes. Despite the steadily increasing interest in micromanufacturing with USP lasers and the first large-scale industrial implementations in recent years, there is still no industrially suitable or even commercially available monitoring system on the market. On the one hand, this is due to the limited availability of the beam sources and the associated short process development time compared to other laser applications. On the other hand, it is due to the complex physical processes caused by the high temporal and spatial energy density, which occur during processing on an ultra-short time scale and space scales of only micrometres. The same also applies to the large area surface structuring processes. Photodiodes detect laser reflection, scattering (in the timescale of a few fs), and plasma formation and propagation (in the timescale of a few μs), offering insights into surface reflectivity changes and plasma intensity during material removal. Despite their effectiveness of the photodiodes, high PRR can cause overlapping of the optical emission events, which highlights the need to use photodiodes with high temporal resolution (see, Table 5) capture picosecond-scale events [105].

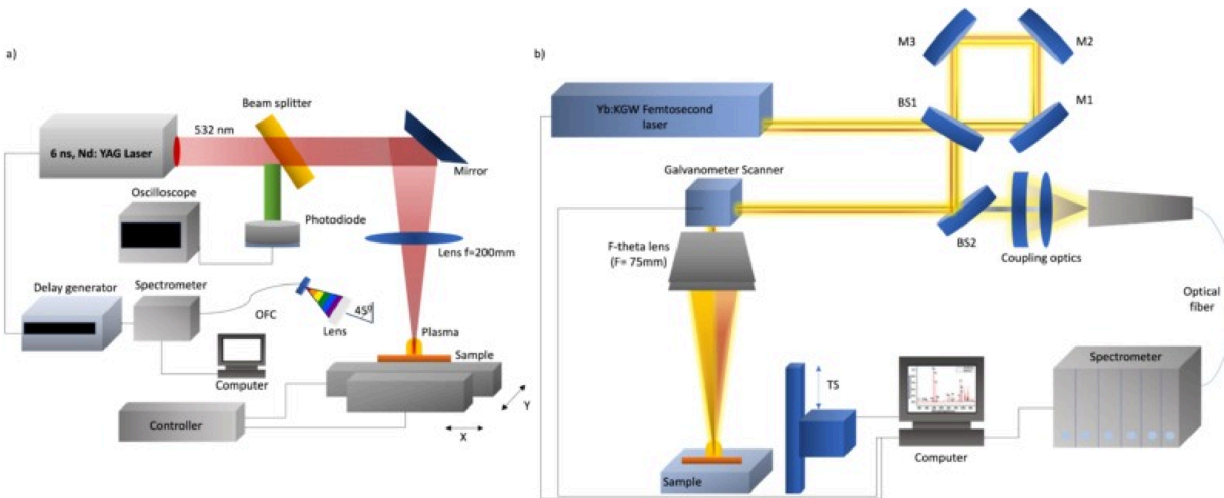
Table 5. Non-exhaustive list of existing photodiode sensors on the market.

Spectral range	Supplier	Cut-off frequency	Rise time
190 to 2600 nm	Hamamatsu	Up to 1 GHz	<1 ns
170 to 2600 nm (VUV to IR)	Alphalas	Up to 30 GHz	<50 ps
340 to 1700 nm	OSI Optoelectronics	Up to 1.25 GHz	<1 ns
350 to 1700 nm	AMS Technologies	Up to 1000 MHz	<24 ns

This chapter presented the use of various photodiodes for process monitoring and feedback management in laser processes, with an emphasis on laser cutting, LPBF, and laser micromachining. Online monitoring of the electromagnetic radiation emitted by the process area around the laser spot has been shown to be an effective method of assessing the process performance and quality. It enables the detection of process defects such as dross, striations, balling, surface roughness degradation, loss of cut, etc. While the process emits a broad spectrum of light (from ultraviolet to infrared), the best results are achieved by monitoring near infrared or thermal radiation, as the described laser operations are mostly thermally driven. Individual point sensors, 2D sensor arrays (e.g. CCD or CMOS camera) or integrated sensors (e.g. large area photodiodes) positioned coaxial or off-axial to the optical path of the laser beam can be used to monitor this radiation. With a coaxial arrangement, the image is not affected by the movement and direction of motion of the laser beam, allowing for improved outcomes. Coaxial configurations, on the other hand, demand more sophisticated optics to separate the primary laser radiation from the emitted process radiation.

Laser-Induced Breakdown Spectroscopy (LIBS)

Laser-material interaction typically induces plasma, which can be studied to provide information about the elemental composition of the ablated material to regulate the machining process, such as stopping the laser beam when a specific material layer is reached [111]. LIBS is an analytical technique to characterize the elemental composition of the laser-induced plasma based on the spectral analysis. During LIBS, powerful laser pulses are used to produce an expanding plasma plume that emits light by recombining ionized particles. The measurement of the plasma emission with spectrally resolved detection provides the elemental composition of the substance being ablated [111,112]. Fig. 12 shows a typical LIBS setup where short pulsed (Fig. 12a) and USP (Fig. 12b) laser sources are generating plasma.



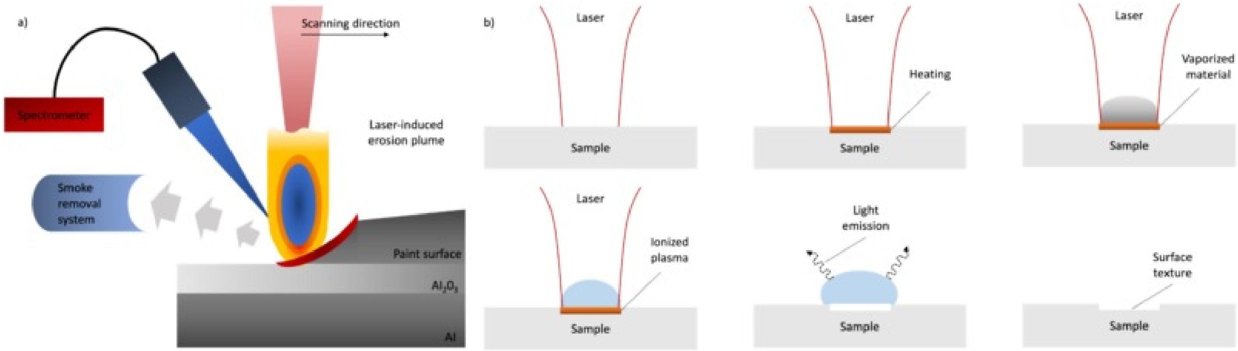
Download: Download high-res image (203KB)

Download: Download full-size image

Fig. 12. Experimental setup of the LIBS-monitored a) coaxial fs laser processing system - reproduced from [116], b) off-axial ns laser processing system – reproduced from [117].

LIBS is a widespread approach that has significant characteristics such as the flexibility to analyse all types of materials independently of their physical forms and aggregate states, the ability to do local microanalysis, high sensitivity, and operational simplicity. LIBS is frequently used for qualitative and quantitative chemical analysis owing to its inherent advantages, such as in-situ analysis, minimal or no sample preparation, capability to identify several elements, and essentially non-invasive measurement procedure due to the small amount of material ablated. LIBS has shown effectiveness for online material identification and sorting, controlled surface cleaning, multi-layer removal monitoring, and thickness or depth measurement [113]. LIBS applications include the material characterization of homogeneous substances and the compositional mapping of heterogeneous substances. LIBS using fs laser sources is a prospective real-time feedback control system that can allow quick, relatively non-destructive sample analysis without direct sample contact. In addition, fs lasers generate LIBS signals with a smaller Bremsstrahlung continuous background than nanosecond lasers. Bremsstrahlung emission depends on the deposited energy, the electron density, and the temperature [114]. Depending on the type of laser sources, laser micromachining generates plasma that can be utilized similarly to typical LIBS applications [115].

The plasma breakdown as the fundamental physical process for the LIBS measurement is a laser-related and process-dependent phenomenon. In the case of laser micromachining, the laser-induced plasma is a by-product of the ablated material and the residual laser energy. The plasma breakdown is essentially a multi-step process that is greatly impacted by the laser pulse irradiation. The energy released by the recombination of ions is an electric magnetic wave in the light spectrum (Fig. 13) [115]. Utilizing these qualities of laser-induced plasma as a by-product of laser ablation procedures, the spectroscopy of plasma emission can identify the substance that was really ablated. It is achieved by the optical diffraction of the emitted plasma spectrum into its wavelength-dependent components. By analysing the spectra, certain substances can be recognized based on their unique plasma emission curves [118].



Download: Download high-res image (125KB)

Download: Download full-size image

Fig. 13. a) Schematical representation of material removal process and recording a spectral signal – reproduced from [119], b) schematic illustration of the plasma expansion mechanism, induced by laser irradiation and subsequent followed by the vaporization and ionization of the ablated material and the recombination of the plasma with the characteristic light emission – reproduced from [115].

LIBS has been applied not only for surface analysis but also the depth-resolved investigation of materials, particularly for coated systems. During depth-profiling, several laser pulses are administered at a specific place on the surface of the sample to create a crater, and the LIBS system collects spectral signals from various depths during the process. By utilizing this technique, it is possible to evaluate the material ablated by a single or series of pulses and correlate the inferred chemical information to the precise depth within the sample [120]. Another potential application of LIBS is the focus position control [147] which is challenging during laser micromachining of large and complex parts [121]. In a pertinent context, it is highlighted that in laser ablation micromachining applications, LIBS may be a useful tool for online monitoring of the laser processing.

The application of LIBS for the monitoring of laser-material interaction has already been demonstrated for various laser-based manufacturing processes. Table 6 summarizes the literature review of LIBS-based monitoring of laser manufacturing. Roozbahani et al. [122]

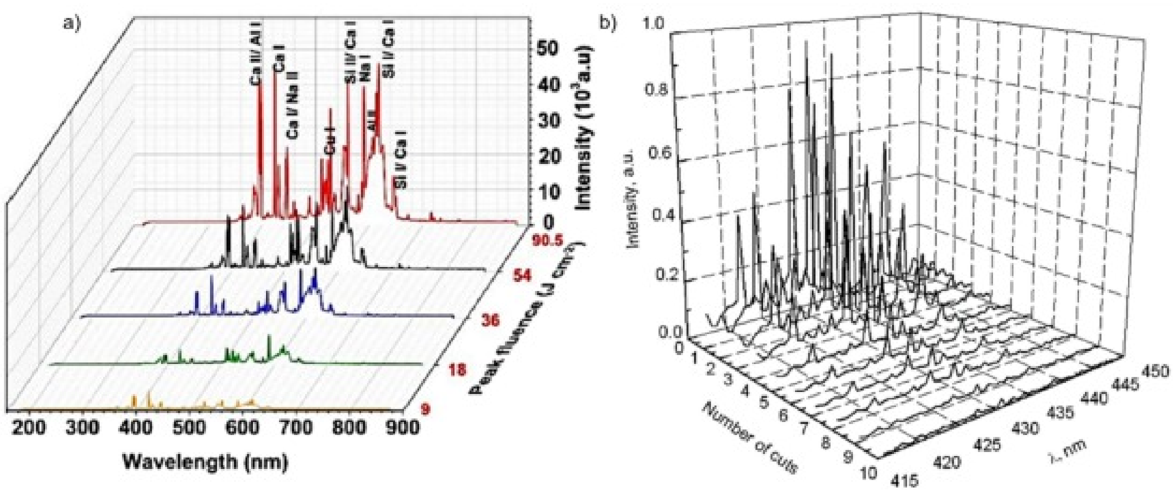
investigated the sensitivity of spectrometers for laser scribing by modifying a number of laser and process parameters and analysing the sensitivity of the spectrometer to the change in radiation emission intensity. As the pulse energy and pulse duration diminished, the intensity of the radiation emission between the observed wavelength range also dropped correspondingly. The spectrometer showed significant sensitivity to the minor change in the focal position. It is evident that the spectrometer is extremely sensitive to the laser beam focus and might be used to locate the focal point. This can be used with a laser control loop to automatically keep the laser beam in focus during processing. Since laser scribing generates fairly comparable radiation in each experiment, the spectrometer data can be utilized to determine whether or not the process is operating optimally [122].

Table 6. Summary of literature review on in-situ sensing of laser-based manufacturing via LIBS.

Reference	Object of investigation	Spectral range	Spectral resolution	Data processing
AM				
Lough et al. [123]	Vapor plume	400–700 nm	0.1 nm	Wavelength – intensity
Ren et al. [124,125]	Plume spectra	310–470 nm	0.5 nm	RF classifier, unsupervised learning
Lednev et al. [126]	Multi-element analysis	<200 nm	0.05 nm	Wavelength – intensity
Laser welding				
Lednev et al. [127]	Elemental analysis	275–290 nm	18 nm	Wavelength – intensity

Reference	Object of investigation	Spectral range	Spectral resolution	Data processing
Huber et al. [128]	Elemental analysis	380–880 nm and 200–800 nm	7 nm	Wavelength – intensity
Laser scribing				
Shiby et al. [117]	Micro-scribing	300–700 nm	1.4 nm	Wavelength – intensity
Das et al. [129]	Elemental analysis	190–900 nm	0.2 nm	Intensity – depth
Diego-Vallejo et al. [130]	Plasma monitoring	300–700 nm	0.316 nm	Wavelength – intensity
Pulsed laser micromanufacturing, laser ablation				
Skruibis et al. [116]	Plasma emission	400–700 nm	0.1 nm	Wavelength – intensity
Tong et al. [112]	Plasma plume	200–850 nm	1.3 nm	Wavelength – intensity
Baskevicius et al. [131]	Plasma plume during laser cutting	400–700 nm	0.1 nm	Wavelength – intensity
Lentjes et al. [132]	Plasma plume	200–1100 nm	2 nm	Wavelength – intensity

Several studies have examined novel possibilities, where in-depth profiling is one of the intended applications that has been successfully tested. In case of laser scribing with multiple passes, the depth of the channel may fluctuate with each laser scan due to varying factors, such as roughness and reflectivity. In multiple-pass laser scribing, it is vital to regulate the material removal depth and remove the substrate film entirely. Additionally, it is essential to comprehend the substrate damage. Consequently, an online monitoring system that can monitor the approximate depth of the microchannel and notify the ablation of the layer beneath the substrate is of great relevance [133]. Shiby et al. [117] reported micro-scribing of Cu-clad substrates and LIBS-based process monitoring. The effect of microchannel roughness and reflectivity on the acquired depth during each laser scan was examined. From the obtained LIBS spectra, the Cu line emission intensity was correlated with the microchannel depth (Fig. 14a). The LIBS was shown to be a proven monitoring technique for estimating the depth of ablation and monitoring the damage to the substrate. Das et al. [129] investigated the viability of qualitatively differentiating between the coating layers, and between the coating and the substrate based on the spectral signals recorded at different sample depths. Monitoring the variation in tungsten spectral intensity allows it to qualitatively identify the coating from the substrate. The depth-resolution was reported to be affected by the mixing of spectral signals from various depths of the conical crater formed by Gaussian laser pulses during depth profiling.



Download: Download high-res image (317KB)

Download: Download full-size image

Fig. 14. Collected LIBS spectra of the fs laser micromachined steel sample related to the number of scanning passes over the same groove [134], b) LIBS spectra during substrate ablation for different fluences [117].

Skruibis et al. [116] demonstrated the determination of optimal focal position using multi-pulse LIBS for monitoring fs laser micromachining of soda-lime glass. During micromachining of glass in air and water with multiple laser pulses at different scanning speeds, the LIBS signal was always stronger and observed for a longer duration compared to that of single-pulse excitation with the same pulse energy. The observed nonlinear increase of LIBS signal in the event of multiple-pulse excitation can be explained by the fact that new and additional pulses contribute to plasma excitation. In conclusion, the proposed experimental setup can be utilized not only for monitoring the laser micromachining of materials, but also for controlling the cutting of layered heterogeneous materials, thin film scribing, thin layer removal, and sensitive material analysis via LIBS methods. Tong et al. [112] proposed an online detection approach for the in-situ control of USP laser micromachining based on the ultrafast LIBS methodology. The challenges related to the laser pulse and the spectral, temporal and spatial information of LIBS signals for identifying various materials were examined. After adequate processing of the spectra, it was demonstrated that the proposed comparison algorithms can produce accurate and consistent detection. Baskevicius et al. reported the use of LIBS-based monitoring of fs laser micromachining of steel [156] and glass [131,159]. These studies investigated the properties of LIBS spectra obtained during fs laser micromachining with various processing parameters (scanning speeds, pulse repetition rates, scanning algorithms, etc.) to assess the viability of this technique for monitoring purposes. For the scanning technique with wider groove spacing, LIBS signals degraded more rapidly and do not exhibit a secondary increase in intensity (Fig. 14b) [134]. These studies indicate that correlation between the LIBS spectra and the process parameters can be identified to realize the laser micromachining process optimization and monitoring.

Diego-Vallejo et al. [130] examined the effect of processing parameters on the produced plasma to identify relevant relationships that may be utilized as monitoring signals during the machining of various materials. A multi-material for photovoltaic applications was used to evaluate the capability of the LIBS system during selective removal of thin films. Deng et al. [135] investigated the plasma emission during fs laser processing, which offers important information about the process nature. The plasma temperature emitted was determined using spectroscopic examination of the ablation plume for a copper sample based on the relative strengths of the spectral lines. By decoding the information provided by the optical spectrometer, it is feasible to construct a practical signal diagnostics tool for monitoring the quality of the process. Lentjes et al. [132] presented a comprehensive analysis of the impact of parameter choices on the distribution of correlation coefficients. The parameters altered including pulse energy, delay time, integration time, grating groove density, object lens distance deviation and crater depth. Experiments revealed that all evaluated characteristics exerted both indirect and direct effects on the correlation coefficient distribution.

The development of online sensing and control systems is vital for enhancing product quality and expanding the AM process for high-value applications, such as in the aerospace sector. Due to the small sampling area and online data, the LIBS approach can be used as an effective monitoring tool during the laser cladding repair of broken parts or tools. Lednev et al. [126] created a small LIBS probe for in-situ quantitative elemental analysis of light and heavy elements within the melt pool generated during coaxial laser cladding. This technique is effective for controlling the synthesis of functionally graded materials during the fabrication of parts or the repair of tools. If a LIBS instrument is equipped with a 3D printer, a complete three-dimensional elemental map of the product/repaired part can be captured to create an “elemental passport”. A similar LIBS system was applied for in-situ monitoring of the laser welding process. In-situ LIBS measurements with the ablation of the weld melt pool demonstrated that the reproducibility of LIBS signals was the same for both defective and optimal laser welding, whereas the emission of atomic and ionic lines was systematically greater for laser welding, allowing both welding regimes to be distinguished in real time [127]. Melt pool tracing by LIBS probe resulted in greater atomic/ionic line plasma emission, while hot solidified clad sampling resulted in an improved LIBS signal repeatability. A good agreement was found between online LIBS analysis and offline XRF measurements, encouraging further applicability of LIBS for AM process control [136].

During LPBF, Lough et al. [47] conducted optical emission spectroscopy (OES) by placing the optics of the spectrometer into the laser beam path. Melt pool characteristics of samples were associated with in-situ OES data corresponding to local processing conditions. This is crucial because integrating OES measurements to melt pool characteristics relates the in-situ gathered data to a crucial component feature that may contain defects due to under- or over-melting and establishes the density and surface roughness of LPBF parts. The intensity of chromium emission correlated well with the size of the melt pool. This work established a correlation between radiometric measurements from OES and a significant sample property, indicating the applicability of LIBS for LPBF part qualification and control [123]. The use of OES as a process monitoring tool offers practical information on the local process conditions in LPBF and assists in system development. In-situ spectroscopic measurements can be utilized in the certification of parts during manufacturing and have the potential for in-process control of LPBF since they efficiently monitor the local melt pool properties.

Ren et al. [124] studied the use of spectrometer for the monitoring of direct energy deposition (DED). The experimental results showed that the aluminium spectral line emissions, baseline intensity, and plume area were sensitive to the irregular deposition quality and have the potential to be used for in-situ quality monitoring. Under conditions of high laser power and rough surfaces, an intense plasma with a columnar structure and active spectra containing primary elemental emission lines were generated. In contrast, situations with relatively low laser power and smooth surfaces yield weak plasma with a water-drop form and mild spectra. Ren et al. also examined in-situ quality monitoring of the DED process by analysing the emission spectrum signals created during the process, a technique known as plasma emission spectroscopy (PES). This study addressed several difficulties associated with the in-situ monitoring of AM, such as the development of robust and effective feature extractors, unsupervised multiple-quality classification, and sequential signal processing. Since the suggested method is generic, it is applicable to the in-situ monitoring of AM in a variety of applications, including the recognition of varied feedstocks and defects [125]. Important spectral characteristics for the identification of porosity were identified and investigated. An RF classifier created and trained using examples containing spectrum characteristics and ground truth deposition attributes. For instances from the training layers or their adjacent layers, the classifier achieved a porosity recognition accuracy of up to 83% [137].

Calibration and filtering are critical components of LIBS for ensuring accurate and reliable monitoring in laser-based manufacturing. Calibration involves establishing a relationship between the intensity of spectral lines and the elemental concentration, typically using standard samples with known compositions. This process is crucial for quantitative analysis and often requires applying a matrix-matched calibration approach to account for variations in plasma characteristics influenced by factors such as laser energy, focal position, and material type [118]. Filtering in LIBS, particularly when used in noisy industrial environments, helps improve signal quality by removing background noise, enabling clear detection of emission lines. While LIBS generally benefits from a high signal-to-noise ratio due to intense plasma emissions, additional hardware filters, like spectral bandpass filters and ND filters, can further isolate relevant spectral lines and prevent detector saturation. Additionally, post-processing filters, such as digital smoothing and baseline correction, can refine the data, enhancing precision and consistency in LIBS measurements [125].

In conclusion, LIBS is an innovative and effective method for in-situ monitoring of laser-based manufacturing processes. LIBS offers in-situ, non-contact, non-destructive monitoring of processed material with high precision and accuracy. The ultrafast LIBS-controlled micromachining technology has several advantages, including less damage to the substrate layer, shorter machining times, more uniform machining features, and lower failure of micromachined devices at high temperatures. With the increasing demand for more efficient and effective manufacturing processes, LIBS has the potential to play a significant role in the monitoring of laser-based manufacturing. The current LIBS market is given in Table 7. However, there is still a need for further research and development to optimize the system performance and to address limitations such as low signal-to-noise ratio. Overall, LIBS presents a promising opportunity for the future of in-situ monitoring of laser-based manufacturing and has the potential to revolutionize the way we approach process control and optimization. LIBS provides elemental and chemical composition data by analysing plasma spectral lines, reflecting electron-lattice energy transfer efficiency. LIBS is highly effective for characterizing plasma formation stages, although intense emissions at high fluences can complicate spectral analysis [114].

Table 7. Existing LIBS sensor market.

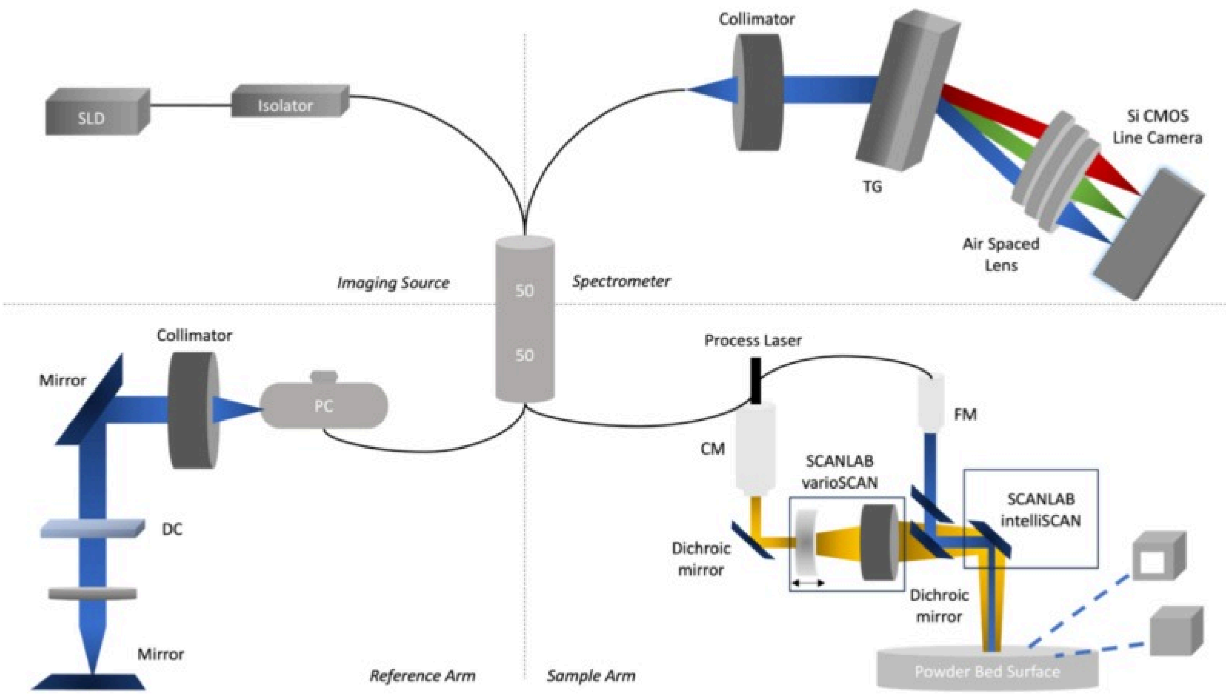
Spectral range	Supplier	A/D resolution	Wavelength resolution	Integration time
200 to 1100 nm	Ocean Optics	16-Bit	10 nm	10 µs to 10 s
200 to 1750 nm	Avantes	16-Bit	7 nm	10 µs to 40 s
190 to 900 nm	LTB Lasertechnik	16-Bit	5 nm	5 ns to 1 ms
500 to 1050 nm	Ibsen Photonics	16-Bit	1.2 nm	1 ms to 20 ms
200 to 2350 nm	Hamamatsu	16-Bit	12 nm	100 µs to 32 s
230 to 1030 nm	Andor Technology	23-Bit	2.3 nm	10s to 1 s

LIBS provides, in contrast to other measurements, the fundamental capability of the material and consequently layer change detection even for thin layers, because LIBS exploits plasma, which is induced by the laser machining process, and is also free of restrictions due to its optical components wide range of applicability. LIBS employs the mechanisms of laser-induced plasma, which enables the measurement of the material composition of the ablated material and the collection of real-time data regarding the laser process. The obtained information is immediately fed back to the laser micromachining system so that its parameters can be adjusted to optimize the current ablation condition. By analysing the plasma emission spectra, the LIBS systems integrated into the laser processing chambers established their capacity for measuring samples and identifying their constituent materials. Furthermore, the adaptability of LIBS configuration to specific boundary conditions makes this technology suitable for a wide range of applications.

Optical coherence tomography (OCT) and laser interferometry

Optical coherence tomography (OCT) is another monitoring technique, based on interferometry, which uses a probe beam to scan the surface and generates in-line information about the structure geometry, [139]. OCT is a suitable technology for imaging ablation structures beneath the surface in the same range as the ablation depth of USP laser and detecting structured geometries. As a result, it is an ideal imaging tool since the typical resolution is in the region of a few μm, which is comparable to the resolution of fs-laser micromachining applications. The topic of integrating an OCT module for in-situ monitoring into an existing laser micromachining system is presented in this chapter. It should be noted that process control with OCT can enhance manufacturing operations and replace sensors in challenging environments to improve measurement precision.

A typical OCT system (as illustrated in Fig. 15) comprises a low-power broadband light source (e.g. super-luminescent diode), a high-speed spectrometer, and a Michelson interferometer based on fibre optics. To prevent back-reflection, the fibre-coupled broadband light emitted by the super-luminescent diode (SLD) must first pass via an optical isolator. A 50:50 beam splitter then separates the light into sample and reference arms. Using a dichroic mirror, light from the sample arm is mixed coaxially with a processing laser beam. The combined beams are then focused using a common laser processing objective on the sample. Backscattered imaging light is collected by the sample arm fibre from the sample [18]. Light in the reference arm travels via a fibre polarization controller (PC) and dispersion matching devices in order to compensate for polarization variations caused by the single mode fibre and dispersion caused by optical equipment in the sample arm. At the beam splitter, reflected light from the sample and reference arms is recombined and delivered to the spectrometer [138]. The relative optical path difference between the sample arm and the reference arm results in electric field phase deviations, which produce fringes in the combined intensity spectrum (i.e., the interferogram). The fringe spacing is determined by the relative difference in optical paths between each arm. The modulation frequency over the interference spectrum increases according to the difference in optical path length between the two arms. Consequently, by spectral analysis of the interferograms, the system measures the distance to every surface point with an accuracy of μm, resulting in a reflection in the imaging region [139]. For the currently-employed low coherence light sources, the coherence length is typically in the range of 1 to 15 μm, allowing OCT to achieve an excellent axial (depth) resolution. A- and B-scans designate single depth scans and cross-sections, respectively. It is feasible to acquire so-called C-scans when imaging at a constant depth. The image plane of C-scans is perpendicular to that of B-scans and corresponds to the orientation of images produced by conventional microscopy [140]. The capacity to perform non-mechanical depth scanning (A-scan) and single-shot measurement is one of the benefits of OCT. Another benefit is that the laser processing can be conducted while simultaneously measuring the structure depth. Therefore, OCT is superior for in-process monitoring of laser processing. In-process monitoring of the depth of the processed structures in laser processing of stainless [141] and glass [142] has been demonstrated using OCT.



Download: Download high-res image (216KB)

Download: Download full-size image

Fig. 15. Schematic diagram of the OCT sensor configuration in the laser-based manufacturing chamber – reproduced from [147].

OCT is classified into time-domain OCT (TD-OCT) [61] and frequency-domain OCT (FD-OCT) [67]. Compared to TD-OCT, FD-OCT is distinguished by faster and greater signal-to-noise ratio measurements. In contrast, the time-domain method permits dynamic focusing and maintains continuous sensitivity throughout the whole depth range. Due to the interferometric detection scheme and the short coherence length of the utilized light sources, a signal is only recognized during TD-OCT measurements if the photons reflected from both interferometer arms have travelled the same distance to the photodetector. Clearly, the mechanical movement of the reference mirror is time-consuming and may result in mechanical instability and noise. The FD-OCT method is one means of accelerating the image acquisition procedure [140]. Here, the reference mirror is stable, and light from both the object and the reference is detected in a spectrally resolved mode. Further classifications of FD-OCT include spectral-domain OCT (SD-OCT) and swept-source OCT (SS-OCT). SD-OCT and SS-OCT improve imaging speed and sensitivity [18]. With better light sources, ultrahigh resolution OCT (UHR-OCT) with depth resolutions as low as 1 μm and enhanced image quality becomes available. The separation of lateral and axial resolution makes it possible to measure structures with a high aspect ratio, such as bore holes. In addition to thickness and distance measurements and surface roughness assessments, it is also feasible to determine the material refraction index by OCT [143]. For these reasons, FD-OCT techniques are preferable for large area use, since they offer faster scan rates and a greater SNR than the classic time-domain method [144]. When comparing the findings from various OCT techniques, the following differences can be summarized in the Table 8.

Table 8. A summary of various OCT techniques.

OCT technology	Advantages	Disadvantages	Typical application areas
TD-OCT [61]	High resolution and sensitivity	Low imaging speed	Quality control and surface characterization
FD-OCT [67]	High imaging speed and resolution	High cost and complexity	Quality control and surface characterization
SD-OCT [18]	High imaging speed and resolution	High cost and complexity	Quality control, surface characterization, and 3D imaging
SS-OCT [18]	High imaging speed and penetration depth	Low lateral resolution	Non-destructive testing, material characterization, and thickness measurements

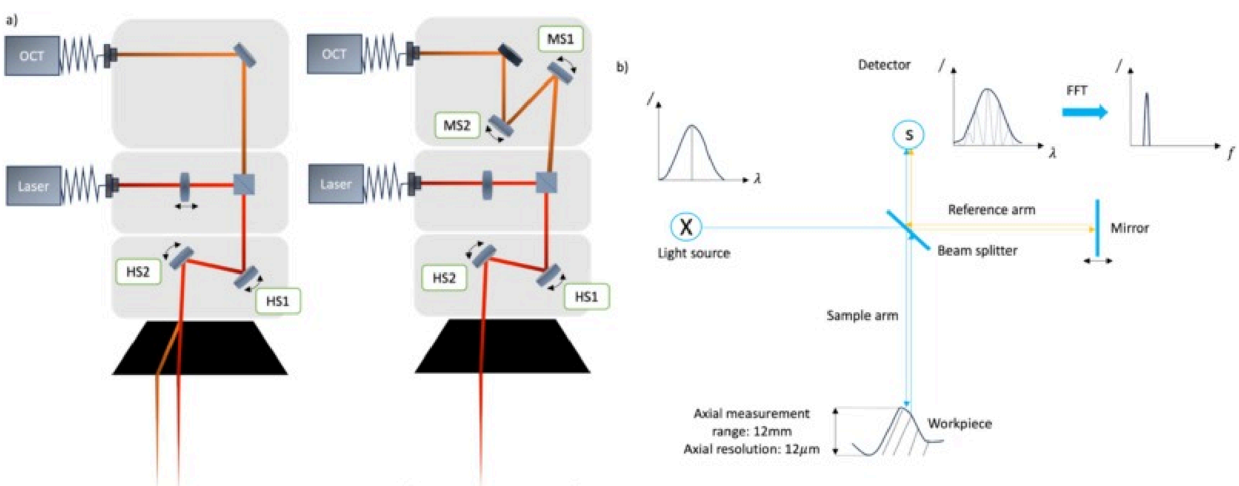
OCT technology	Advantages	Disadvantages	Typical application areas
UHR-OCT [144]	High resolution and sensitivity	High cost and complexity, need for high-quality optical components	Surface characterization, microscopy and material science
PS-OCT [140]	Provides birefringence information	Low penetration depth and sensitivity	Material characterization, birefringence measurements, and stress analysis

Since OCT can offer surface profile and depth information, it enables a variety of high-resolution, non-destructive inspection techniques during laser ablation. OCT systems with an associated OCT scanner, data acquisition and digital signal processing unit can be simply included into current laser heads by connecting to the existing camera interface, while the laser, process fibre, and head configurations remain unmodified [57]. Recent works [150,151] have demonstrated that OCT is a rapid and non-destructive method for evaluating the shape and depth of laser-induced (sub)surface structures in a range of materials. The high sensitivity of OCT system enables the imaging of extremely sharp edges, and it can also provide feedback on the optimal machining parameters. This can be done during post-processing or in in-situ during the machining process to enable access to dynamic processes [140]. Table 9 summarizes the findings of OCT monitoring in laser-based manufacturing. In comparison to non-coherent techniques based on laser profiling or camera vision systems, interferometry as a coherent optical technique allows for simple in-process measurements even in the harsh laser-based manufacturing environment. Interferometric techniques are less impacted by harsh process condition since they utilize phase information that can be recovered from scattered light even at low intensities, and the coherent operating principle results in a significant rejection of signals caused by non-coherent arc light [145].

Table 9. Summary of literature review on in-situ sensing of laser-based manufacturing via OCT.

Literature	Object of investigation	Axial resolution	Scan rate	Sensor wavelength
AM				
Neef et al. [146]	LPBF defect detection	10 µm	70 kHz	880 nm
DePond et al. [147]	LPBF bulk behaviour	10 µm	300 kHz	840 nm
Kanko et al. [138]	Melt pool morphology	7 µm	200 kHz	860 nm
Laser welding				
Blecher et al. [148]	Keyhole depth measurement	22 µm	200 kHz	840 nm
Boley et al. [149]	Vapor capillary – seam depth	10 µm	70 kHz	650 nm
Webster et al. [12]	Weld depth measurement	–	230 kHz	–
Pulsed laser micromanufacturing, laser ablation				
Zechel et al. [151]	Laser micro structuring	3.6 µm	70 kHz	1030 nm
Hasegawa et al. [18]	Laser micro structuring	4.51 µm	2 kHz	1030 nm
Webster et al. [141]	Laser ablation monitoring	5–10 µm	20 kHz	830 nm
Wiesner et al. [143]	Laser micromachining	1 µm	1.4 kHz	930 µm
Ji et al. [152]	Pulse morphology changes	7–8 µm	230 kHz	860 kHz

The preferred concept for integrating the developed FD-OCT into a laser micromachining machine is based on the use of an optical filter or a dichroic beam splitter as an optical coupler. The coupling is accomplished by utilizing this optical element in the reflective region for the measuring wavelengths and in the transmissive area for the structuring laser wavelength. The measuring laser beam transmission efficiency is a critical requirement for this setup. The laser beam coupling performance is directly proportional to the ablation coefficient of the machining process as well as the total heat generation of the coupling system. To ensure a stable system with minimal long-term deviations, such as component wear, and minimal energy losses, an optical element close to 100% transmission must be selected. Beam alignment precision is a further essential system requirement. A misalignment between the machining and measuring laser beams results in a measurement error, as seen in Fig. 16a [150]. In order to ensure coupling of the machining and measuring beam paths, an additional pair of galvanometer mirrors (MS1 and MS2, as shown in Fig. 16a) could be added which additionally deflects the measuring beam.

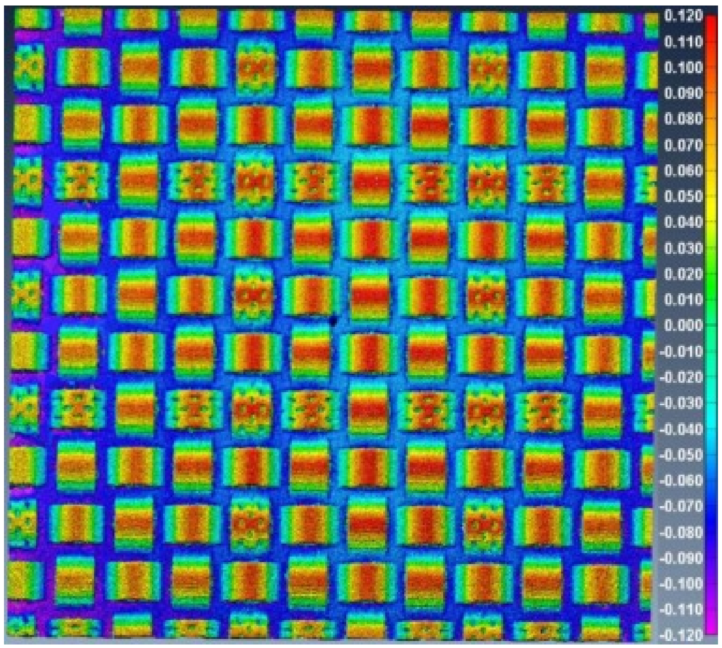


Download: Download high-res image (168KB)

Download: Download full-size image

Fig. 16. a) OCT system comparison without (left) and with beam manipulation (right) – reproduced from [151], b) Schematic layout of the OCT sensor setup.

Schmitt et al. developed an OCT-based system for in-situ monitoring of laser micromachining which can achieve in-line depth measurements on glass, plastics, metals, and composites. Towards demonstrating its applicability for laser-structured tools and moulds, measurement of laser-machined structure (20 mm×20 mm) was demonstrated (Fig. 17), which illustrates the reliability of OCT for inline inspection for surfaces [150]. The depth measurements showed a standard deviation of <218 nm and a nonlinearity of <1% throughout a measurement range of 1 mm. On the basis of this process monitoring, a machining control can be established, leading to completely automated process adjustment and production. Zechel et al. integrated an offline OCT with a laser machining system for measuring the laser-structured lines [151].



Download: Download high-res image (267KB)

Download: Download full-size image

Fig. 17. 3D measurement of a laser structured surface (unit in mm) [150].

Hasegawa et al. [18] demonstrated process monitoring of structures produced with line-shaped fs laser pulses by a spatial light modulator (SLM). SS-OCT was utilized for the in-process monitoring due to the fact that the line-shaped beam processes a large area at a high rate. In addition, multi-ring drilling was performed to verify the viability of OCT for in-process monitoring of the structure depth during laser processing. The debris deposited on the bottom of the structure significantly reduced the SNR of the A-scan profile in the case of low pulse energy drilling; however, in-process monitoring of the depth of the processed structure was effectively performed. From the findings, it is possible to monitor not only the structure depth but also its shape and the distribution of deposited debris in the processed groove. The proposed method is useful for precise laser processing with feedback control of the laser parameters based on monitoring the structure during processing. Webster et al. [141] imaged percussion drilling of industrial materials in-situ using SD-OCT. It has a fast integration time (8 μs), a high sensitivity (>90 dB), a high resolution (14 μm), a long field of view (>1 mm) and can image features with a high aspect ratio. The designed system is capable of guiding blind hole percussion drilling and monitoring melt pool dynamics in holes hundreds of microns deep. Ji et al. [152] merged inline coherent imaging (ICI) into a scanning laser processing system and discussed its applications and constraints. The link between ablation rate, average power, and scanning speed was investigated by computer processing of recorded depth data without sample post-processing. They demonstrated micron-scale (7–8 μm axial resolution), real-time (up to 230 kHz), depth monitoring of fast scanning laser machining of various types of materials (steel and silicon). Morphological variations identified for steel trench ablation provide direct experimental comparisons with theoretical and numerical models to assist in gaining a deeper comprehension of the underlying mechanisms.

High speed, high resolution imaging of laser-powder interactions provides a better physics-based understanding of the LPBF process by illustrating the complex dynamics of spatter and the vapor plume, but does not capture surface structure at the component scale [153,154]. Although optical metrology tools such as structured light microscopy have been developed to identify defects during processing or between layers, they still present significant challenges to integration in a full-scale LPBF machine because they require ex-situ platforms, are time consuming, or have a limited measurement area [155]. SD-OCT has been recommended as a monitoring tool for AM due to its long working distance, high sampling rate, and axial and lateral resolution on the micrometre scale [147]. Since OCT can provide absolute height measurements, it is possible to quantify the absolute height of the powder layer after recoating as well as the consolidation height, which is not constant layer-to-layer and different from the layer thickness but follows a geometric progress towards steady state [156].

Neef et al. [146] demonstrated an adaptive control strategy for the LPBF process. The control system for the LPBF process uses a scanning sensor based on low coherence interferometry (LCI). This coaxially-attached sensor is capable of detecting surfaces and, at its current stage of development, promises to be able to detect faults and manufacturing process irregularities that could lead to defects in long term. The LCI-based sensor can provide layer-based information regarding the quality of the final part. This would be accomplished without removing the specimen from the equipment. Moreover, parameter window could be found rapidly if the iterative cycle of processing and qualifying the results occurs within the machine. The first experimental results indicate that powder and LPBF structures can be differentiated using the sensor acquired topographies. Large area SD-OCT surface profiling was introduced by DePond et al. [147] as an in-situ process monitoring between each layer of metal LPBF structures. The amount of spatter retained between layers was displayed and the effect of power density on the geometrical tolerance of unsupported overhang structures was evaluated based on these observations. Spatter particles, which serve as defect nucleation sites during welding or powder spreading, were detected and analysed by SD-OCT. Kanko et al. [138] demonstrated the application of SD-OCT, referred as ICI, on metal L-PBF process and evaluated measurement reliability on a static interface. Single track heights were measured before, during, and after scanning to correlate roughness as a function of scanning parameters on stainless steel. ICI measurements of solidified track morphology immediately following laser processing allow the influence of various process parameters on track quality to be evaluated in-situ, enabling more efficient parameter space optimization for new materials and geometries.

Webster et al. [12] demonstrated that by combining high-speed CMOS technology with ICI imaging, ICI can be utilized to solve a significant industrial issue, namely the direct measurement and control of laser weld depth, as well as to provide novel laser ablation control for automatic 3D subtractive processing. In this work, researchers proved that ICI is an accurate tool for high-speed, real-time assessment of weld depth, even for kW-class keyhole welding. The effects of the ablation plume, which could adversely affect the depth measurement, were insignificant due to the short lifetime of the plume in comparison to the ICI integration time. Blecher et al. [148] established the applicability of ICI for in-situ measurements of keyhole depth in stainless steel, Ti–6Al–4V, and aluminium alloys. Real-time keyhole depth measurements from autogenous bead-on-plate welding of five alloys were compared to keyhole depths derived from metallographic transverse cross-sections. The good agreement between the two sets of data indicates that the technique can be applied to a wide variety of alloys while maintaining the capacity to monitor the keyhole depth in real time. Boley et al. [149] demonstrated that the optically measured length of the vapor capillary correlates reliably to the weld seam depth when employing a two-stage approach to evaluate the OCT signal, including a first step to remove the noise and a second step to acquire the weld seam depth. Hallam et al. [157] presented a range-resolve interferometry (RRI) method and its incorporation into the WAAM system. Subsequently, the findings from a sequence of arc torch layers used to construct a wall along a single axis with the RRI instrument were compared to independent reference measurements. This study reports the first time the integration of galvanometer with OCT to achieve lateral scanning of large areas. The large depth of field enables layer height measurements to be directly referenced to the build plate, thereby considerably lowering measurement uncertainty by keeping the measurement independent of the motion system and build plate bending. This technology is well-positioned to perform in-process control based on the surface features the arc torch is approaching, and it can also provide quality assurance for the preceding layer of the wall.

Calibration is a critical step in ensuring the accuracy and reliability of OCT systems for in-situ monitoring of laser-based micromachining. Geometric calibration aligns the spatial dimensions of OCT images with physical coordinates, often using structured phantoms or precision stage movements to validate lateral and axial accuracy [140]. Spectral calibration addresses wavelength non-linearities, employing techniques such as Fabry-Pérot interferometers for reference. Depth calibration ensures accurate z-axis measurements by using step-height standards and refractive index corrections to adjust for material variations. Intensity calibration normalizes signal amplitudes through reflectivity measurements of known materials, is achieved via numerical compensation algorithms or dispersion-matching glass in the reference arm [145]. These methods collectively enhance the precision and robustness of OCT-based monitoring, ensuring reliable performance in dynamic micromachining environments. Filtering reconstructed OCT images is crucial for enhancing image quality by reducing noise while preserving structural details. Speckle noise can be mitigated using median filters, Gaussian smoothing, or advanced techniques like non-local means filtering, which averages pixel intensities based on similarity. Bilateral filtering and anisotropic diffusion smooth homogeneous regions while maintaining sharp boundaries. Frequency-domain approaches, including high-pass and low-pass filters, emphasize edges or suppress noise, respectively. Wavelet-based denoising decomposes images into multiple scales, selectively removing noise in fine components while preserving features. These techniques enhance OCT image clarity and reliability for applications like in-situ micromachining monitoring [154].

Several use cases of OCT in laser-based manufacturing were reviewed in this section. Following an introduction to the OCT technology and some considerations regarding its usage in non-destructive testing, a brief literature study of OCT cases in the field of in-situ monitoring was provided. In contrast, reports on the application of OCT in industrial environments and for in-situ monitoring of industrial processes are currently limited. OCT monitors surface and subsurface changes, such as ablation depth and melting-induced surface roughness, making it suitable for USP laser processes despite challenges posed by strong scattering from plasma and debris [140]. Nevertheless, the literature review, as discussed in this chapter, highlights the need for faster and more robust OCT devices in the future. The current OCT monitoring systems in the market is listed in Table 10.

Table 10. Non-exhaustive list of existing OCT sensors in the market.

Central wavelength	Supplier	Axial resolution	A-scan rate
1060 nm	Santec	<9 μm	>100 kHz
840 nm	Laser depth dynamics	10 μm	300 kHz
900 or 1060 nm	Precitec	4 to 12,600 μm	70 kHz
880, 900, or 930 nm	Thorlabs	3 to 6 μm	Up to 248 kHz

The measurement technique based on low-coherence interferometry significantly enhances in-situ monitoring in laser micromanufacturing applications. Due to this enhanced monitoring capability, laser micromachining can be precisely regulated and modified in real-time during the machining process. In addition, the technology facilitates an automated modification of the laser process settings, a precise alignment of the to-be-machined workpiece, and an immediate quality control following the process. The high accuracy of OCT makes it a useful tool for quality assurance in industrial applications, and it can be further enhanced to achieve depth control for laser micromachining. It is shown that OCT has the capability to detect joints and defects that are inaccessible to in-direct process monitoring techniques. These characteristics are beneficial not just due to the decreased overall process time, but also they allow for greater process stability and product quality with reduced part rejections. OCT can improve the entire production process by enabling direct and independent measurements in real-time that are unaffected by ambient conditions and laser processing. OCT provides a solution for high-performance, high-throughput laser-based production at low cost per unit by introducing new options for modern online laser guidance with simultaneous process monitoring.

Confocal microscopy

Due to the high number of process and material related factors such as laser defocusing, generation of laser-induced plasmas, and various material specifications, the ablation rate during laser micromachining is extremely nonlinear with respect to depth and machining time. Despite the availability of several reliable and repeatable methods for in-situ laser ablation depth measurements, limitations in the measurement methods still persist. Laser machining profiles, i.e. the ablation depth under specified laser settings, are usually determined through trial-and-error experimentation in the current industrial practice. Nevertheless, during laser micromachining, ablation conditions may be influenced by unexpected factors, such as laser power fluctuation, misalignment of the focusing system, optical misalignments caused by mechanical vibrations, and human impacts. Online, high-sampling-rate examinations are vital for the quality assurance process, ensuring a thorough assessment of components. However, their high throughput demands can result in increased costs [158].

Confocal microscopy employs point illumination and a pinhole in an optically conjugate plane in front of the detector to eliminate unfocused picture data. Only light emerging from the focal plane is detected, resulting in much higher image quality in comparison to wide-field fluorescence images. The thickness of the focal plane is inversely proportional to the square of the numerical aperture of the objective lens employed, as well as dependent on the optical properties of the specimen and the ambient refraction index. This confocal microscopy model extends to depth detection of laser ablated structure. Only the signal reflected from the focal plane of the objective in a specimen is received by a confocal microscope (Fig. 18). By scanning the pinhole, the position of maximum signal strength corresponding to the depth of a hole bottom surface during laser machining can be determined [159]. These optical systems have a lateral resolution of around 0.5 μm and a vertical resolution in the sub-nanometre range [160]. Due to the complexity of alignment methods, however, the majority of optical profilers come in relatively compact size, making them impracticable for in-situ measurements [161]. During in-line confocal microscopy measurement, most reflected light passes through the pinhole only when the target point is on the focal plane. Within the depth of field (DOF) range of a confocal system, the reflected light intensity detected forms a depth response curve (DRC). The photodiode detection of the DRC peak reveals the target point focus plane on the measured surface [162]. With the high-resolution encoder of the confocal system, the surface height of the target point can be measured. Therefore, by recording the height, a surface profile and roughness can be determined. In addition to the pinhole scanning approach, certain confocal microscopes employ the slit scanning technique. The slit scanning method offers the benefit of enhancing signal intensity and speed, but the lateral cross-talk greatly degrades the vertical resolution [163].

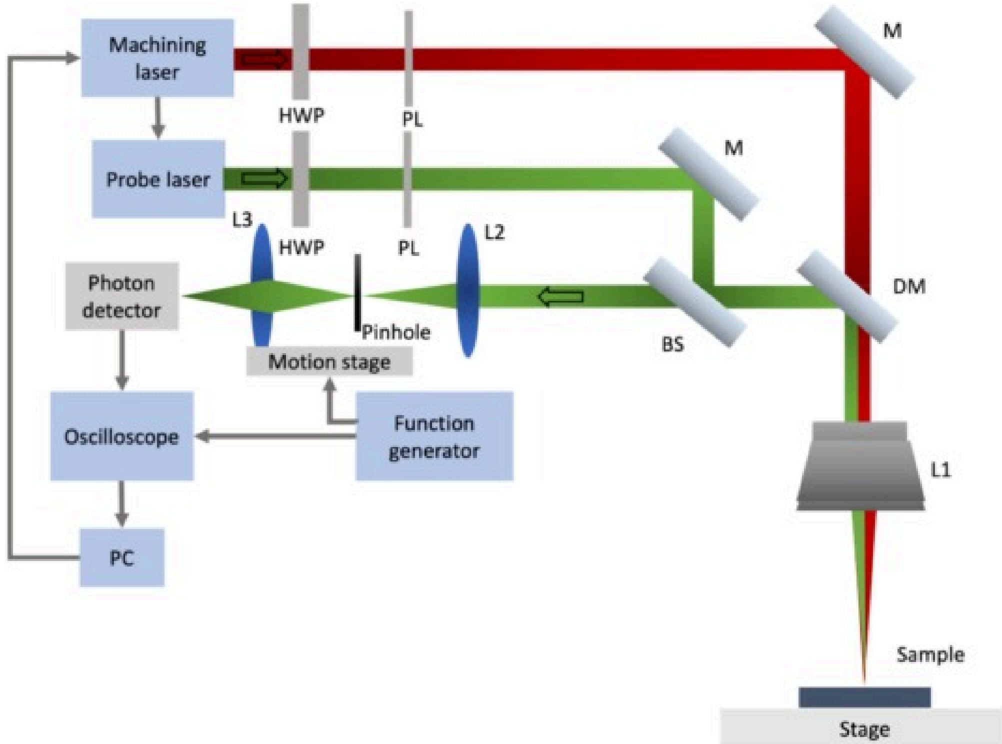


Fig. 18. Schematic of the optical layout of confocal microscopy for real-time depth monitoring. HWP: Half-Wave Plate; PL: Polarizer; DM: Dichroic Mirror; BS: Beam Splitter; M: Mirror; and PC: Personal Computer – reproduced from [171].

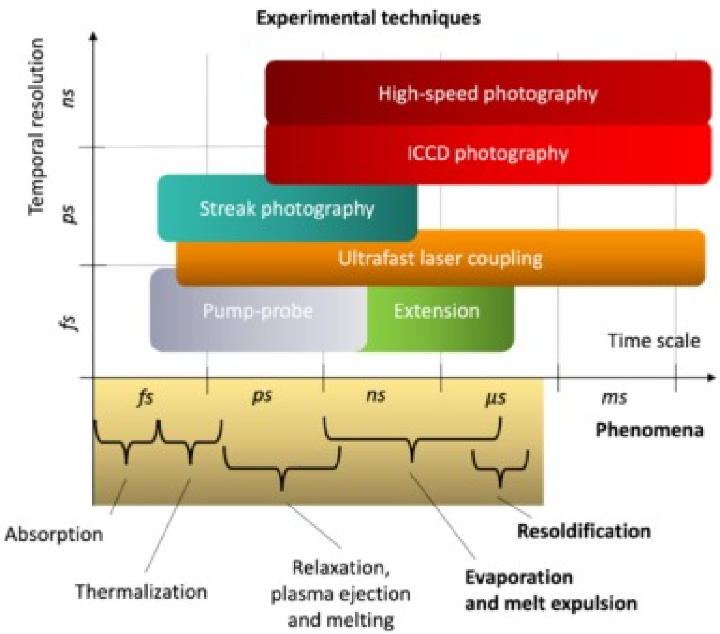
Chen et al. [158] proposed a confocal microscopy technique for in-line monitoring of laser ablation depth as it applies to blind-hole laser drilling for single and multilayer bulk heterogeneous structures as well as for various materials, an alloy and a silicon wafer. Reflection of the probe beams from the bottom surfaces of the drilled holes is the basis for the detection method. Using a confocal optical structure, the method achieves a precision of 1 μm and an error of 5% for 100 μm-deep holes. This level of precision and accuracy is adequate for real-time monitoring of machining depth during laser micro-hole drilling, and the system is robust enough for usage in industrial and research settings. In addition, this approach is independent of the wavelengths of the lasers used for machining and is compatible with all commercial laser machining systems. On the sample surfaces, a low-divergence-angle beam could achieve detection depths of >100 μm. Using laser confocal technology, Fu et al. [161] created a non-contact and in-situ surface topography measurement device. The purpose of their study was to investigate roughness measurement technology that can be employed in the manufacturing environment, as opposed to the conventional laboratory desktop measurement. A high precision 2-axis linear stage and robot arm have been coupled with a laser confocal sensor. Experimentation demonstrates that the suggested system can measure surfaces with Ra values ranging from 0.2 to 7 μm.

Calibration ensures the accuracy of confocal microscopy measurements by aligning system outputs with physical dimensions and intensities of the sample. Lateral resolution is calibrated using certified grids or micrometres to determine pixel-to-length conversions, while vertical resolution employs step height standards to validate and adjust scaling. Intensity or reflectance calibration involves reference materials with known optical properties to align intensity profiles. Focus calibration refines Z-axis measurements by using flat surfaces or standards minimizing depth errors and ensuring precise characterization [159].

In conclusion, confocal microscopy offers precise offline analysis of post-ablation surface features, capturing ablation crater morphology and heat-induced surface changes, though its application to in-situ monitoring remains limited by slow scanning speeds [162]. Although there is limited research conducted on the implementation of confocal microscopy in the laser manufacturing machines, confocal microscopy has proven to be an effective tool for in-situ monitoring of various manufacturing processes. In-line monitoring with confocal microscopy is particularly useful for high-precision manufacturing, such as in the medical device and electronics industries, where even minor deviations from specifications can result in significant product failures. However, its implementation into the laser manufacturing chamber is complicated, comes with high cost and requires specialized equipment. Its ability to provide high-resolution, 3D images in real-time has revolutionized the way we understand processes in detail. With continued advancements in both laser technology and confocal microscopy, the potential for even further improvements in the quality and efficiency of laser-based manufacturing is immense.

Camera-based ablation monitoring

Observing and comprehending ultrafast dynamic processes, such as fs laser ablation, is essential for precise machining and manufacturing. As a result, ultrafast optical imaging, which can achieve blur-free visualization of transient dynamics (Fig. 19), has caught the interest of researchers in physics, chemistry, optical engineering, industrial manufacturing, and materials science. Monitoring the cumulative ablation rate, as well as the processed depth or volume, is crucial for USP micromachining of holes or grooves [164]. Interactions between fs laser and material have been extensively studied in the past. However, a clear understanding of morphological responses is still lacking; for example, direct visualization of the complete ultrafast structural dynamics in metals has never been achieved. Interestingly, fs laser-induced morphological changes quite often rapidly alter the physical properties of a material [165]. Hence, one of the primary objectives of visual-based monitoring is to reveal some of the fundamental characteristics of laser-matter interaction such as plasma generation, evaporation, melt expulsion, and their dynamics. Due to the varied time scales of these phenomena, research has focused on the development and deployment of novel experimental techniques as well as the systematic analysis of the observed dynamics [166].

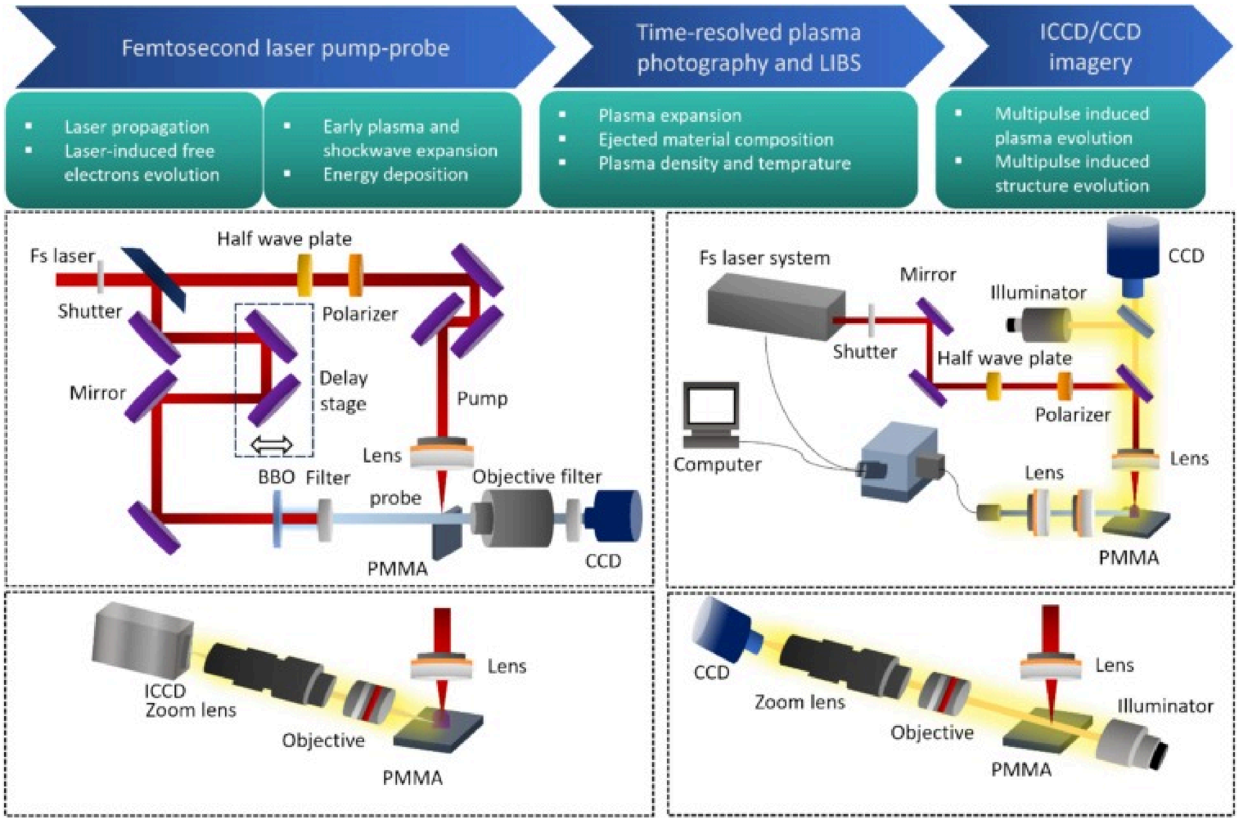


Download: Download high-res image (147KB)

Download: Download full-size image

Fig. 19. Map of phenomena during ablation of materials with ultrafast laser radiation and optical-imaging based measurement techniques for its detection – reproduced from [166].

High-speed cameras are the common tools used for the monitoring of industrial production processes. In order to see the object at the micrometre-to-nanometre scale, a camera must not only have high spatial resolution, but also high temporal resolution. The need for high temporal resolution arises from the principle that even a slow event appears to occur "fast" on a small length scale. This necessity for optical imaging of transient events is widespread in various fields, including photochemistry, plasma physics, microfluidic biotechnology, and semiconductor physics [167]. These transient events, which occur in two-dimensional (2D) space and at fs to ns time scales, reveal many fundamentally significant mechanisms [168]. To verify the theories, multiscale observation systems (Fig. 20) based on a fs laser pump-probe combined with continuous imaging technology, time-resolved plasma imaging technology, and CCD/ICCD imaging technology have been developed to visually monitor dynamic changes in materials throughout the process at various temporal and spatial scales [16].



Download: Download high-res image (494KB)

Download: Download full-size image

Fig. 20. Schematic of multiscale model and observation system – reproduced from [171].

The sampling interval and temporal resolution of time-resolved imaging based on pump-probe techniques are on the order of fs, which makes it ideal for capturing reproducible transient dynamics with a high sampling frequency [169]. However, this approach is ineffective for events with a low repetition rate or that are nonrepeatable. Numerous methods for single-shot optical imaging have replaced the pump–probe techniques [168].

Traditional high-speed cameras are constrained by the response time of electronic components. Some film-based cameras have achieved a frame rate of 25 million frames per second (Mfps). Nevertheless, numerous ultrafast phenomena are either complex or difficult to reproduce. In these instances, pump-probe methods would result in significant inaccuracy and low time efficiency. In recent years, numerous single-shot ultrafast optical imaging techniques have been developed to overcome the limitations of pump-probe methods. In this context, “single-shot” refers to capturing the entirety of a dynamic process in real time (i.e., the actual duration of an event) without repeating the event. Extending the existing classification, researchers define “ultrafast” as imaging speeds of 100 million frames per second (Mfps) or higher, corresponding to interframe time intervals of 10 ns or less. “Optical” refers to the detection of photons in the spectral range of extreme ultraviolet to far infrared [170].

To image a slowly moving object ($v = 1\,m/s$) with a spatial resolution of $1\,\mu m$, a high temporal resolution of $(1\,\mu m)/(1\,m/s) = 1\,\mu s$ is required, which corresponds to a high frame rate of 1 Mfps. Evidently, conventional cameras based on electronic image sensors, such as the charge-coupled device (CCD) and complementary metal-oxide-semiconductor (CMOS), are incapable of providing such high frame rates under desirable imaging conditions, as their image acquisition speed is severely constrained by their electrical operation and limited storage [167]. Several novel and unconventional approaches to high-speed optical imaging have been reported in recent years to circumvent these technical obstacles. Serial time-encoded amplified imaging/microscopy (STEAM) is a continuous imaging method with an integrated optical image amplifier that enables image acquisition at a high frame rate of 100 Mfps without compromising sensitivity [172]. Sequentially timed all-optical mapping photography (STAMP) is an ultrafast burst imaging method that can achieve motion-picture femto-photography at a frame rate of 1 trillion frames per second (Tfps) by overcoming the electrical limitations of conventional image sensors with its all-optical operation [173]. Single-shot multispectral tomography (SMT) is a high-speed tomographic imaging technique based on spectral multiplexing to illuminate the imaging target from multiple angles of incidence simultaneously [174].

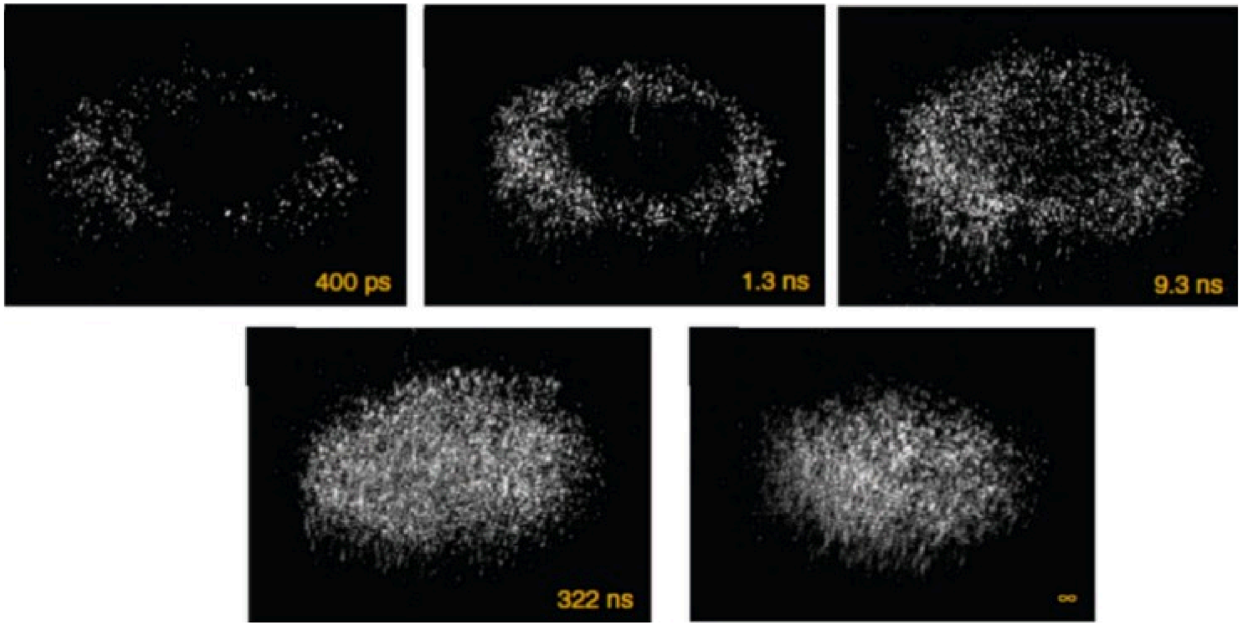
Fluorescence imaging by radiofrequency-tagged emission (FIRE) is a high-speed fluorescence imaging technique that reaches 100 times higher frame rates than standard techniques by spatially frequency-modulating the excitation light and demodulating the fluorescence signal after detecting it with a sensitive single-pixel photodetector [175]. Compressed ultrafast photography (CUP) is an ultrafast receive-only streak imaging technique that incorporates a compressive-sensing structure into the data capture of a streak camera and, as a result, achieves two-dimensional receive-only image acquisition at 100 Gfps [176] and a temporal resolution of 50 ps [177].

Although these multiple shot imaging techniques have achieved fs temporal resolution and passive detection, they are dependent on the precise repetition of the ultrafast event of interest during temporal or spatial scanning. Consequently, these imaging techniques are inapplicable when temporal focusing must be captured in a single measurement [178]. Due to the non-repeatable nature of such transient phenomena (e.g., micromachining using temporal focusing in glass), visualizing temporal focusing in real-time (i.e., at its actual time of occurrence) is required for their investigation and further application. In addition, a fs-level exposure time is required to clearly resolve the evolving instantaneous spatiotemporal details since the width and intensity of the laser pulse undergo dramatic changes during temporal focusing. In addition, the nm-to-μm spatial scales of these transient events necessitate ultrafast imaging for blur-free observation; for example, for imaging a light-speed event, an imaging speed of 1 Tfps is required for a spatial resolution of 300 μm [167]. As these events are frequently self-luminous, a passive detector is highly desirable for direct recording [179]. A variety of single-shot ultrafast imaging techniques have been developed recently. Active-illumination-based techniques have achieved frame rates of one trillion frames per second [180]. However, these methods are incapable of imaging luminescent transient events; so direct imaging of evolving light patterns in temporal focusing is not possible.

To enable real-time, ultrafast, passive imaging of temporal focusing, Liang et al. [181] developed single-shot trillion-frame-per-second compressed ultrafast photography (T-CUP), which can image non-repeatable transient events at up to 10 Tfps in a receive-only manner. At frame rates of up to 100 billion fps, the resulting CUP system can passively receive photons scattered or emitted from dynamic scenes. One-shot real-time imaging of temporal focusing is anticipated to contribute immediately to the investigation of nonlinear light–matter interactions. By improving the frame rate by two orders of magnitude compared with the previous state-of-the-art, T-CUP demonstrated that the ever-lasting pursuit of a higher frame rate is far from ending. T-CUP is the only detection solution currently available for passively probing dynamic self-luminescent events at fs timescales in real-time, revealing spatiotemporal details of transient scattering events that have been previously inaccessible using previous systems [168]. The unprecedented capability of T-CUP for real-time, wide-field, fs-level imaging from the visible to the near infrared will pave the way for future microscopic studies of time-dependent optical and electronic properties of novel materials under transient out of equilibrium conditions.

Using single-shot measurements based on fs time-resolved optical polarography (FTOP) and an echelon, Wang et al. [179] investigated the propagation dynamics of fs laser pulses in fused silica. The FTOP technique can be used to directly observe the instantaneous intensity distributions of optical pulses propagating with a two-dimensional spatial distribution. The single-shot measurements have been used to visualize the effect of pulse-energy fluctuation on the spatial and temporal distribution of a single laser pulse. From the single-shot images, it is possible to directly observe the shot-to-shot variations of the pump pulse propagation profiles resulting from the pulse energy fluctuation. Zeng et al. [177] developed a new single-shot ultrafast optical imaging technique, known as framing imaging by noncollinear optical parametric amplification (NCOPA) or framing imaging based on noncollinear optical parametric amplification (FINCOPA). FINCOPA is a candidate for single-shot imaging with high spatial and temporal resolutions due to its high frame rate and large frame count. Using a laser-induced air plasma grating as a target, FINCOPA has experimentally captured images with a frame rate of 10 Tfps and a resolution of 50fs. This work has provided the highest experimental record of the effective frame rate with the temporal resolution of 50fs. FINCOPA is a powerful single-shot ultrafast optical imaging technique for accurate temporal and spatial observations of single transient events occurring in the fs region, including atomic or molecular dynamics in photonic materials, plasma physics, and USP laser ablation for precise micromachining.

Utilizing scattered light rather than specular reflections, Fang et al. [182] developed an ultrafast optical imaging technique for examining the structural dynamics of metals. The scattered-light imaging provides a near-zero background and high contrast, enables to obtain high-quality real-space images of the structural dynamics of metals and resolve the entire temporal and spatial evolution of fs laser-induced surface structure formation from beginning to end in metals. Transient surface structures first appear in the central portion of the ablated spot at lower laser fluences that favour nanostructure formation. At higher laser fluences, which favour microstructure formation, nanostructures emerge first at the periphery, and microstructures eventually populate the central region of the laser ablated spot. For example, the predicted cooling time for zinc is two orders of magnitude shorter than the time found in the observations (Fig. 21), and researchers speculate that this slower cooling process is due to a previously observed enhanced thermal coupling phenomenon. The visualization and control of surface structural dynamics are not only critical for understanding laser-induced material responses, but they also pave the way for the design of new material functionalities via surface structuring.

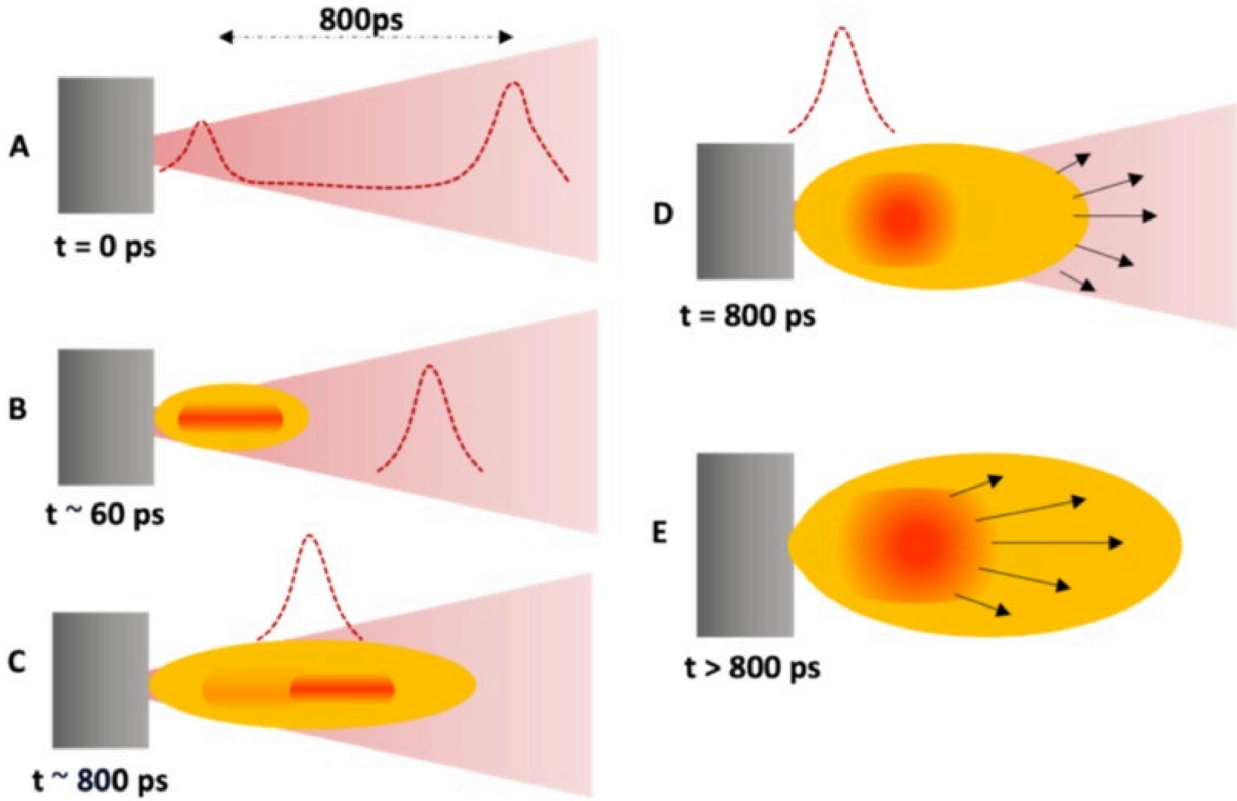


Download: Download high-res image (138KB)

Download: Download full-size image

Fig. 21. A comparison of CCD images of transient Zn surface structures observed at various delay times and final solidified structures following a pump pulse at a fluence of $1.0 J cm^{-2}$ [182].

Machining a sample with a fs laser generates plasma. Plasma luminescence produces a plasma spot image. Laser-produced plasmas (LPPs) are useful in a wide range of applications, including extreme ultraviolet generation. The nature of LPP expansion is primarily determined by the density and temperature of the plume, which are influenced by laser parameters such as wavelength, pulse duration, energy, and spot size, as well as the nature and pressure of the ambient gas and material properties [54]. To comprehend the effect of laser–plasma, laser–target, and plasma–plasma interactions, Smijesh et al. [183] compared the structural modifications of the plasma plume using ultrafast pump-probe imaging technique in the double pulse (DP) scheme to its single pulse (SP) counterpart (Fig. 22). Wang et al. [165] used the two-dimensional maximum class square error method to segment the spot image to extract characteristics such as pixel area, perimeter, and brightness of the image, and then derive the relationship between the characteristic parameters, and the machining parameters, such as laser power, processing speed, and processing depth (the perimeter of the laser spot image increases with increasing laser power; the brightness distribution of the laser spot image increases with increasing laser power).

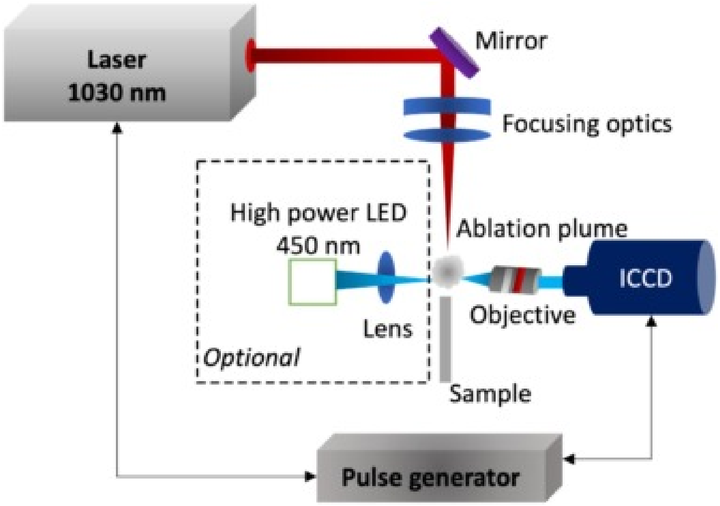


Download: Download high-res image (205KB)

Download: Download full-size image

Fig. 22. (a) Schematic for generation of plasma in double pulse geometry. Here (b) the first pulse produces a plasma and the second pulse, delayed by 800ps, then interacts with the plasma produced by the first laser pulse via mechanisms such as (c) laser–plasma, (d) laser–target, and (e) plasma–plasma interactions – reproduced from [183].

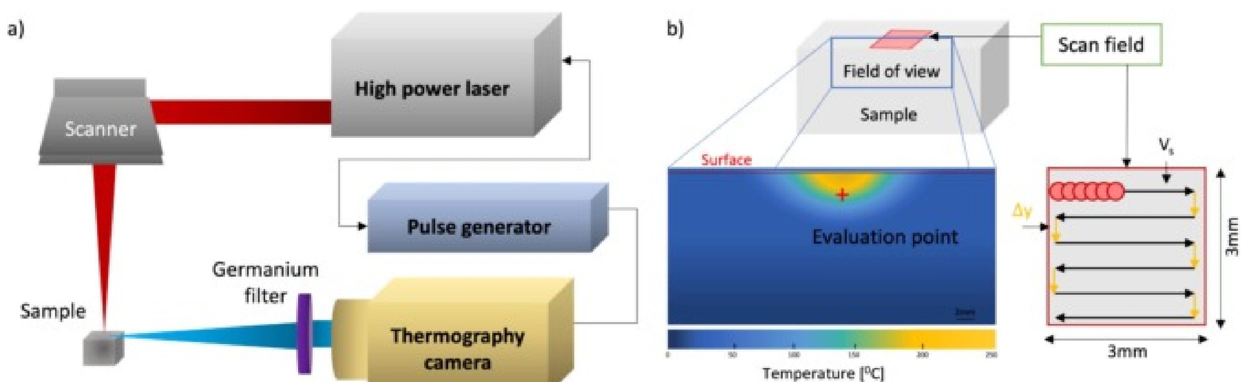
To gain a better understanding of the ablation and shielding effects during processing with USP bursts, in-situ imaging of the process luminescence is performed by Bornschlegel et al. [184]. Shielding effects prevent the laser pulses from reaching the surface with their full pulse energy, thereby decreasing the ablation rate. This effect is the result of the interaction between the incident laser beam and the ablation plumes of previous pulses, in which the radiation is reflected, scattered, or absorbed by particles, vapor, or plasma originating from the ablated matter. The ICCD sensor with an internal upstream multichannel plate is capable of exposure times as short as 200 ps but it was set to 1 ns in the experimental studies presented (Fig. 23). It is shown that there is no correlation between the shielding effects and the ablated volume per burst pulse and the intensity of the process luminescence. The surface of the workpiece can reach the melting point of the material locally as a result of the thermal accumulation of residual energy from laser pulses. In another work, Bornschlegel et al. [185] verified the previously reported results by investigating the heating effect in a time-resolved manner. To do so, the evolution of the workpiece temperature is measured with a thermographic system during the USP ablation process. With this setup, it is possible to determine the residual heat as a percentage based on a simulation and to identify changes in the heat input during the process time (Fig. 24). As a result of these findings, it is possible to understand processes and estimate the heat input using a constant heat source in an operating region near the optimal fluence.



Download: Download high-res image (96KB)

Download: Download full-size image

Fig. 23. Setup for in-situ investigation of the ablation process with optical background illumination for shadow-graphical images – reproduced from [184].

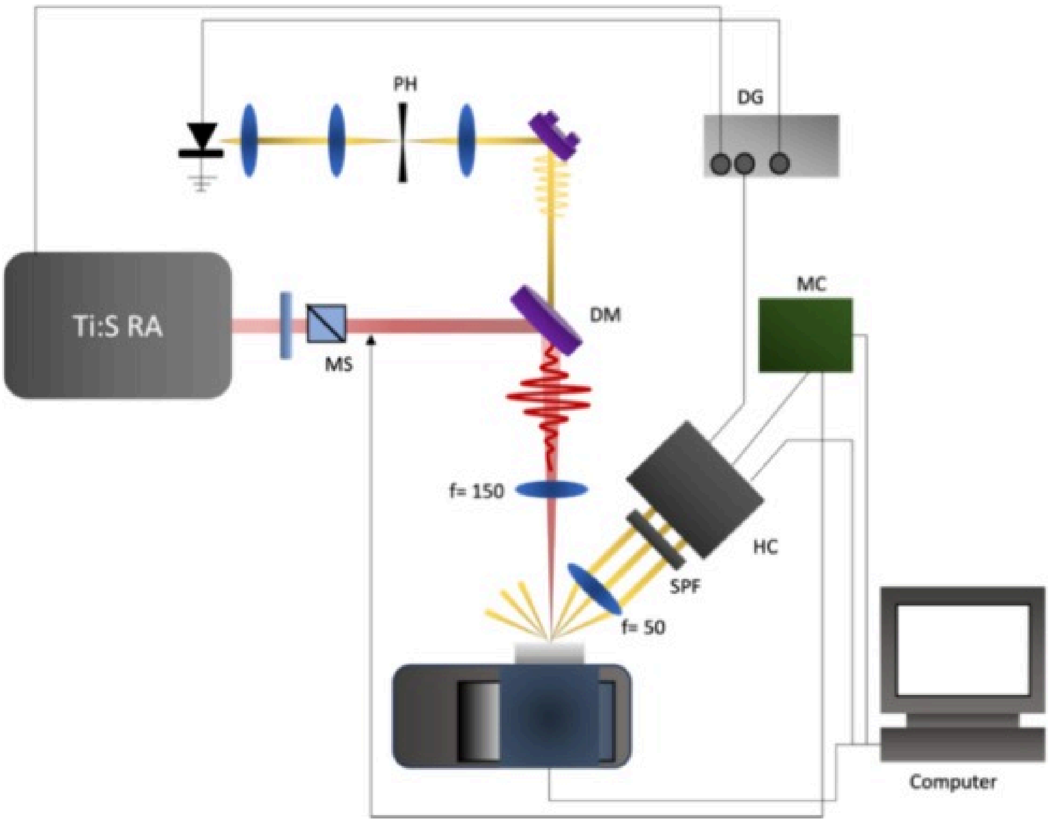


Download: Download high-res image (142KB)

Download: Download full-size image

Fig. 24. a) Setup for in-situ investigation of the heat accumulation of ablation processes, b) schematic of evaluation point for the investigation over time and the scan strategy with illustrated scan direction on the sample surface – reproduced from [185].

Tani et al. [164] demonstrated the extraction of multiple data from three consecutive frames of measured in-situ speckle patterns during laser processing. When spatially coherent light is scattered by a rough surface and projected onto a screen, the scattered light on the screen displays a laser speckle pattern, which is characterized by a highly irregular intensity profile. Fig. 25 depicts the concept of the method. Using an in-situ neural network analysis of speckle patterns, these researchers created a method for determining the progress of laser processing. The neural network can accurately predict the ablation depth of a groove to within 2 μm. Unlike conventional interferometric monitoring techniques, this method is simple to implement. In addition, the prediction time of <0.1 milliseconds aligns well for real-time feedback for complex processing procedures.



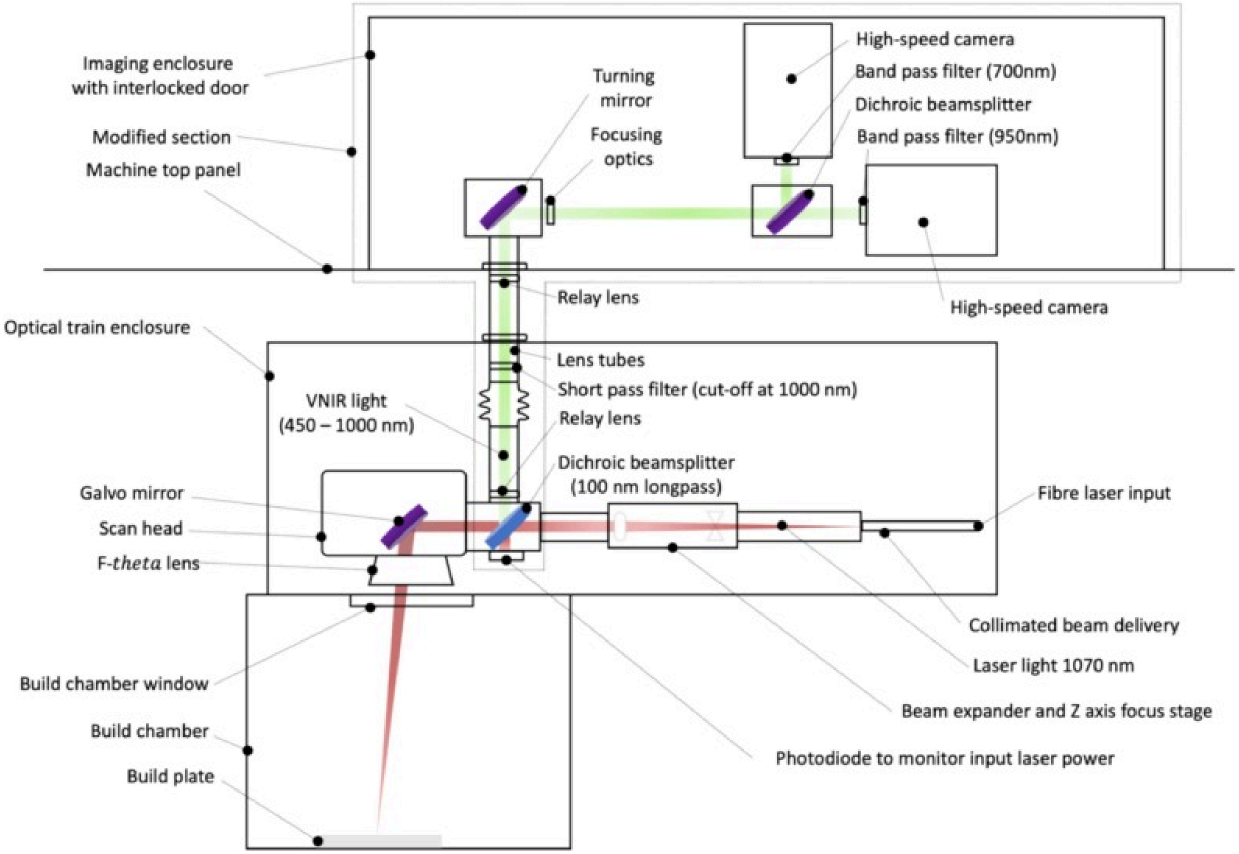
Download: Download high-res image (137KB)

Download: Download full-size image

Fig. 25. Experimental setup of in-situ speckle pattern imaging. The abbreviations are as follows: PH: pinhole, DG: delay generator, MS: mechanical shutter, DM: dichroic mirror, MC: micro-computer, HC: high speed camera, and SPF: short pass filter - reproduced from [164].

During AM, a wide range of material discontinuities are known to occur, the most typical of which are voids located in the bulk of the fused material, either between layers (elongated pores) or within the layer (gas pores) [186,187]. Instead of relying on a best-guess approach, a deeper understanding of process principles provides additional information to accelerate development [188]. The scan head and a semi-transparent mirror is designed to convey electromagnetic radiation released by the melt pool to a high-speed camera. The camera measured the melt pool dimensions. In this instance, only the active LPBF build region is monitored, enabling 10μm per pixel resolution. The photodiode sensor alone can identify temperature gradients over the build region, but for effective management, a camera with high temporal and local resolution of the melt pool is necessary. Closed-loop feedback could be incorporated to stabilize the melt pool and maintain temperatures within a predetermined range, which reduces the formation of over-melted zones and resultant gas holes [189].

Mercelis et al. [190] demonstrated LPBF monitoring and feedback control possibilities by utilizing a coaxial optical system. This pioneering work showcased the significant advantage of obtaining spatially resolved information from the high-speed camera, alongside a basic analog output indicating the overall melt pool area. By simultaneously activating both camera and photodiode information sources, it is possible to use the camera images to understand the photodiode signal. The images of the melt pool help to interpret the influence of changes in process parameters and powder composition on the melt pool behaviour. A feedback mechanism can mitigate the effects of changing boundary conditions. Hooper et al. [188] described the development of a new coaxial imaging setup with two high-speed cameras to give both good spatial and temporal resolution of the melt pool (Fig. 26). The two-wavelength imaging setup developed resolves temperature and accounts for unknown emissivity values during the material transition from powder to liquid to a bulk solid. Temperatures in the approximate range of 1000–4000K were captured at 100kHz with a resolution of 20μm. The technique was validated against measurements from a type C thermocouple. Thermal gradients were measured and found to vary with parameter used, component geometry and location in the scan path.



Download: Download high-res image (338KB)

Download: Download full-size image

Fig. 26. Schematical representation of optical layout of commercial LPBF machine and modification to allow coaxial imaging of the melt – reproduced from [188].

Calibration involves establishing a precise relationship between the camera pixel coordinates and the physical dimensions of the observed scene, which is essential for quantitative analysis of phenomena like ablation depth, plasma plume expansion, and melt pool dynamics. Geometric calibration techniques, such as pinhole camera models or nonlinear lens distortion corrections, are implemented to account for optical aberrations and ensure accurate spatial resolution [167]. Radiometric calibration is also critical, especially when high dynamic range (HDR) imaging is used to capture transient luminescence or thermal gradients. This involves correcting the camera response curve to ensure linearity between the recorded pixel intensities and the actual light intensities [168]. Furthermore, temporal calibration of high-speed cameras is essential for synchronizing frame acquisition with ultrafast events. Techniques such as temporal gating and external trigger synchronization are used to achieve ns or fs precision [170].

To ensure the effective analysis of high-speed imaging data, advanced filtering techniques are employed to mitigate noise and enhance the relevant features of transient phenomena. Digital image processing techniques, such as Gaussian smoothing, wavelet transforms, and adaptive thresholding, are frequently used to enhance SNR and extract meaningful patterns from the images [172]. Additionally, spatial and temporal filtering, including Fourier-based high-pass and low-pass filters, is applied to remove irrelevant background signals and isolate key transient features like plasma dynamics or melt pool variations. Recent advancements in machine learning have also enabled the development of data-driven filtering algorithms that adaptively refine images based on learned patterns, providing better accuracy in identifying subtle morphological changes during laser-matter interaction [176].

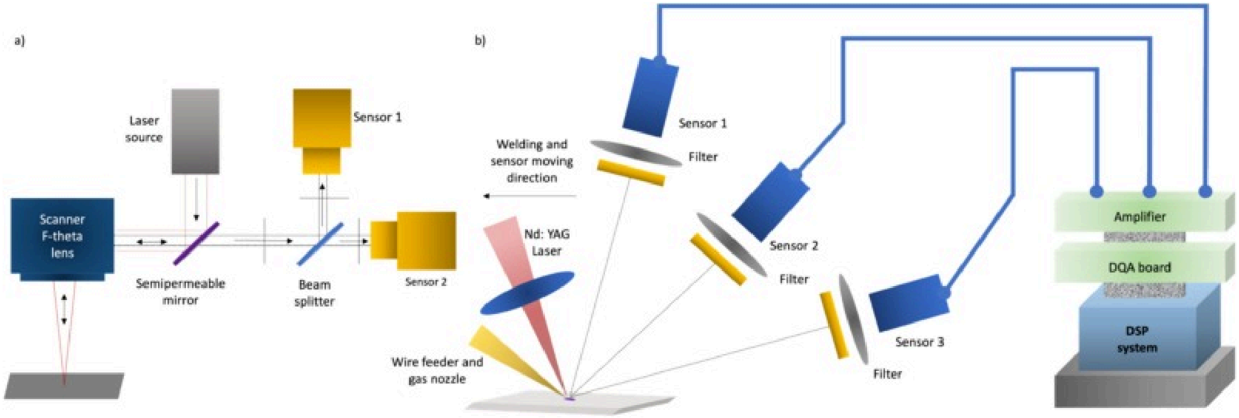
The camera-based monitoring approach described in this chapter can be used as a soft-sensing technique to develop closed-loop micromachining technology by processing the acquired laser-matter interaction images. Development of a process monitoring system is the initial step towards a closed-loop process control module for the prevention of failures. Such a system must be capable of time- and space-resolved observation of the melt pool and its dynamics in order to detect the emergence of build defects. In contrast to conventional imaging systems, high-resolution imaging at fast scanning speeds requires an external illumination source. In conclusion, the use of ultra-high-speed cameras in USP laser ablation monitoring is a powerful tool that provides real-time visualization of the ablation process. It offers a unique opportunity to understand the complex interaction between the laser and the target material and can aid in the optimization of laser ablation processes. With its ability to capture high-speed and high-resolution images, ultra-high-speed cameras have become an indispensable tool for the study of laser ablation and will continue to play an important role in advancing this field. Lastly, high-speed camera-based monitoring visualizes plasma formation, material ejection dynamics, and shock wave propagation. These systems require advanced filtering techniques and high frame rates to manage intense emissions and capture ultrafast phenomena accurately. Together, these monitoring techniques provide comprehensive insights into USP laser processes, each addressing unique aspects of ablation, plasma dynamics, and material removal behaviour [172].

Multi-sensor fusion

Modern instrumentation systems are integrated with several sensors, each of which serves a unique purpose. Each sensor in the monitoring system can independently measure a particular parameter and utilize a specialized signal-processing algorithm to merge the independent readings into a single measurement value. The term for this type of system is multi-sensor system [191]. By installing a multi-sensor system for monitoring and controlling laser-based manufacturing operations, for instance, it is possible to significantly reduce machining errors and improve surface finishes. The multi-sensor technique is increasingly being utilized by low-cost, high-performance microprocessors. In the meantime, sensing instruments make signal processing systems and digital closed-loop control very cost-effective [192]. Consequently, a multi-sensor system is essential for achieving greater machining precision.

The increasing use of sensors in industrial processes has driven a growing demand for methods to leverage the large volumes of heterogeneous data generated, necessitating the development and application of fusion techniques to integrate such data and enhance the insights obtained. There are two basic forms of fusion in the area of machine learning: early and late fusion. Early fusion is implemented in the initial stages of an application and refers to the combination of raw data or extracted features (feature fusion). Late fusion refers to the combination of the outcomes of multiple algorithms and is the final step in the analytic pipeline. Techniques that combine the estimated class probabilities of each algorithm can be used to achieve late fusion. Advanced approaches, like stacking and boosting, utilize the results of other classifiers as input to a new ensemble algorithm [192].

Regarding multi-sensor fusion for laser-based manufacturing processes, extensive research has been performed using a variety of sensors and algorithms to investigate process dynamics and product quality. Each type of sensor input is limited to one or more process characteristics. As a technique for merging information from multiple sources, sensor fusion attracted the interest of numerous researchers. Considering industrial requirements, the majority of the available in-situ monitoring technologies have adopted sensors with high speed and cheap cost, such as photoelectric sensors, acoustic sensors, and temperature sensors. As the cost of cameras decreases, visual sensing becomes more cost-effective and faster. Manufacturers are increasingly turning to automated processing technologies, which can decrease their dependence on operators during production. The realization of automated processing systems is predicated mostly on dependable and strong monitoring systems, which are employed for online and offline monitoring of essential machining processes [193]. Each sensor in the monitoring system can independently measure a particular parameter and utilize a specialized signal-processing algorithm to merge the independent readings into a single measurement value. By implementing a multi-sensor system coaxially or off-axially (Fig. 27) for monitoring and managing laser-based machining operations and machine condition, it is possible to reduce machining errors significantly and improve laser-based manufacturing.



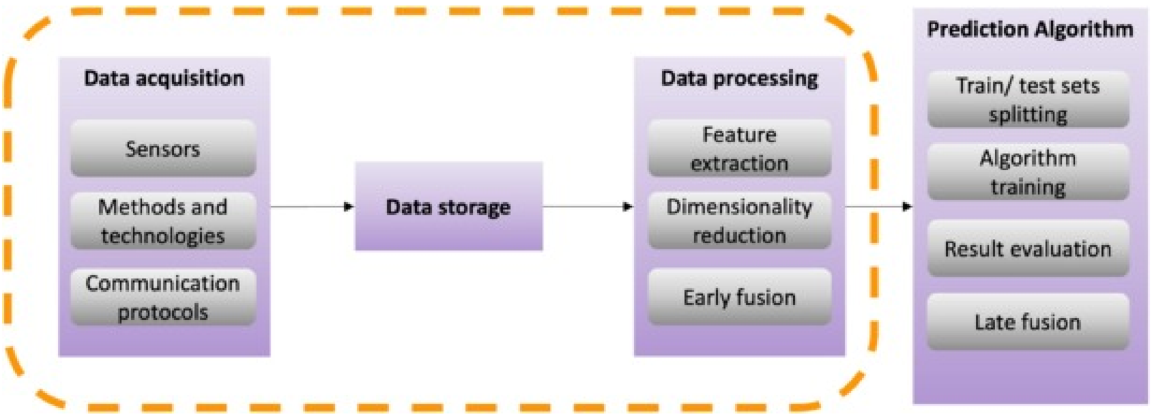
Download: Download high-res image (171KB)

Download: Download full-size image

Fig. 27. Schematic of a multi-sensor monitoring system, a) co-axial multi-sensor system, b) off-axial multi-sensor system.

The analysis of data obtained from various sensors in a typical manufacturing process includes specific methodologies, which are comparable to the steps in any analytic task involving prediction (Fig. 28). Initially, raw data must be acquired and stored from sensors. In order to examine the data, cleaning and filtering processes are typically applied. In signal processing, feature engineering is essential, and the transformation methods used, and features extracted depend on the type of sensor and the purpose of the analysis. Applying feature selection methods, which select smaller sets of variables for input to a classification or regression algorithm, reduces the enormous number of signal features produced. Fusion can be applied at many points of the aforementioned pipeline, including the first stage for merging the raw data, after feature extraction for combining the extracted features, or at the final stage for combining the prediction algorithm outputs [201]. In a multi-sensor system, the coordination of all sensors is essential to attaining system operation and the sensors transmit measurement data to the data fusion layer, which combines the information using sophisticated signal processing techniques. Meanwhile, the control application transmits adequate control signals to the actuator

[202]. Using specified standards, data fusion can provide a consistent interpretation of a tested object, hence improving the performance of the information system in comparison to subset-based systems. Compared to the independent processing of a single source, the benefits of data fusion include enhancing detectability and reliability, expanding the range of spatial-temporal perception, enhancing detection accuracy, increasing the dimension of target features, and improving the fault-tolerance of the system [203].



Download: Download high-res image (150KB)

Download: Download full-size image

Fig. 28. Flowchart of typical analytic pipeline followed in laser-based manufacturing applications.

A summary of multi-sensor fusion monitoring in laser-based manufacturing processes from the literature is provided in Table 11. Although numerous approaches to process monitoring in USP laser micromachining have been published, none of them have been able to accurately and unrestrictedly forecast the quality of the emerging structure. There is a compromise between directly collecting 3D-topography information with low sample rates, resulting in low spatial resolution, and recording process emissions, which is faster but more difficult to interpret. When paired with suitable analysis methodologies, the multi-sensor fusion strategy shows considerable promise. By monitoring many process emissions simultaneously, researchers anticipate gaining a greater quantity of relevant process information. In order to maintain continuous processing conditions and eliminate environmental disturbances, experts also monitor and analyse machine, laser, and environmental errors and drifts [66].

Table 11. List of multi-sensors monitoring setups for laser manufacturing processes.

Literature	Sensor type	Aspect of monitoring	Fusion level
Zhang et al. [194]	Vision+AE	Welding condition: 1) Burning through 2) okay 3) under penetration	Feature (SVM)
Snow et al. [195]	Vision+thermal camera+AE	Defect localization and quality assessment using AI	Data and feature (CNN)
Zamiela et al. [196]	Thermal camera+AE	Porosity detection	Feature (U-net fusion)
Zhang et al. [197]	Photodiode+AE	Welding penetration state	Feature (CNN)
You et al. [198]	Visual+photodiode	Plasma and keyhole features	Data
Deng et al. [199]	Photodiode+thermal camera	Plasma-light-temperature	Data
Li et al. [200]	Photodiode+AE	Porosity defect localization	Feature (classification)
Zuric et al. [66]	Photodiode+AE	Micromachining processing condition	Data
Conesa et Al. [19]	Visual+AE	Plasma formation and plume expansion	Data

Literature	Sensor type	Aspect of monitoring	Fusion level
Xie et al. [119]	Visual+AE	Material removal and surface characteristics	Data

You et al. [198,204] designed multiple optics sensing system of four sensors, including two photodiodes and two vision sensors, to monitor laser welding of stainless steel. Photodiode sensors detected light emission signals, including visible light and laser reflection. The images obtained by the high-speed camera depicted the primary characteristics of the laser welding process, plasma and keyhole characteristics, which highlighted fundamental physical mechanism of laser welding. By quantitatively assessing the feature extracted from various optics sensors, the suggested system provides a better understanding and more precise evaluation of the laser welding process, as demonstrated by experimental findings. A multi-sensing system provides more specific data for recognizing the welding process for online adaptive control. Zhang et al. [197] detected the welding penetration state using an IR sensor, an UV sensor, and a sound sensor. The sound sensor is used to detect the sounds generated during the welding process, while the infrared sensor is used to detect the heat radiation of the molten pool and the UV sensor is used to detect the optical radiation of plasma. Deng et al. [199] created a compact laser welding process monitoring platform that combines multi-sensor data fusion to detect in-process defects. This system captures the time series of plasma intensity, light intensity, and temperature data simultaneously. The system then evaluates these signal characteristics and determines the types of in-process errors. Li et al. [200] presented a feature-level multi-sensor fusion methodology to combine acoustic emission and optical signals for in-situ monitoring of LPBF quality. During the multi-track and multi-layer melting process, an off-axial in-situ monitoring device equipped with a microphone and a photodiode is created to capture the sound and light signals. A signal-to-image approach is created to convert 1D sensing signals to 2D images by taking into account the laser scanning data. In order to perform in-situ quality monitoring utilizing multiple sensors, the processed images are then utilized to train a CNN model to extract and combine the characteristics acquired from two distinct sensors. In exchange for a modest increase in processing time, the suggested multi-sensor fusion method greatly increases classification accuracy over single-sensor-based approaches.

This section outlines multi-sensor process monitoring systems and data fusion technology. In general, single sensors can only receive a portion of process related information. The four characteristics of integrated and fused multi-sensor information are abundance, complementarity, real time, and low cost. Intelligent manufacturing has the potential to make extensive use of multi-sensor information fusion technologies. In this examination on multi-sensor systems, numerous applications of multi-sensor systems in laser-based manufacturing research field were identified. Furthermore, with the development of modern technology, such as sensors, big data and cloud processing, computer software and hardware, and artificial intelligence, multi-sensor data fusion will become the focus of future target recognition research.

Conclusion

Monitoring of laser micromachining processes is imperative due to a multitude of critical factors including process quality verification, understanding the process dynamics, and developing closed-loop process control. Due to the spatial and temporal scales of the process dynamics involved, monitoring of ultrashort pulse laser machining is more complex. This paper systematically reviews various types of sensor techniques used for real-time monitoring of laser-based manufacturing processes as summarized in Table 12. Monitoring systems based on acoustic emission, optical emission, laser-induced breakdown spectroscopy, optical coherence tomography, confocal microscopy and vision-based have been reviewed in detail. The goal is to understand the methodologies used in conventional laser-based manufacturing processes towards adapting the monitoring systems for the ultrashort pulsed laser machining.

Table 12. Summary of comparison of different monitoring techniques for laser micromachining.

Technique	Principle	Spatial resolution (lateral and axial)	Temporal resolution	Advantages	Limitations	Required modificat for fs-lase processin
AE	Detecting transient waves of energy when material interacts with laser	AE sensor can detect surface displacement as small as 10^{-3} µm	Capturing events in the order of microseconds to milliseconds; SBAE: 100–1500 kHz, ABAE; 20 Hz–20 kHz	-Provides immediate feedback on structural changes or damage propagation, enabling prompt decision-making, -Reduced downtime and cost	-Too sensitive to the noise of environment, -Cannot detect structure size, only amplitude of released energy	Minimal
Photodiode	Processing zone light intensity to electrical current	Single channel PDs show low spatial resolution in the range of 10 to 100 µm	Rapid response time down to 10^{-9}–10^{-12} s	-High sensitivity for high-speed laser application, -Low cost, -Fast response time	-Low efficiency in identifying minor effects, -Data acquisition is challenging	Some adjustmer

Technique	Principle	Spatial resolution (lateral and axial)	Temporal resolution	Advantages	Limitations	Required modificat for fs-lase processin
LIBS	Laser-induced sparks	Resolution down to a few micrometres	Time resolution of 5 ms per spectral point	-Depth-profiling or depth-resolved analysis, -An easy, fast, real-time and in-situ measurement	-Available for heterogeneous or layered materials, -Accuracy depending on the plume behaviour	Some adjustmer
OCT	Depth-based light interference	Resolution down to a few micrometres	Hundreds to several thousand frames per second	-Same optical path with laser machine, no need to adjust the sample	-High implementation cost -Limitation of monitoring high aspect ratio features	Significan adjustmer
Confocal microscopy	Pinpointed light detection	Sub-micrometre to a few micrometres	Milliseconds to seconds	-High-resolution, 3D images in real-time	-High initial cost and complex instrumentation, -Complicated implementation into the laser chamber	Significan adjustmer

Technique	Principle	Spatial resolution (lateral and axial)	Temporal resolution	Advantages	Limitations	Required modificat for fs-lase processin
Vision-based	Image-based laser-material interaction monitoring	High-resolution (1.4 megapixel)	Up to 100 billion fps with special vision-based arrangements	-Gain better insight into the ablation and shielding effects during processing, -Gain depth profile of the temperature	-Requirement for additional component setup, -Low sampling speed for fs laser material interaction and high price	Some adjustmer
Multi-sensor	Combination of multiple techniques	–	–	-Increased reliability due to multiple sensors data, -Combining data from multiple sensors can improve accuracy and reduce uncertainties	-Challenges in merging and interpreting data from disparate sources, -Multiple sensors increase equipment costs	Some adjustmer

A comprehensive overview of diverse monitoring techniques has been provided in this review paper, outlining the comparative analysis of various methodologies utilized in laser micromachining. All the monitoring systems can do inline measurements however implementation of optical-based techniques (e.g. optical coherence tomography and confocal microscopy) are complex compared to the single-channel techniques (e.g. acoustic emission and photodiodes). Based on the nature of the techniques, they can be used for process quality verification (optical coherence tomography and confocal microscopy), studying the process dynamics (spectroscopy and vision-based), process control – either closed loop (acoustic emission and photodiode) or open-loop (optical coherence tomography and confocal microscopy). Another classification consists of surface-based (e.g. vision-based) vs sub-surface based process quality verification (e.g. acoustic emission, laser-induced breakdown spectroscopy and photodiode). Furthermore, it is important to consider surface limitations and the nature of direct and indirect responses when selecting monitoring techniques. For instance, acoustic emission presents challenges when attaching to freeform surfaces, making optical coherence tomography and optical emission more suitable alternatives for freeform surface machining purposes. In addition, acoustic emission, optical emission, and laser-induced breakdown spectroscopy provide indirect process information requiring correlation with the ongoing process. On the other hand, techniques like optical coherence tomography and confocal microscopy offer real-time, direct data that is directly relevant to the process, thereby improving monitoring accuracy and efficiency.

For widespread implementation of in-situ process monitoring in laser-based manufacturing processes, numerous obstacles must be overcome: a) Real-time identification and closed-loop control of parameters, is limited by poor spatial resolution, limited field of view and high temporal load. Development of new sensors with high spatial and temporal resolution is needed, b) Another difficulty involves the calibration of sensors, with non-contact readings and machine vision, c) Management of a large amount of monitoring data is another challenge which requires adequate data infrastructures consisting of networked systems and centralized servers for data storage and analysis, d)

In a statistical process monitoring framework, substantial research efforts are required to design and apply data mining and analysis methodologies for automated defect detection, e) The utilization of advanced machinery technology, along with the implementation of online monitoring and control, is currently restricted when it comes to 3D freeform surface processing using ultrafast lasers. Very few foundational studies on the evolution of reactive, corrective, or feedback control actions have been published to date.

In transitioning from monitoring to closed-loop feedback control in femtosecond laser micromachining, it is crucial to precisely measure the time it takes for the system to respond and assess the ability of monitoring techniques to provide real-time information. This assessment should take into account variables such as the rate at which data is collected, the amount of time it takes to process the data, and the delay in communication. Integrating monitoring techniques with control algorithms is essential for making quick decisions, using methods such as PID controllers or machine learning-based approaches. Furthermore, it is essential to take into account the constraints imposed by both hardware and software, such as the restrictions in sensor technology and computational sources. Strategies for improving control performance may include optimizing the placement of sensors or refining control algorithms. In conclusion, while the challenges of monitoring intense light-matter interaction and process quality during ultrashort pulse laser micromachining are significant, the advancements in acoustical and optical emission sensing techniques hold promise for overcoming these obstacles and paving the way for more precise and efficient monitoring in a wide range of applications.

CRediT authorship contribution statement

Kerim Yildirim: Writing – review & editing, Writing – original draft, Visualization, Project administration, Conceptualization. **Balasubramanian Nagarajan:** Writing – review & editing. **Tegoeh Tjahjowidodo:** Writing – review & editing. **Sylvie Castagne:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was partially funded by the KU Leuven Interdisciplinary Project Miracle (IDN/20/011).

References

[1] S. Nanthakumar, D. Thangaraju, R. Kumar
Experimental investigation of 2d-SnS_2 hexagonal nanosheets as lubricating additive in the microextrusion process
Proc Inst Mech Eng C J Mech Eng Sci (2022), 10.1177/09544062221123691 ↗
Google Scholar ↗

[2] L. Calabrese, M. Azzolini, F. Bassi, E. Gallus, S. Bocchi, G. Maccarini, *et al.*
Micro-milling process of metals: a comparison between femtosecond laser and EDM techniques
Journal of Manufacturing and Materials Processing (2021), 10.3390/jmmp5040125 ↗
Google Scholar ↗

[3] D.E. Lee, I. Hwang, C.M.O. Valente, J.F.G. Oliveira, D.A. Dornfeld
Precision manufacturing process monitoring with acoustic emission
Condition monitoring and control for intelligent manufacturing (2006), pp. 33-54, 10.1007/1-84628-269-1_2 ↗
Google Scholar ↗

[4] X. Lu, M.-D. Wang, P. Zhang, C. Li
Femtosecond laser machining of flexible printed circuit boards

J Phys Conf Ser (2022), 10.1088/1742-6596/2185/1/012082 ↗

Google Scholar ↗

[5] E. Mazur, K.C. Phillips, H.H. Gandhi, S.K. Sundaram
Ultrafast laser processing of materials: a review
Advances in Optics and Photonics, 7 (4) (2015), pp. 684-712, 10.1364/AOP.7.000684 ↗

Google Scholar ↗

[6] Le Harzic R, Sommer S, Le Harzic R, Breitling D, Weikert M, Sommer S, et al. Pulse width and energy influence on laser micromachining of metals in a range of 100 fs to 5 ps. Elsevier n.d. doi:https://doi.org/10.1016/j.apsusc.2004.12.027 ↗.

Google Scholar ↗

[7] A.H. Hamad
Effects of different laser pulse regimes (nanosecond, picosecond and femtosecond) on the ablation of materials for production of nanoparticles in liquid solution
IntechOpen, London, UK (2016 Sep 7)

Google Scholar ↗

[8] J. Ion
Laser processing of engineering materials: Principles, procedure and industrial application
Elsevier (2005 Mar 22)

Google Scholar ↗

[9] A. Ostendorf, F. Korte, G. Kamlage, U. Klug, J. Koch, J. Serbin, *et al.*
Applications of femtosecond lasers in 3D machining
3D laser microfabrication (2006), pp. 341-377, 10.1002/352760846X.ch12 ↗

View in Scopus ↗ Google Scholar ↗

[10] E.N. Glezer, E. Mazur

Ultrafast-laser driven micro-explosions in transparent materials
Appl Phys Lett, 71 (7) (1997), pp. 882-884, 10.1063/1.119677 ↗
View in Scopus ↗ Google Scholar ↗

[11] W. Wiesemann
2.8 process monitoring and closed-loop control: 2 production engineering
Laser Applications (2004), pp. 243-275
Crossref ↗ Google Scholar ↗

[12] P.J. Webster, L.G. Wright, Y. Ji, C.M. Galbraith, A.W. Kinross, C. Van Vlack, *et al.*
Automatic laser welding and milling with in situ inline coherent imaging
Opt Lett, 39 (21) (2014), pp. 6217-6220, 10.1364/OL.39.006217 ↗
View in Scopus ↗ Google Scholar ↗

[13] G. Miloshevsky
Ultrafast laser matter interactions: modeling approaches, challenges, and prospects
Modelling and Simulation in Materials Science and Engineering, 30 (8) (2022), Article 083001, 10.1088/1361-651X/ac8abc ↗
View in Scopus ↗ Google Scholar ↗

[14] L. Orazi, L. Romoli, M. Schmidt, L. Li
Ultrafast laser manufacturing: from physics to industrial applications
CIRP Annals, 70 (2) (2021 Jan 1), pp. 543-566, 10.1016/j.cirp.2021.05.007 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[15] P. Vanwersch, S. Schildermans, B. Nagarajan, A. Van Bael, S. Castagne
Three-dimensional modelling of femtosecond laser ablation of metals
Lasers in Manufacturing and Materials Processing, 9 (2022), pp. 515-531, 10.1007/s40516-022-00187-1 ↗
View in Scopus ↗ Google Scholar ↗

[16] B. Guo, J. Sun, Y. Hua, N. Zhan, J. Jia, K. Chu
Femtosecond laser micro/nano-manufacturing: theories, measurements, methods, and applications
Nanomanufacturing and Metrology, 3 (2020), pp. 26-67, 10.1007/S41871-020-00056-5 ↗
View in Scopus ↗ Google Scholar ↗

[17] X. Li, Y. Guan
Theoretical fundamentals of short pulse laser-metal interaction: a review
Nanotechnology and Precision Engineering, 3 (2020), pp. 105-125, 10.1016/j.npe.2020.08.001 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[18] S. Hasegawa, M. Fujimoto, T. Atsumi, Y. Hayasaki
In-process monitoring in laser grooving with line-shaped femtosecond pulses using optical coherence tomography
Light: Advanced Manufacturing, 3 (2022), p. 1, 10.37188/LAM.2022.033 ↗
Google Scholar ↗

[19] L. Mei, G. Guan
Profilometry and atomic force microscopy for surface characterization
Nano TransMed, 2 (1) (2023), Article e9130017
View PDF View article Crossref ↗ View in Scopus ↗ Google Scholar ↗

[20] H. Roozbahani, M. Alizadeh, H. Handroos, A. Salminen
Design and development of a multisensory real-time monitoring platform for ultrafast laser engraving process
IEEE Access (2022), 10.1109/access.2022.3185131 ↗
Google Scholar ↗

[21] R. Schmitt, G. Mallmann
Process monitoring in laser micro machining
Photonik International, 11 (2013), pp. 57-59

Google Scholar ↗

[22] A. Miotello, R. Kelly
Laser-induced phase explosion: new physical problems when a condensed phase approaches the thermodynamic critical temperature
Applied Physics A, 69 (Suppl. 1) (1999), pp. S67-S73, 10.1007/s003399900296 ↗
View in Scopus ↗ Google Scholar ↗

[23] Q. Lu, S.S. Mao, X. Mao, R.E. Russo
Delayed phase explosion during high-power nanosecond laser ablation of silicon
Appl Phys Lett, 80 (17) (2002), pp. 3072-3074, 10.1063/1.1473862 ↗
View in Scopus ↗ Google Scholar ↗

[24] D. Von der Linde, K. Sokolowski-Tinten
The physical mechanisms of short-pulse laser ablation
Appl Surf Sci, 154 (2000), pp. 1-10, 10.1016/S0169-4332(99)00440-7 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[25] G.M. Robinson, M.J. Jackson
Femtosecond laser micromachining of aluminum surfaces under controlled gas atmospheres
Journal of Materials Engineering and Performance, 15 (2006), pp. 155-160, 10.1361/105994906X95805 ↗
View in Scopus ↗ Google Scholar ↗

[26] B. Ali, I. Litvinyuk, M. Rybachuk
Femtosecond laser micromachining of diamond: current research status, applications and challenges
Carbon N Y (2021), 10.1016/j.carbon.2021.04.025 ↗
Google Scholar ↗

[27] D.W. Förster, B. Jäggi, A. Michalowski, B. Neuenschwander
Review on experimental and theoretical investigations of ultra-short pulsed laser ablation of metals with burst pulses
Materials (2021), 10.3390/ma14123331 ↗
Google Scholar ↗

[28] R. Weber, T. Graf, P. Berger, V. Onuseit, M. Wiedenmann, C. Freitag, *et al.*
Heat accumulation during pulsed laser materials processing
Opt Express (2014), 10.1364/oe.22.011312 ↗
Google Scholar ↗

[29] V. Gartiser
Femtosecond fiber delivery for industrial applications
(2023), 10.1117/12.2650326 ↗
Google Scholar ↗

[30] V. Schulze, P. Weber
Precise ablation milling with ultrashort pulsed Nd: YAG lasers by optical and acoustical process control
Laser-based micro-and nanopackaging and assembly IV, vol. 7585, SPIE (2010), pp. 172-180
[23]
Crossref ↗ Google Scholar ↗

[31] P. Ziehl, A. Pollock, P. Ziehl, A. Pollock
Acoustic emission for civil structures
Acoustic emission (2012), 10.5772/32652 ↗
Google Scholar ↗

[32] M. Grasso, B.M. Colosimo
Process defects and in situ monitoring methods in metal powder bed fusion: a review

Measurement Science and Technology, 28 (4) (2017), Article 044005, 10.1088/1361-6501/aa5c4f ↗

View in Scopus ↗ Google Scholar ↗

[33] H. Zeng
Ultrafast nonlinear optics
vol. 3 (2015), 10.1515/9783110304497.191 ↗

Google Scholar ↗

[34] P.J. Webster, X.Z. Joe, B.Y. Leung, L.G. Wright, K.D. Mortimer, J.M. Fraser
Inline coherent imaging of laser micromachining
2010 international symposium on optomechatronic technologies, IEEE (2010), pp. 1-4, 10.1109/ISOT.2010.5687305 ↗
[25]

Google Scholar ↗

[35] K. Sokolowski-Tinten, J. Bialkowski, A. Cavalleri, D. Von der Linde, A. Oparin, J. Meyer-Ter-Vehn, *et al.*
Transient states of matter during short pulse laser ablation
Phys Rev Lett, 81 (1998), pp. 224-227, 10.1103/PHYSREVLETT.81.224 ↗

View in Scopus ↗ Google Scholar ↗

[36] T. Herzinger
Prozessüberwachung beim Laserbohren von Turbinenschaufeln. Utz
(2004)

Google Scholar ↗

[37] C. Gehrke
Überwachung der Struktureigenschaften beim Oberflächenstrukturieren mit ultrakurzen Laserpulsen
(2013)

Google Scholar ↗

[38] A. Sun, E. Kannatey-Asibu, M. Gartner
Sensor systems for real-time monitoring of laser weld quality
J Laser Appl, 11 (1999), pp. 153-168, 10.2351/1.521893 ↗
View in Scopus ↗ Google Scholar ↗

[39] J. Shao, Y. Yan
Review of techniques for on-line monitoring and inspection of laser welding
Journal of physics: conference series, IOP Publishing (2005), 10.1088/1742-6596/15/1/017 ↗
(Vol. 15, No. 1, p. 101)
Google Scholar ↗

[40] C. Wei, L. Li
Acoustic emission and ultrasound monitoring in laser micro/nanofabrication
Handbook of laser micro- and nano-engineering (2021), pp. 1979-2002, 10.1007/978-3-030-63647-0_47 ↗
View in Scopus ↗ Google Scholar ↗

[41] C.B. Scruby, L.E. Drain
Laser ultrasonics
Laser ultrasonics (2019), 10.1201/9780203749098/LASER-ULTRASONICS-SCRUBY-DRAIN ↗
Google Scholar ↗

[42] A. Sarrafzadeh, R.J. Churchill, M.G. Niimura
Laser generated ultrasound
Acousto-ultrasonics (1988), pp. 201-207, 10.1007/978-1-4757-1965-9_15 ↗
Google Scholar ↗

[43] H. Hébert, F. Vidal, F. Martin, J.C. Kieffer, A. Nadeau, T.W. Johnston
Ultrasound generated by a femtosecond and a picosecond laser pulse near the ablation threshold

J Appl Phys (2005), p. 98, 10.1063/1.1999827 ↗

Google Scholar ↗

[44] D. Schneider
Laser-induced surface acoustic waves for material testing
Handbook of advanced nondestructive evaluation (2019), pp. 171-234, 10.1007/978-3-319-26553-7_38 ↗

Google Scholar ↗

[45] L.M. Lyamshev
Optoacoustic sources of sound
Soviet Physics - Uspekhi, 24 (1981), pp. 977-995, 10.1070/PU1981V024N12ABEH004757/META ↗

View in Scopus ↗ Google Scholar ↗

[46] T. Pezeril
Laser generation and detection of ultrafast shear acoustic waves in solids and liquids
Opt Laser Technol, 83 (2016), pp. 177-188, 10.1016/J.OPTLASTEC.2016.03.019 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[47] L. Li
A comparative study of ultrasound emission characteristics in laser processing
Appl Surf Sci, 186 (1–4) (2002), pp. 604-610, 10.1016/S0169-4332(01)00695-X ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[48] D.D. Vallejo
Spectroscopic investigations of plasma emission induced during laser material processing
epubli (2014), p. 15

View in Scopus ↗ Google Scholar ↗

[49] V. Deutsch, M. Platte, M. Vogt
Ultraschallprüfung: Grundlagen und industrielle Anwendungen
Springer-Verlag (2013), p. 12
Google Scholar ↗

[50] P. Weber
Steigerung der Prozesswiederholbarkeit mittels Analyse akustischer Emissionen bei der Mikrolaserablation mit UV-Pikosekundenlasern
(2014)
[Doctoral dissertation, Zugl.: Karlsruhe, Karlsruher Institut für Technologie (KIT), Diss.]
Google Scholar ↗

[51] X. Xie, Y. Zhang, Q. Huang, Y. Huang, W. Zhang, J. Zhang, *et al.*
Monitoring method for femtosecond laser modification of silicon carbide via acoustic emission techniques
J Mater Process Technol, 1 (290) (2021), Article 116990, 10.1016/j.jmatprotec.2020.116990 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[52] J.G. Webster, H. Eren (Eds.), Measurement, instrumentation, and sensors handbook: spatial, mechanical, thermal, and radiation measurement, CRC Press (2017)
[19]
Google Scholar ↗

[53] M.C. Lu, S.J. Chiou, B.S. Kuo, M.Z. Chen
Analysis of acoustic emission (AE) signals for quality monitoring of laser lap microwelding
Applied Sciences, 11 (15) (2021), p. 7045, 10.3390/app11157045 ↗
View in Scopus ↗ Google Scholar ↗

[54] S. Soloman
Sensors and control systems in manufacturing

McGraw-Hill, Inc. (1994)

Google Scholar ↗

[55] N.S. Choi, T.W. Kim, K.Y. Rhee

Kaiser effects in acoustic emission from composites during thermal cyclic-loading

Ndt & E International, 38 (4) (2005), pp. 268-274

View PDF View article View in Scopus ↗ Google Scholar ↗

[56] C. Rubio-González, J.L. Ocaña, G. Gomez-Rosas, C. Molpeceres, M. Paredes, A. Banderas, *et al.*

Effect of laser shock processing on fatigue crack growth and fracture toughness of 6061-T6 aluminum alloy

Mater Sci Eng A, 386 (1–2) (2004), pp. 291-295, 10.1016/j.msea.2004.07.025 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[57] M. Stafe, C. Negutu, I.M. Popescu

Real-time determination and controlof the laser-drilled holes depth

Shock Waves, 14 (1) (2005), pp. 123-126, 10.1007/s00193-004-0240-7 ↗

View in Scopus ↗ Google Scholar ↗

[58] D. Ye, G.S. Hong, Y. Zhang, K. Zhu, J.Y. Fuh

Defect detection in selective laser melting technology by acoustic signals with deep belief networks

The International Journal of Advanced Manufacturing Technology, 96 (2018), pp. 2791-2801, 10.1007/s00170-018-1728-0 ↗

View in Scopus ↗ Google Scholar ↗

[59] V. Pandiyan, R. Drissi-Daoudi, S. Shevchik, G. Masinelli, R. Logé, K. Wasmer

Analysis of time, frequency and time-frequency domain features from acoustic emissions during laser powder-bed fusion process

Procedia CIRP, 94 (2020), pp. 392-397, 10.1016/j.procir.2020.09.152 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[60] N. Eschner, L. Weiser, B. Häfner, G. Lanza
Classification of specimen density in laser powder bed fusion (L-PBF) using in-process structure-borne acoustic process emissions
Addit Manuf, 34 (2020), Article 101324, 10.1016/j.addma.2020.101324 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[61] B.S. Kuo, M.C. Lu
Analysis of a sound signal for quality monitoring in laser microlap welding
Applied Sciences, 10 (6) (2020), p. 1934, 10.3390/app10061934 ↗
View in Scopus ↗ Google Scholar ↗

[62] M. Bastuck, H.G. Herrmann, B. Wolter, D. Böttger, P.C. Zinn
AkuProLas: acoustic inline process monitoring for laser welding applications
19th world conf. non-destructive testing (2016)
Google Scholar ↗

[63] T. Takata, M. Enoki, P. Chivavibul, A. Matsui, Y. Kobayashi
Acoustic emission monitoring of laser shock peening by detection of underwater acoustic wave
Mater Trans, 57 (5) (2016), pp. 674-680, 10.2320/matertrans.M2015401 ↗
View in Scopus ↗ Google Scholar ↗

[64] J.J. Wu, Zhao J. Bin, H.C. Qiao, X.J. Liu, Y.N. Zhang, T.Y. Hu
Acoustic wave detection of laser shock peening
Opto-Electronic Advances, 1 (2018), pp. 1-5, 10.29026/OEA.2018.180016 ↗
Google Scholar ↗

[65] X. Xie, Q. Huang, J. Long, Q. Ren, W. Hu, S. Liu

A new monitoring method for metal rust removal states in pulsed laser derusting via acoustic emission techniques
J Mater Process Technol, 1 (275) (2020), Article 116321, 10.1016/j.jmatprotec.2019.116321 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[66] M. Zuric, O. Nottrodt, P. Abels
Multi-sensor system for real-time monitoring of laser micro-structuring
Journal of Laser Micro/Nanoengineering, 14 (3) (2019), 10.2961/jlmn.2019.03.0008 ↗
Google Scholar ↗

[67] A. Stournaras, G. Chryssolouris
On acoustic emissions in percussion laser drilling
The International Journal of Advanced Manufacturing Technology, 46 (2010), pp. 611-620, 10.1007/s00170-009-2144-2 ↗
View in Scopus ↗ Google Scholar ↗

[68] V. Schulze, P. Weber
Acoustical focal positioning for micromachining using a picosecond pulsed UV-laser
30th international congress on applications of lasers and electro-optics, ICALEO 2011 (2011), pp. 966-971, 10.2351/1.5062357 ↗
View in Scopus ↗ Google Scholar ↗

[69] A. Kacaras, M. Bächle, M. Schwabe, F. Zanger, F.P. León, V. Schulze
Acoustic emission-based characterization of focal position during ultra-short pulse laser ablation
Procedia CIRP, 1 (81) (2019), pp. 270-275, 10.1016/j.procir.2019.03.047 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[70] S. Palanco, J. Laserna
Spectral analysis of the acoustic emission of laser-produced plasmas
Appl Optics, 42 (30) (2003), pp. 6078-6084, 10.1364/AO.42.006078 ↗

View in Scopus ↗ Google Scholar ↗

[71] S. Conesa, S. Palanco, J.J. Laserna
Acoustic and optical emission during laser-induced plasma formation
Spectrochimica Acta Part B: Atomic Spectroscopy, 59 (9) (2004), pp. 1395-1401, 10.1016/j.sab.2004.06.004 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[72] E. Manikanta, P. Venkateshwarlu, S.S. Shiva, V.R. Kumar, C. Leela, S.P. Tewari, *et al.*
Dynamic response of metals and alloys to laser-induced shock waves
Laser sources and applications, vol. 8433, SPIE (2012, May), pp. 546-551
Google Scholar ↗

[73] E.V. Bordatchev, S.K. Nikumb
Effect of focus position on informational properties of acoustic emission generated by laser–material interactions
Appl Surf Sci, 253 (3) (2006), pp. 1122-1129, 10.1016/j.apsusc.2006.01.047 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[74] B. Fischer, W. Rohringer, N. Panzer, S. Hecker
Acoustic process control for laser material processing: optical microphone as a novel “ear” for industrial manufacturing
Wiley Online Library, 14 (2017), pp. 21-25, 10.1002/latj.201700029 ↗
Google Scholar ↗

[75] C. Prieto, R. Fernandez, C. Gonzalez, M. Diez, J. Arias, R. Sommerhuber, *et al.*
In situ process monitoring by optical microphone for crack detection in Laser Metal Deposition applications
11th CIRP conference on photonic technologies [LANE 2020] (2020 Sep)
Google Scholar ↗

[76] S. Lee, S. Ahn, C. Park

Analysis of acoustic emission signals during laser spot welding of SS304 stainless steel
J Mater Eng Perform, 23 (2014), pp. 700-707, 10.1007/S11665-013-0791-9 ↗
View in Scopus ↗ Google Scholar ↗

[77] M.Z. Chen, M.C. Lu, P.N. Wang, S.J. Chiou
Experimental study of quality monitoring system integrated with a microphone array in laser microlap welding
Int J Adv Manuf Technol, 121 (2022), pp. 2305-2316, 10.1007/S00170-022-09459-8 ↗
View in Scopus ↗ Google Scholar ↗

[78] J. Trapp, A.M. Rubenchik, G. Guss, M.J. Matthews
In situ absorptivity measurements of metallic powders during laser powder-bed fusion additive manufacturing
Appl Mater Today, 1 (9) (2017), pp. 341-349, 10.1016/j.apmt.2017.08.006 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[79] R. Acharya, J.A. Sharon, A. Staroselsky
Prediction of microstructure in laser powder bed fusion process
Acta Mater, 1 (124) (2017), pp. 360-371, 10.1016/j.actamat.2016.11.018 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[80] K. Wasmer, T. Le-Quang, B. Meylan, S.A. Shevchik
In situ quality monitoring in AM using acoustic emission: a reinforcement learning approach
J Mater Eng Perform, 28 (2019), pp. 666-672, 10.1007/S11665-018-3690-2 ↗
View in Scopus ↗ Google Scholar ↗

[81] D. Ye, Y. Zhang, K. Zhu, G. Hong, J. YingHsi
Characterization of acoustic signals during a direct metal laser sintering process

Advances in energy science and equipment engineering II (2017), pp. 1315-1320, 10.1201/9781315116174-89 ↗

Google Scholar ↗

[82] H. Gaja, F. Liou
Defects monitoring of laser metal deposition using acoustic emission sensor
The International Journal of Advanced Manufacturing Technology., 90 (2017), pp. 561-574, 10.1007/s00170-016-9366-x ↗

View in Scopus ↗ Google Scholar ↗

[83] W. Jiajun, Z. Jibin, Q. Hongchao, L. Ying, S. Boyu, H. Taiyou, *et al.*
The application status and development of laser shock processing
Opto-Electronic Engineering, 45 (2) (2018), 10.12086/OEE.2018.170690 ↗

Google Scholar ↗

[84] M. Enoki, K. Kobayashi, T. Takata, A. Matsui, Y. Kobayashi
Quantitative acoustic emission measurement of laser peening
30th European conference on acoustic emission testing and 7th international conference on acoustic emission, vol. 1 (2012), pp. 12-15

Google Scholar ↗

[85] D. Birtalan, W. Nunley
Optoelectronics: infrared-visable-ultraviolet devices and applications
CRC Press (2018), p. 8, 10.1201/9781315218847/OPTOELECTRONICS-DAVE-BIRTALAN-WILLIAM-NUNLEY ↗

Google Scholar ↗

[86] I. Smurov
Laser process optical sensing and control
IV international WLT-conference on lasers in manufacturing (2007), pp. 537-546
[18]

Google Scholar ↗

[87] J.P. Kruth, J. Duflou, P. Mercelis, J. Van Vaerenbergh, T. Craeghs, J. De Keuster
On-line monitoring and process control in selective laser melting and laser cutting
Proceedings of the 5th lane conference, laser assisted net shape engineering, vol. 1 (2007), pp. 23-37
Google Scholar ↗

[88] T. Purtonen, A. Kalliosaari, A. Salminen
Monitoring and adaptive control of laser processes
Physics Procedia, 1 (56) (2014), pp. 1218-1231, 10.1016/j.phpro.2014.08.038 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[89] P. Yadav, O. Rigo, C. Arvieu, E. Le Guen, E. Lacoste
In situ monitoring systems of the SLM process: on the need to develop machine learning models for data processing
Crystals, 10 (6) (2020), 10.3390/cryst10060524 ↗
Google Scholar ↗

[90] G. Held
Introduction to light emitting diode technology and applications
Introduction to light emitting diode technology and applications (2016), 10.1201/9781420076639/INTRODUCTION-LIGHT-EMITTING-DIODE-TECHNOLOGY-APPLICATIONS-GILBERT-HELD ↗
Google Scholar ↗

[91] P. Norman, H. Engström, P. Gren, Kaplan AFH
Correlation between photodiode monitoring and high speed imaging of the dynamics causing laser welding defects
ICALEO 2008 - 27th international congress on applications of lasers and electro-optics, congress proceedings (2008), pp. 829-837, 10.2351/1.5061270 ↗

View in Scopus ↗ Google Scholar ↗

[92] S. Jayasinghe, P. Paoletti, C. Sutcliffe, J. Dardis, N. Jones, P.L. Green
Automatic quality assessments of laser powder bed fusion builds from photodiode sensor measurements
Progress in Additive Manufacturing, 7 (2022), pp. 143-160, 10.1007/S40964-021-00219-W ↗
View in Scopus ↗ Google Scholar ↗

[93] Z. Mao, W. Feng, H. Ma, Y. Yang, J. Zhou, S. Liu, *et al.*
Continuous online flaws detection with photodiode signal and melt pool temperature based on deep learning in laser powder bed fusion
Optics & Laser Technology, 1 (158) (2023), Article 108877, 10.1016/j.optlastec.2022.108877 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[94] J. Yu, Y. Sohn, Y.W. Park, J.S. Kwak
The development of a quality prediction system for aluminum laser welding to measure plasma intensity using photodiodes
Journal of Mechanical Science and Technology, 30 (2016), pp. 4697-4704, 10.1007/s12206-016-0940-9 ↗
View in Scopus ↗ Google Scholar ↗

[95] J. De Keuster, J. Duflou, J.P. Kruth
Real-time, adaptive control and optimisation of high-power CO2 laser cutting using photodiodes
Proceedings of the 5th lane conference, laser assisted net shape engineering (2007), pp. 979-992
Google Scholar ↗

[96] B. Adelmann, M. Schleier, B. Neumeier, R. Hellmann
Photodiode-based cutting interruption sensor for near-infrared lasers
Appl Optics, 55 (7) (2016), pp. 1772-1778, 10.1364/AO.55.001772 ↗
View in Scopus ↗ Google Scholar ↗

[97] S.M. Garcia, J. Ramos, J.I. Arrizubieta, J. Figueras
Analysis of photodiode monitoring in laser cutting
Applied Sciences, 10 (18) (2020), 10.3390/app10186556 ↗
Google Scholar ↗

[98] Y.W. Park, S. Rhee
Study of a line width estimation model for laser micro material processing using a photodiode
Optics & Laser Technology, 39 (7) (2007), pp. 1461-1471, 10.1016/j.optlastec.2006.09.013 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[99] J. Martan, D. Moskal, L. Prokešová, M. Honner
Detection of heat accumulation in laser surface texturing by fast infrared detectors
The laser in manufacturing (LiM2019), JLPS-Japan Laser Processing Society, Osaka, Japan (2019), pp. 1-7
Google Scholar ↗

[100] C.C. Ho, Y.J. Chang, J.C. Hsu, C.M. Chiu, C.L. Kuo
Optical emission monitoring for defocusing laser percussion drilling
Measurement, 1 (80) (2016), pp. 251-258
View PDF View article View in Scopus ↗ Google Scholar ↗

[101] D.Y. You, X.D. Gao, S. Katayama
Review of laser welding monitoring
Science and Technology of Welding and Joining, 19 (3) (2014), pp. 181-201, 10.1179/1362171813Y.0000000180 ↗
View in Scopus ↗ Google Scholar ↗

[102] Postma S. Weld pool control in Nd: YAG laser welding.
Google Scholar ↗

[104] C.-F. Lin, S.-H. Chen, T.-Y. Yu, Y.J. Park, M.W. Lee, *et al.*
On-line depth measurement for laser-drilled holes based on the intensity of plasma emission
IopscienceIopOrg, 25 (2014), Article 94007, 10.1088/0957-0233/25/9/094007 ↗
Google Scholar ↗

[105] P.P. Pronko, S.K. Dutta, D. Du, R.K. Singh
Thermophysical effects in laser processing of materials with picosecond and femtosecond pulses
J Appl Phys, 78 (10) (1995), pp. 6233-6240, 10.1063/1.360570 ↗
View in Scopus ↗ Google Scholar ↗

[106] M. Honer
Prozesssicherungsmaßnahmen beim Bohren metallischer Werkstoffe mittels Laserstrahlung. Utz
(2004)
Google Scholar ↗

[107] A. Stournaras, K. Salonitis, G. Chryssolouris
Acoustic and optical sensing for monitoring of blind laser drilling geometrical features
ICALEO 2009 - 28th international congress on applications of lasers and electro-optics, congress proceedings, 102 (2009), pp. 1515-1524, 10.2351/1.5061523 ↗
View in Scopus ↗ Google Scholar ↗

[108] D. Walter
Online-Qualitätssicherung beim Bohren mittels ultrakurz gepulster Laserstrahlung
Herbert Utz Verlag (2010)
Google Scholar ↗

[109] J. De Keuster, J.R. Duflou, J.P. Kruth

Monitoring of high-power CO 2 laser cutting by means of an acoustic microphone and photodiodes

The International Journal of Advanced Manufacturing Technology., 35 (2007), pp. 115-126, 10.1007/s00170-006-0695-z ↗

View in Scopus ↗ Google Scholar ↗

[110] N. Levichev, G.C. Rodrigues, J.R. Duflou

Real-time monitoring of fiber laser cutting of thick plates by means of photodiodes

Procedia CIRP, 1 (94) (2020), pp. 499-504, 10.1016/j.procir.2020.09.171 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[111] T. Tong, J. Li, J.P. Longtin

Ultrafast laser-induced breakdown spectroscopy and application to real-time control of ultrafast laser micromachining process

Heat transfer summer conference, vol. 36959 (2003), pp. 53-61, 10.1115/HT2003-47068 ↗

View in Scopus ↗ Google Scholar ↗

[112] T. Tong, J. Li, J.P. Longtin

Real-time control of ultrafast laser micromachining by laser-induced breakdown spectroscopy

Appl Optics, 43 (9) (2004), pp. 1971-1980, 10.1364/AO.43.001971 ↗

View in Scopus ↗ Google Scholar ↗

[113] Y. Song, G. Hu, Z. Zhang, Y. Guan

Real-time spectral response guided smart femtosecond laser bone drilling

Opt Lasers Eng, 128 (2020), Article 106017, 10.1016/j.optlaseng.2020.106017 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[114] R.K. Gill, F. Knorr, Z.J. Smith, M. Kahraman, D. Madsen, D.S. Larsen, *et al.*

Characterization of femtosecond laser-induced breakdown spectroscopy (fsLIBS) and applications for biological samples

Appl Spectrosc, 68 (2015), pp. 949-954, 10.1366/13-07293 ↗

View in Scopus ↗ Google Scholar ↗

[115] R. Kunze, G. Mallmann, R. Schmitt
Inline plasma analysis as tool for process monitoring in laser micro machining for multi-layer materials
Physics Procedia, 83 (2016), pp. 1329-1338, 10.1016/j.phpro.2016.08.140 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[116] J. Skruibis, O. Balachninaite, S. Butkus, V. Vaicaitis, V. Sirutkaitis
Multiple-pulse laser-induced breakdown spectroscopy for monitoring the femtosecond laser micromachining process of glass
Optics & Laser Technology., 1 (111) (2019), pp. 295-302, 10.1016/j.optlastec.2018.10.005 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[117] S. Shiby, N.J. Vasa
Nanosecond laser-assisted micro-scribing of a copper film on a dielectric material with laser-induced breakdown spectroscopy based monitoring
Optics & Laser Technology., 1 (2022), p. 147, 10.1016/j.optlastec.2021.107685 ↗

Google Scholar ↗

[118] J.L. Gottfried, F.C. De Lucia Jr
Laser-induced breakdown spectroscopy: capabilities and applications
Defense Technical Information Center document (2010)

Google Scholar ↗

[119] A.A. Voznesenskaya, D.S. Gusev, D.A. Kochuev, A.A. Polyakova, K.A. Frolov
Laser-induced breakdown spectroscopy monitoring of aluminum alloys laser cleaning process
J Phys Conf Ser (2022), p. 2316, 10.1088/1742-6596/2316/1/012006 ↗

Google Scholar ↗

[120] H. Balzer, M. Hoehne, R. Noll, V. Sturm
New approach to online monitoring of the Al depth profile of the hot-dip galvanised sheet steel using LIBS
Anal Bioanal Chem, 385 (2006), pp. 225-233, 10.1007/S00216-006-0347-Z ↗
View in Scopus ↗ Google Scholar ↗

[121] A.A. Voznesenskaya, D.S. Gusev, D.A. Kochuev, A.A. Polyakova, K.A. Frolov
Laser-induced breakdown spectroscopy monitoring of aluminum alloys laser cleaning process
Journal of physics: conference series, IOP Publishing (2022 Aug 1), 10.1088/1742-6596/2316/1/012006 ↗
[vol. 2316, no. 1, p. 012006]
Google Scholar ↗

[122] H. Roozbahani, A. Salminen, M. Manninen
Real-time online monitoring of nanosecond pulsed laser scribing process utilizing spectrometer
J Laser Appl (2017), p. 29, 10.2351/1.4983520 ↗
Google Scholar ↗

[123] C.S. Lough, L.I. Escano, M. Qu, C.C. Smith, R.G. Landers, D.A. Bristow, *et al.*
In-situ optical emission spectroscopy of selective laser melting
Journal of Manufacturing Processes., 1 (53) (2020), pp. 336-341, 10.1016/j.jmapro.2020.02.016 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[124] W. Ren, Z. Zhang, Y. Lu, G. Wen, J. Mazumder
In-situ monitoring of laser additive manufacturing for Al7075 alloy using emission spectroscopy and plume imaging
IEEE Access, 21 (9) (2021), pp. 61671-61679, 10.1109/ACCESS.2021.3074703 ↗
View in Scopus ↗ Google Scholar ↗

[125] W. Ren, G. Wen, Z. Zhang, J. Mazumder
Quality monitoring in additive manufacturing using emission spectroscopy and unsupervised deep learning
Materials and Manufacturing Processes, 37 (11) (2022), pp. 1339-1346, 10.1080/10426914.2021.1906891 ↗
View in Scopus ↗ Google Scholar ↗

[126] V.N. Lednev, P.A. Sdvizhenskii, R.D. Asyutin, R.S. Tretyakov, M.Y. Grishin, A.Y. Stavertiy, *et al.*
In situ multi-elemental analysis by laser induced breakdown spectroscopy in additive manufacturing
Addit Manuf, 25 (2019), pp. 64-70, 10.1016/j.addma.2018.10.043 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[127] V.N. Lednev, P.A. Sdvizhenskii, A.Y. Stavertiy, M.Y. Grishin, R.S. Tretyakov, R.D. Asyutin, *et al.*
Online and in situ laser-induced breakdown spectroscopy for laser welding monitoring
Spectrochimica Acta Part B: Atomic Spectroscopy, 175 (2021), Article 106032, 10.1016/j.sab.2020.106032 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[128] S. Huber, J. Glasschroeder, M.F. Zaeh
Analysis of the metal vapour during laser beam welding
Physics Procedia, 12 (2011), pp. 712-719, 10.1016/j.phpro.2011.03.089 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[129] D.K. Das, J.P. McDonald, S.M. Yalisove, T.M. Pollock
Depth-profiling study of a thermal barrier coated superalloy using femtosecond laser-induced breakdown spectroscopy
Spectrochimica Acta Part B: Atomic Spectroscopy., 63 (1) (2008), pp. 27-36, 10.1016/j.sab.2007.10.048 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[130] D. Diego-Vallejo, D. Ashkenasi, G. Illing, H.J. Eichler
Plasma monitoring during laser material processing
Physics Procedia., 1 (12) (2011), pp. 404-410, 10.1016/j.phpro.2011.03.150 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[131] A. Baskevicius, A. Alesenkov, G. Chozevskis, J. Litvaityte, O. Balachninaite, D. Paipulas, *et al.*
Optimization of laser-ablation micromachining by choice of scanning algorithms and use of laser-induced-breakdown spectroscopy
JlpsGrJp (2013), p. 8, 10.2961/jlmn.2013.01.0006 ↗
Google Scholar ↗

[132] M. Lentjes, K. Dickmann, J. Meijer
Influence of process parameters on the distribution of single shot correlation coefficients obtained by correlating LIB-spectra
Applied Physics A, 88 (2007), pp. 661-666, 10.1007/s00339-007-4023-8 ↗
View in Scopus ↗ Google Scholar ↗

[133] J.A. Aguilera, C. Aragon, F. Penalba
Plasma shielding effect in laser ablation of metallic samples and its influence on LIBS analysis
Appl Surf Sci, 127 (1998), pp. 309-314, 10.1016/S0169-4332(97)00648-X ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[134] A. Baskevicius, O. Balachninaite, M. Karpavicius, S. Butkus, D. Paipulas, V. Sirutkaitis
Monitoring of the femtosecond laser micromachining process of materials immersed in water by use of laser-induced breakdown spectroscopy
JlpsGrJp (2016), p. 11, 10.2961/jlmn.2016.03.0018 ↗
Google Scholar ↗

[135] Y.Z. Deng, H.Y. Zheng, V.M. Murukeshan, X.C. Wang, G.C. Lim, B.K.A. Ngoi
In-process monitoring of femtosecond laser material processing
Int J Nanosci, 4 (2005), pp. 761-766, 10.1142/S0219581X05003632 ↗

View in Scopus ↗ Google Scholar ↗

[136] V.N. Lednev, P.A. Sdvizhenskii, R.D. Asyutin, R.S. Tretyakov, M.Y. Grishin, A.Y. Stavertiy, *et al.*
In situ elemental analysis and failures detection during additive manufacturing process utilizing laser induced breakdown spectroscopy
Opt Express, 27 (4) (2019), pp. 4612-4628, 10.1364/OE.27.004612 ↗
View in Scopus ↗ Google Scholar ↗

[137] W. Ren, J. Mazumder
In-situ porosity recognition for laser additive manufacturing of 7075-Al alloy using plasma emission spectroscopy
Sci Rep, 10 (1) (2020), Article 19493, 10.1038/s41598-020-75131-4 ↗
View in Scopus ↗ Google Scholar ↗

[138] J.A. Kanko, A.P. Sibley, J.M. Fraser
In situ morphology-based defect detection of selective laser melting through inline coherent imaging
J Mater Process Technol, 231 (2016), pp. 488-500, 10.1016/j.jmatprotec.2015.12.024 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[139] N. Dupriez
Journal CT-LT, 2016 undefined. OCT for efficient high quality laser welding: high-speed, high-resolution online seam tracking, monitoring and quality control
Wiley Online Library, 13 (2016), pp. 37-41, 10.1002/latj.201600020 ↗
[undefined]
Google Scholar ↗

[140] A. Nemeth, G. Hannesschläger, E. Leiss-Holzinger, K. Wiesauer, M. Leitner
Optical coherence tomography–applications in non-destructive testing and evaluation
Optical Coherence Tomography, 6 (6) (2013), pp. 163-186, 10.5772/53960 ↗

View in Scopus ↗ Google Scholar ↗

[141] P.J. Webster, L.G. Wright, K.D. Mortimer, B.Y. Leung, J.X. Yu, J.M. Fraser
Automatic real-time guidance of laser machining with inline coherent imaging
J Laser Appl, 23 (2) (2011), 10.2351/1.3567955 ↗
Google Scholar ↗

[142] P.H. Tomlins, G.N. Smith, P.D. Woolliams, J. Rasakanthan, K. Sugden
Femtosecond laser micro-inscription of optical coherence tomography resolution test artifacts
Biomed Opt Express, 2 (5) (2011), pp. 1319-1327, 10.1364/BOE.2.001319 ↗
View in Scopus ↗ Google Scholar ↗

[143] M. Wiesner, J. Ihlemann, H.H. Müller, E. Lankenau, G. Hüttmann
Optical coherence tomography for process control of laser micromachining
Review of Scientific Instruments, 81 (3) (2010), 10.1063/1.3356080 ↗
Google Scholar ↗

[144] K. Goda, A. Fard, O. Malik, G. Fu, A. Quach, B. Jalali
High-throughput optical coherence tomography at 800 nm
Opt Express, 20 (18) (2012), pp. 19612-19617, 10.1364/OE.20.019612 ↗
View in Scopus ↗ Google Scholar ↗

[145] S. Takushima, D. Morita, N. Shinohara, H. Kawano, Y. Mizutani, Y. Takaya
Optical in-process height measurement system for process control of laser metal-wire deposition
Precis Eng, 62 (2020), pp. 23-29, 10.1016/j.precisioneng.2019.11.007 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[146] A. Neef, V. Seyda, D. Herzog, C. Emmelmann, M. Schönleber, M. Kogel-Hollacher

Low coherence interferometry in selective laser melting
Physics Procedia, 56 (2014 Jan 1), pp. 82-89, 10.1016/j.phpro.2014.08.100
View PDF View article View in Scopus Google Scholar

[147] P.J. DePond, G. Guss, S. Ly, N.P. Calta, D. Deane, S. Khairallah, *et al.*
In situ measurements of layer roughness during laser powder bed fusion additive manufacturing using low coherence scanning interferometry
Materials & Design, 154 (2018), pp. 347-359, 10.1016/j.matdes.2018.05.050
View PDF View article View in Scopus Google Scholar

[148] J.J. Blecher, C.M. Galbraith, Vlack C. Van, T.A. Palmer, J.M. Fraser, P.J.L. Webster, *et al.*
Real time monitoring of laser beam welding keyhole depth by laser interferometry
19, Taylor & Francis (vol. 2014), pp. 560-564, 10.1179/1362171814Y.0000000225
View in Scopus Google Scholar

[149] M. Boley, F. Fetzer, R. Weber, T. Graf
Statistical evaluation method to determine the laser welding depth by optical coherence tomography
Opt Lasers Eng (2019), 10.1016/j.optlaseng.2019.03.014
Google Scholar

[150] R. Schmitt, T. Pfeifer, G. Mallmann
Machine integrated telecentric surface metrology in laser structuring systems
Acta Imeko (2013), 10.21014/acta_imeko.v2i2.106
Google Scholar

[151] F. Zechel, R. Kunze, P. Widmann, R.H. Schmitt
Adaptive process monitoring for laser micro structuring of electrical steel using optical coherence tomography with non-colour corrected lenses
Procedia CIRP (2020), 10.1016/j.procir.2020.09.141

Google Scholar ↗

[152] Y. Ji, A.W. Grindal, P.J. Webster, J.M. Fraser
Real-time depth monitoring and control of laser machining through scanning beam delivery system
J Phys D Appl Phys, 48 (15) (2015), Article 155301, 10.1088/0022-3727/48/15/155301 ↗
View in Scopus ↗ Google Scholar ↗

[153] S. Ly, A.M. Rubenchik, S.A. Khairallah, G. Guss, M.J. Matthews
Metal vapor micro-jet controls material redistribution in laser powder bed fusion additive manufacturing
Sci Rep (2017), 10.1038/s41598-017-04237-z ↗
Google Scholar ↗

[154] P. Lott, H. Schleifenbaum, W. Meiners, K. Wissenbach, C. Hinke, J. Bültmann
Design of an optical system for the in situ process monitoring of selective laser melting (SLM)
Physics Procedia (2011), 10.1016/j.phpro.2011.03.085 ↗
Google Scholar ↗

[155] B. Zhang, J. Ziegert, F. Farahi, A. Davies
In situ surface topography of laser powder bed fusion using fringe projection
Addit Manuf (2016), 10.1016/j.addma.2016.08.001 ↗
Google Scholar ↗

[156] P. Bidare, R.R. Maier, R.J. Beck, J.D. Shephard, A.J. Moore
An open-architecture metal powder bed fusion system for in-situ process measurements
Addit Manuf (2017), 10.1016/j.addma.2017.06.007 ↗
Google Scholar ↗

[157] J.M. Hallam, T. Kissinger, T.O. Charrett, R.P. Tatam
In-process range-resolved interferometric (RRI) 3D layer height measurements for wire+ arc additive manufacturing (WAAM)
Measurement Science and Technology., 33 (4) (2022), Article 044002, 10.1088/1361-6501/ac440e ↗
View in Scopus ↗ Google Scholar ↗

[158] X. Chen, Y. Xu, N.K. Chen, S. Shy, H.C. Chui
In-situ depth measurement of laser micromachining
Photonics, vol. 8, no. 11, MDPI (2021), p. 493, 10.3390/photonics8110493 ↗
View PDF View article Google Scholar ↗

[159] R.H. Webb
Confocal optical microscopy
Rep Prog Phys (1996), 10.1088/0034-4885/59/3/003 ↗
Google Scholar ↗

[160] T.V. Vorburger, H.G. Rhee, T.B. Renegar, J.F. Song, A. Zheng
Comparison of optical and stylus methods for measurement of surface texture
The International Journal of Advanced Manufacturing Technology. (2007), 10.1007/s00170-007-0953-8 ↗
Google Scholar ↗

[161] S. Fu, F. Cheng, T. Tjahjowidodo, Y. Zhou, D. Butler
A non-contact measuring system for in-situ surface characterization based on laser confocal microscopy
Sensors (2018), 10.3390/s18082657 ↗
Google Scholar ↗

[162] A.K. Ruprecht, C. Pruss, H.J. Tiziani, W. Osten, P. Lucke, A. Last, *et al.*
Confocal micro-optical distance sensor: principle and design

Optical measurement systems for industrial inspection IV, vol. 5856, SPIE (2005), pp. 128-135, 10.1117/12.612008 ↗

View in Scopus ↗ Google Scholar ↗

[163] Giusca CL, Leach RK. Calibration of the metrological characteristics of Imaging Confocal Microscopes (ICMs).

Google Scholar ↗

[164] S. Tani, Y. Aoyagi, Y. Kobayashi
Neural-network-assisted in situ processing monitoring by speckle pattern observation
Opt Express (2020), 10.1364/OE.400785 ↗

Google Scholar ↗

[165] F.B. Wang, L.H. Zhao, P. Tu, Y. Liu, J.X. Chen
Image feature analysis of plasma spot produced from femtosecond laser ablation for silicon wafer
Journal of Micro/Nanolithography, MEMS, and MOEMS (2017), 10.1117/1.JMM.16.2.025003 ↗

Google Scholar ↗

[166] I. Mingareev
Ultrafast dynamics of melting and ablation at large laser intensities
Cuvillier Verlag (2009)

Google Scholar ↗

[167] H. Mikami, L. Gao, K. Goda
Ultrafast optical imaging technology: principles and applications of emerging methods
Nanophotonics (2016), 10.1515/nanoph-2016-0026 ↗

Google Scholar ↗

[168] J. Liang, L.V. Wang

Single-shot ultrafast optical imaging
Optica (2018 Sep 20), 10.1364/OPTICA.5.001113 ↗
Google Scholar ↗

[169] A. Velten, T. Willwacher, O. Gupta, A. Veeraraghavan, M.G. Bawendi, R. Raskar
Recovering three-dimensional shape around a corner using ultrafast time-of-flight imaging
Nat Commun (2012), 10.1038/ncomms1747 ↗
Google Scholar ↗

[170] P.W. Fuller
An introduction to high speed photography and photonics
The Imaging Science Journal (2009), 10.1179/136821909X12490326247524 ↗
Google Scholar ↗

[171] H.L. Tsai, L. Jiang
Multiscale modeling of ultrafast laser-material interactions
Pacific international conference on applications of lasers and optics, vol. 2008, no. 1, Laser Institute of America (2008), pp. 755-760, 10.2351/1.5057121 ↗
View in Scopus ↗ Google Scholar ↗

[172] K. Goda, K.K. Tsia, B. Jalali
Serial time-encoded amplified imaging for real-time observation of fast dynamic phenomena
Nature, 458 (7242) (2009), pp. 1145-1149
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[173] K. Nakagawa, A. Iwasaki, Y. Oishi, R. Horisaki, A. Tsukamoto, A. Nakamura, *et al.*
Sequentially timed all-optical mapping photography (STAMP)
Nature Photonics (2014), 10.1038/NPHOTON.2014.163 ↗
Google Scholar ↗

[174] N.H. Matlis, A. Axley, W.P. Leemans
Single-shot ultrafast tomographic imaging by spectral multiplexing
Nat Commun (2012), 10.1038/ncomms2120 ↗
Google Scholar ↗

[175] E.D. Diebold, B.W. Buckley, D.R. Gossett, B. Jalali
Digitally synthesized beat frequency multiplexing for sub-millisecond fluorescence microscopy
Nature Photonics (2013), 10.1038/NPHOTON.2013.245 ↗
Google Scholar ↗

[176] L. Gao, J. Liang, C. Li, L.V. Wang
Single-shot compressed ultrafast photography at one hundred billion frames per second
Nature (2014), 10.1038/nature14005 ↗
Google Scholar ↗

[177] X. Zeng, S. Zheng, Y. Cai, Q. Lin, J. Liang, X. Lu, *et al.*
High-spatial-resolution ultrafast framing imaging at 15 trillion frames per second by optical parametric amplification
Advanced Photonics (2020), 10.1117/1.AP.2.5.056002 ↗
Google Scholar ↗

[178] J. Liang, L. Zhu, L.V. Wang
Single-shot real-time femtosecond imaging of temporal focusing
Light: Science & Applications (2018), 10.1038/s41377-018-0044-7 ↗
Google Scholar ↗

[179] X. Wang, L. Yan, J. Si, S. Matsuo, H. Xu, X. Hou
High-frame-rate observation of single femtosecond laser pulse propagation in fused silica using an echelon and optical polarigraphy technique

Appl Optics (2014), 10.1364/AO.53.008395 ↗

Google Scholar ↗

[180] A. Ehn, J. Bood, Z. Li, E. Berrocal, M. Aldén, E. Kristensson
FRAME: femtosecond videography for atomic and molecular dynamics
Light: Science & Applications (2017), 10.1038/lsa.2017.45 ↗

Google Scholar ↗

[181] D.N. Vitek, E. Block, Y. Bellouard, D.E. Adams, S. Backus, D. Kleinfeld, *et al.*
Spatio-temporally focused femtosecond laser pulses for nonreciprocal writing in optically transparent materials
Opt Express (2010), 10.1364/OE.18.024673 ↗

Google Scholar ↗

[182] R. Fang, A. Vorobyev, C. Guo
Direct visualization of the complete evolution of femtosecond laser-induced surface structural dynamics of metals
Light: Science & Applications (2017), 10.1038/lsa.2016.256 ↗

Google Scholar ↗

[183] N. Smijesh, K.H. Rao, D. Chetty, I.V. Litvinyuk, R.T. Sang
Plasma plumes produced by laser ablation of Al with single and double pulse schemes
Opt Lett (2018), 10.1364/OL.43.006081 ↗

Google Scholar ↗

[184] B. Bornschlegel, J. Finger
In-situ analysis of ultrashort pulsed laser ablation with pulse bursts
Journal of Laser Micro Nanoengineering (2019), 10.2961/jlmn.2019.01.0015 ↗

Google Scholar ↗

[185] B. Bomschlegel, J. Koller, J. Finger

In-situ analysis of heat accumulation during ultrashort pulsed laser ablation
Journal of Laser Micro Nanoengineering (2020), 10.2961/jlmn.2020.01.2010 ↗
Google Scholar ↗

[186] Van Elsen M. Complexity of Selective Laser Melting: a new optimisation approach.
Google Scholar ↗

[187] S.K. Everton, M. Hirsch, P. Stravroulakis, R.K. Leach, A.T. Clare
Review of in-situ process monitoring and in-situ metrology for metal additive manufacturing
Materials & Design (2016), 10.1016/j.matdes.2016.01.099 ↗
Google Scholar ↗

[188] P.A. Hooper
Melt pool temperature and cooling rates in laser powder bed fusion
Addit Manuf (2018), 10.1016/j.addma.2018.05.032 ↗
Google Scholar ↗

[189] S. Berumen, F. Bechmann, S. Lindner, J.P. Kruth, T. Craeghs
Quality control of laser-and powder bed-based Additive Manufacturing (AM) technologies
Physics Procedia (2010), 10.1016/j.phpro.2010.08.089 ↗
Google Scholar ↗

[190] P. Mercelis, J.P. Kruth, J. Van Vaerenbergh
Feedback control of selective laser melting
Proceedings of the 15th international symposium on electromachining, UNIV NEBRASKA-LINCOLN (2007), pp. 421-426
View in Scopus ↗ Google Scholar ↗

[191] B. Khaleghi, A. Khamis, F.O. Karray, S.N. Razavi

Multisensor data fusion: a review of the state-of-the-art
Information Fusion (2013), 10.1016/j.inffus.2011.08.001 ↗
Google Scholar ↗

[192] R.C. Luo, C.C. Chang, C.C. Lai
Multisensor fusion and integration: theories, applications, and its perspectives
IEEE Sensors Journal (2011), 10.1109/JSEN.2011.2166383 ↗
Google Scholar ↗

[193] L. Kong, X. Peng, Y. Chen, P. Wang, M. Xu
Multi-sensor measurement and data fusion technology for manufacturing process monitoring: a literature review
International Journal of Extreme Manufacturing (2020), 10.1088/2631-7990/ab7ae6 ↗
Google Scholar ↗

[194] Z. Zhang, G. Wen, S. Chen
Multisensory data fusion technique and its application to welding process monitoring
2016 IEEE workshop on advanced robotics and its social impacts (ARSO), IEEE (2016), pp. 294-298, 10.1109/ARSO.2016.7736298 ↗
View in Scopus ↗ Google Scholar ↗

[195] Z. Snow, E.W. Reutzel, J. Petrich
Correlating in-situ sensor data to defect locations and part quality for additively manufactured parts using machine learning
J Mater Process Technol (2022), 10.1016/j.jmatprotec.2021.117476 ↗
Google Scholar ↗

[196] C.E. Zamiela
Deep multi-modal U-net fusion methodology of infrared and ultrasonic images for porosity detection in additive manufacturing

Mississippi State University (2021)

Google Scholar ↗

[197] P. Zhang, L. Kong, W. Liu, J. Chen, K. Zhou
Real-time monitoring of laser welding based on multiple sensors
2008 Chinese control and decision conference, IEEE (2008 Jul 2), pp. 1746-1748, 10.1109/CCDC.2008.4597620 ↗

View in Scopus ↗ Google Scholar ↗

[198] D. You, X. Gao, S. Katayama
Multiple-optics sensing of high-brightness disk laser welding process
Ndt & E International (2013), 10.1016/j.ndteint.2013.07.005 ↗

Google Scholar ↗

[199] F. Deng, Y. Huang, S. Lu, Y. Chen, J. Chen, H. Feng, *et al.*
A multi-sensor data fusion system for laser welding process monitoring
IEEE Access (2020), 10.1109/ACCESS.2020.3015529 ↗

Google Scholar ↗

[200] J. Li, X. Zhang, Q. Zhou, F.T. Chan, Z. Hu
A feature-level multi-sensor fusion approach for in-situ quality monitoring of selective laser melting
Journal of Manufacturing Processes (2022), 10.1016/j.jmapro.2022.10.050 ↗

Google Scholar ↗

[201] A. Tsanousa, E. Bektsis, C. Kyriakopoulos, A. Gómez González, U. Leturiondo, I. Gialampoukidis, *et al.*
A review of multisensor data fusion solutions in smart manufacturing: systems and trends
MdpiCom (2022), 10.3390/s22051734 ↗

Google Scholar ↗

[202] B.D. Majumder, J.K. Roy, S. Padhee
Recent advances in multifunctional sensing technology on a perspective of multi-sensor system: a review
IEEE Sensors Journal (2018), 10.1109/JSEN.2018.2882239 ↗
Google Scholar ↗

[203] F. Castanedo
A review of data fusion techniques
The Scientific World Journal, 2013 (1) (2013), Article 704504
View in Scopus ↗ Google Scholar ↗

[204] D. You, X. Gao, S. Katayama
Multisensor fusion system for monitoring high-power disk laser welding using support vector machine
IEEE Trans Industr Inform (2014), 10.1109/TII.2014.2309482 ↗
Google Scholar ↗

Microstructure manipulation by laser-surface remelting of a full-Heusler compound to enhance thermoelectric properties

Leonie Gomell, ..., Baptiste Gault

Acta Materialia • Volume 223 • 2022 • Article 117501

View PDF

Research article Open access

Laser intensity profile as a means to steer microstructure of deposited tracks in Directed Energy Deposition

Scholte J.L. Bremer, ..., Gert-willem R.B.E. Römer

Materials & Design • Volume 227 • 2023 • Article 111725

View PDF

Research article Full text access

Influence of focus positions on underwater femtosecond laser dicing of silicon wafer

Wang Chengjin, ..., Zheng Hongyu

Journal of Manufacturing Processes • Volume 92 • 2023 • Pages 189-195

View PDF

Research article Open access

A physics-motivated geometric method for overheating prevention in topology optimization for additive manufacturing

Manabendra Nath Das, ..., Can Ayas

Computer Methods in Applied Mechanics and Engineering • Volume 447 • 2025 • Article 118363

View PDF

Mini review Full text access

Dermacentor variabilis (American dog tick)

Sarah Myers, Kathryn Duncan

Trends in Parasitology • Volume 40, Issue 3 • 2024 • Pages 273-274

View PDF

Cited by (21)

Research article

Femtosecond laser ablation of SiC fiber-reinforced titanium composites: Mechanisms of ablation and surface morphology evolution

Wang L., ..., Yuan S.

Composites Part B Engineering • Volume 311 • 2026 • Article 113266

Show abstract

Research article

Exploring material removal dynamics during femtosecond laser micromachining by in-situ acoustic emission monitoring with physics-based and data-driven analysis

Yildirim K., ..., Castagne S.J.

Optics and Laser Technology • Volume 192 • 2025 • Article 113721

Show abstract

Research article

GHz-selectable passive harmonic mode-locking fiber laser based on phosphorus-assisted hybrid mechanism

Pan R., ..., Dong K.

Optics and Laser Technology • Volume 189 • 2025 • Article 113064

Show abstract

Research article

Petri Net-Based Information Enhancement for Quality Monitoring in the Cigarette Manufacturing Process

Chen L., ..., Yu Z.

Measurement Journal of the International Measurement Confederation • Volume 255 • 2025 • Article 117999

Show abstract

Research article

Mechanism and technology of laser adaptive cleaning for aircraft skin multilayer coatings of airbus A320

Yang G.Y., ..., Zhao S.